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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 227 — Past Questions & Materials
by Renance

Course: EMT 227

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 7

Levels: 200

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Paper 1 of 7 — EMT 227 Limnology and oceanography 2023/2024

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

SECOND SEMESTER 2023/2024 DEGREE EXAMINATION

EMT 227: Limnology and Oceanography (2 units) Time: 2 Hours

INSTRUCTION: Answer two (2) questions from each Section.

Section A

Kh. With the aid of a diagram, describe the water cycle.

ib. List five types of aquatic organism.

le. Give ten importance of water.

2a. Explain in details the two facets of hydrobiology.

2b. List five characteristics of the world ocean and five resources found in the ocean.

20: How are lakes formed?

a. Why are artificial lakes constructed? ;

3b. What is lake zonation? With the aid of a diagram, explain the different zones in a lake and its importance in understanding the lake's ecological balance.

Ue What are the vital roles of lakes in maintaining ecological balance.

ecological balance.

Section B

Ja. What are ocean currents, and what are the main types of currents based on their causes?

1b. Describe the role of wind in the formation of ocean waves and identify different types of waves.

1c. Describe the vertical temperature distribution in the ocean and explain the significance of the thermocline.

2a. How do ocean currents influence global climate and marine navigation?

2b. How does light penetration affect marine ecosystems, and what is the significance of the euphotic zone?

2c. What is the relationship between seawater conductivity and salinity, and how does temperature affect conductivity?

3a. What causes tides, and what are the main effects of tidal movements on coastal areas?

3b. How does solar radiation affect the temperature of seawater, and why are sea surface temperatures higher near the equator compared to the poles?

3c. What adaptations do marine organisms have to cope with changes in salinity?

Paper 2 of 7 — EMT 227, LIMNOLOGY AND OCEANOGRAPHY past question

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
SECOND SEMESTER 2022/2023 DEGREE EXAMINATION

EMT 227: Limnology and Oceanography (2 units) **Time: 2 hours**

INSTRUCTION: Answer four (4) questions in all.

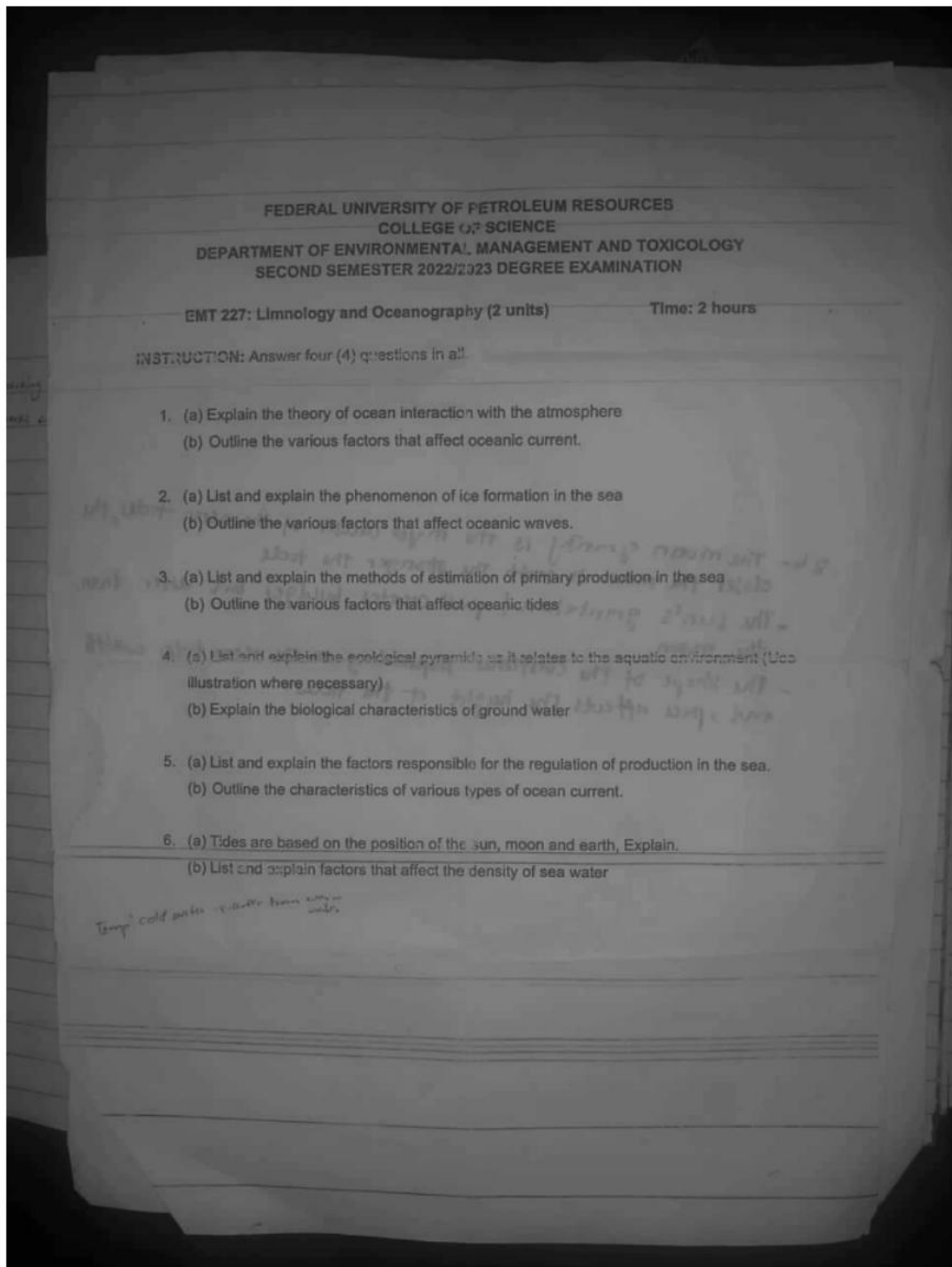
1. (a) Explain the theory of ocean interaction with the atmosphere
 (b) Outline the various factors that affect oceanic current.
2. (a) List and explain the phenomenon of ice formation in the sea
 (b) Outline the various factors that affect oceanic waves.
3. (a) List and explain the methods of estimation of primary production in the sea
 (b) Outline the various factors that affect oceanic tides
4. (a) List and explain the ecological pyramid as it relates to the aquatic environment (Use illustration where necessary)
 (b) Explain the biological characteristics of ground water
5. (a) List and explain the factors responsible for the regulation of production in the sea.
 (b) Outline the characteristics of various types of ocean current.
6. (a) Tides are based on the position of the sun, moon and earth, Explain.
 (b) List and explain factors that affect the density of sea water

is denser than water

Paper 3 of 7 — EMT 227

Level 200 • Second Semester • PastQuestion

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Paper 4 of 7 — EMT 227 Notes

Level 200 • Second Semester • Handout

Limnology & Oceanography

INTRODUCTION

Hydrobiology is the science of life and life processes in water.

Much of modern hydrobiology can be viewed as a sub-discipline of ecology but the sphere of hydrobiology includes taxonomy, economic biology, industrial biology, morphology, physiology etc. The one distinguishing aspect is that all relate to aquatic organisms. Much work is closely related to limnology and oceanography and can be divided into lotic system ecology (flowing waters) and lentic system ecology (still waters).

Hydrobiology can also be defined as the study of water populations, their interrelations with habitat and significance for the transformation of energy and matter, and the biological productivity of the ocean, seas, and inland waters.

Hydrobiology is primarily an ecological science. The living conditions in water are determined by the physico-geographic characteristics of the body of water. Many of these features—for example, the chemical composition of the water (in particular, the composition and amount of biogenic elements and dissolved gases, the nature of bottom sediments, and the transparency of the water)—are strongly influenced by aquatic organisms and are often determined by their life processes. Therefore, to the extent to which hydrobiology studies the role of living phenomena in the context of the aggregate of interdependent processes of the aquatic medium, it has problems in common with the combined disciplines of limnology and oceanography. Such matters as the biological structure of the ocean, the bio-limnological and bio-oceanological typology of bodies of water and water masses, and the regular patterns of the recycling of substances and the flow of energy are treated at this level of research.

Hydrobiology is greatly concerned with:

- ❖ Establishing a scientific basis for the rational exploitation of biological resources of the waters. This is bound up in many ways with the requirements of the marine and freshwater fishing industries, farm pond fishery, and the use of aquatic invertebrates and mammals (food-fish hydrobiology).
- ❖ The use of the continental surface fresh water for drinking purposes
- ❖ Industrial supply of water.
- ❖ The protection of natural water against pollution
- ❖ The self-purification of polluted water.
- ❖ The role of aquatic organisms as agents of self-purification
- ❖ The biological methods of treating waste waters.
- ❖ Presence of certain indicator organisms (biological analysis of water quality) through hydro-biological methods which is used to evaluate the extent of water pollution.
- ❖ Technical hydrobiology - concerns with the problems of biological interference with the water supply and operation of ships (e.g. the fouling by microorganisms and sessile animals)

of the hulls of ships, various apparatus and hydraulic structures, and the pipes and water supply lines of electro-thermal power plants; the overgrowth of aquatic plants in reservoirs, and the damage to ships and port installations by wood borers and rock borers).

- ❖ Navigational hydrobiology - the study of biological interference (for example, bioluminescence) with naval activities;
- ❖ Agricultural hydrobiology includes the study of the role of hydrobionts in the fertilization of rice paddies and the possibilities of fish breeding in these waters.

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- ❖ New problems arising from water supply, such as the need to determine the effect of plankton on the absorption and scattering of sound—information indispensable to specialists in underwater acoustics.

Hydrobiology used to be a branch of biology!

But the natural communities of aquatic organisms that constitute the water population did not receive systematic study until the latter part of the 19th century. This study led to the eventual separation of hydrobiology from botany and zoology, which had long been engaged in the study of both terrestrial and aquatic organisms.

Of great importance to the progress of hydrobiology as a science with its own objects of study, methods, and tasks were the first quantitative investigations of plankton (chiefly tiny water-dwelling organisms and a life-form specific to the aquatic environment), initiated in the 1880's by the German scientist W. Hansen. Using the example of Kiel Bay, he showed the need for quantitative data on plankton as a source of food for commercial fish and as the basis of the biological productivity of the sea. Quantitative studies were subsequently begun on benthos, organisms that live at the bottom of bodies of water (again primarily for the sake of promoting the fishing industry). The quantitative study of benthos became widespread after bottom grabs (sample-taking devices) came into use; these were first suggested for marine research in 1911 by the Danish scientist C. Petersen and for freshwater studies by the Swedish scientist S. Eckmann.

Quantitative methods for investigating natural communities of aquatic organisms in order to determine the number (density) of individuals of various species and their biomass have received very broad attention in hydrobiology. Many specialized hydrobiological devices, such as plankton nets, plankton catchers, plankton samplers, and bottom grabs of various designs, are used for this purpose.

Aquatic Organisms

1. Plankton: microscopic water-dwelling plants and animals
2. Benthos: organisms that live at the bottom of water bodies
3. Nekton: the larger free-swimming animals that are capable of resisting the current, such as fish, squid, aquatic mammals, etc.
4. Periphytons or epibionts: The communities of organisms living on the surface of submerged objects.
5. Neuston: communities of animals and plants characteristically of the surface of the water

6. Pleuston: semi-aquatic submerged organisms
7. Epineuston: organisms that skim over or lie upon the surface film
8. Hyponeuston: organisms that live under the surface film but remain in close contact with it.

The first, primarily floristic, faunistic, and biogeographical state in hydrobiological research arose from the need to study the composition and distribution of the species of organisms populating the seas and inland waters. This task, especially in regard to less known regions and taxonomic groups, is still important. The English marine expedition on the Challenger (December 1872 to May 1876) was of exceptional importance in laying the foundation for the study of life at great depths.

As information accumulated on the composition of aquatic populations, attention was focused on elucidating the ecological conditions under which particular biocenoses and habitats of individual species of aquatic organisms were formed.

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Much attention is devoted to furthering the understanding of the importance of biological phenomena in the classification of natural waters, as well as to the theory of biological productivity, the patterns of the biotic cycle, and the flow of energy in aquatic communities.

The next task of hydrobiological research is to clarify the function of aquatic organisms in the processes at work in their environment. This information is needed in order to control biological productivity and self-purification as well as to make rational use of biological resources. The functional characteristics of aquatic organisms can be determined only by experimental investigations of their metabolism, growth, nutrition, and chemical and biochemical composition.

The solution of a number of hydrobiological problems often requires research at the most varied levels—from the molecular, cellular, and organismic to the populational and biocenotic. For example, to determine the causes of excessive development of phytoplankton (so-called water-bloom), it is necessary to take into account both the interaction of different algal and microbial species through specific metabolites excreted into the water and the cycle of biogenic elements (for example, nitrogen and phosphorus). This cycle depends upon the properties of the body of water as a whole and upon the runoff from its drainage basin.

The regular pattern of the interdependence of all phenomena in a body of water, which is an integrated natural object, was emphasized at the turn of the century in the classic works of the Swiss limnologist F. Forel. In the 1920's, A. Thienemann of Germany and E. Naumann of Sweden demonstrated the possibility of subdividing lakes, as well as other bodies of water, into biolimnological types (for example, into oligotrophic and eutrophic). The problems of typology and the classification of natural bodies of water are still under study.

The great complexity and heterogeneity of the natural phenomena with which hydrobiology deals have led to the use of many research techniques. Examples of these techniques include measurement through radioactive carbon of the intensity of plankton photosynthesis (proposed by the Danish scientist E. Stiemann-Nielsen), a method which already has yielded data for gauging the primary production of the ocean and of the hydrosphere as a whole;

spectrophotometric methods of determining the chlorophyll content of plankton; and methods of studying the role of aquatic bacteria (principally by the Soviet scientists E. L. Isachenko, V. S. Butkevich, A. S. Razumov, S. I. Kuznetsov, Iu. and I. Sorokin). Sampling and observation in connection with marine and some freshwater hydrobiological research are aided by aqualungers; underwater television and photography are used at great depths, and the distribution of plankton and other aquatic organisms is traced with the help of echo-sounding devices. The latest physical methods are used in the study of deep-sea bioluminescence and to gain an understanding of the interconnections between the processes at work in aquatic ecosystems. Mathematical simulation and electronic computers are also employed.

The growing influence of theoretical studies upon the solution of practical problems is characteristic of hydrobiology. Hydrobiological knowledge and methods are widely used to evaluate food reserves of bodies of water as bases for their fish yield. These methods are also used in commercial prospecting and in fish breeding. The great progress of hydrobiology in the USSR has made it possible to begin to influence the biological productivity of bodies of water. Before the war, L. A. Zenkevich directed experimental work on enriching the bottom fauna of the Caspian Sea, to which the polychaeteworm *Nereis*, an important source of food for sturgeon, was introduced. Organisms used as foods (chiefly crustaceans, such as mysids) have been successfully adapted to many reservoirs and some lakes (for example, Lake Balkhash). As a result of hydrobiological research, new methods have been proposed for increasing the fish

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yield of ponds through the addition of inorganic fertilizers. These fertilizers are now regularly used in pond fish farming, the productivity of which has thereby been greatly increased.

The field of sanitary hydrobiology studies the effect upon aquatic organisms and their communities of the toxic substances in industrial effluent, the mechanism of the biological self-purification of water, and other matters relating to the pressing problem of supplying man's growing need for pure water.

Broad-scale research in marine hydrobiology is conducted by the Nigerian Institute of Oceanography Marine Research (NIOMR), Lagos.

The most important international organizations concerned with hydrobiology are the permanent International Council for the Exploration of the Sea (Copenhagen), which publishes the *Journal de Conseil* (since 1926), and the International Association of Theoretical and Applied Limnology, which was founded in 1922 and regularly sponsors congresses of limnologists (the 18th congress was held in 1971). The oldest international journal of hydrobiology is the *Archiv für Hydrobiologie* (Stuttgart, since 1906). Other publications include the *Internationale Revue der gesamten Hydrobiologie und Hydrographie* (Leipzig, since 1908) and *Limnology and Oceanography* (USA, since 1956).

Classification Of Waterbodies

Aquatic ecosystems can be classified into 2 based on the presence or absence of flow in its water. These are lentic and lotic aquatic ecosystems.

a. Based on presence or absence of flow in its water

- i. Lentic aquatic ecosystems are those which contain stagnant water e.g. ponds, lakes, etc.
- ii. Lotic aquatic ecosystems are those which contain flowing water e.g. streams, rivers, creeks, etc.

b. Based on salt content

- i. Freshwater e.g. rivers, stream, lakes, bogs, etc.
- ii. Marine water: Oligohaline, mesohaline & polyhaline

Water salinity

Fresh water Brackish water Saline water Brine

< 0.05% 0.05 – 3% 3 – 5% > 5%

< 0.5 ‰ 0.5 – 30 ‰ 30 – 50 ‰ > 50 ‰

Lentic Ecosystem

Lentic refers to standing or still water. It is derived from the Latin lentus, which means sluggish. Lentic systems are diverse, ranging from a small, temporary rainwater pool a few inches deep to Lake Baikal, which has a maximum depth of 1740 m. The general distinction between pools/ponds and lakes is vague, but Brown states that ponds and pools have their entire bottom surfaces exposed to light, while lakes do not. In addition, some lakes become seasonally stratified. Ponds and pools have two regions: the pelagic open water zone, and the benthic zone, which comprises the bottom and shore regions. Since lakes have deep bottom regions not exposed to light, these systems have an additional zone, the profundal. These three areas can

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have very different abiotic conditions and, hence, host species that are specifically adapted to live there.

Lotic Ecosystem

Lotic refers to flowing water, from the Latin lotus, past participle of lavare, to wash. Lotic ecosystems can be contrasted with lentic ecosystems, which involve relatively still terrestrial waters such as lakes and ponds. Together, these two fields form the more general study area of freshwater or aquatic ecology.

Lotic waters can be diverse in their form, ranging from a spring that is only a few centimeters wide to a major river that is kilometers in width. Despite these differences, the following unifying characteristics make the ecology of running waters unique from that of other aquatic habitats.

- Continuous unidirectional
- Presence of water current, with the velocity of water changing with changes in the water volume.
- Thorough mixing of contents, therefore stratification and stagnation are altogether absent.
- The physic-chemical parameters of water quality are in a state of perpetual change
- The amount of DO is usually less than that of ponds or lakes because they receive additional organic matter that adds to their BOD
- Turbidity usually limits light penetration to deeper zone of such systems
- Productivity is low

- Most algae grow attached to rocks and other organisms on the bottom

Importance of Water

1. Drinking
2. Domestic use (cooking, washing, etc)
3. Industrial use (cooling, washing, etc)
4. Transportation
5. Irrigation
6. Source of food (periwinkle, fishes, mollusks, etc)
7. Research
8. Relaxation (skiing, boating, yachting, water polo, swimming, etc)
9. Power generation (Hydro-electricity)
10. Major component of protoplasm
11. Lakes and sea also moderates the surrounding region's temperature and climate because water has a very high specific heat capacity ($4,186 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$). In the daytime, a lake can cool the land beside it with local winds, resulting in a sea breeze; in the night, it can warm it with a land breeze.

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FACETS OF HYDROBIOLOGY

Hydrobiology can be subdivided into two, Oceanography and Limnology. Oceanography is the study of the oceans, limnology deals with inland waters.

Limnology

Limnology is the study of inland bodies of water (lakes, ponds, reservoirs, streams, rivers, wetlands, estuaries) and their related ecosystems.

Limnology deals with freshwaters within continental boundaries. Limnology plays a major role in water use and distribution as well as in wildlife habitat protection. Limnologists work on lake and reservoir management, water pollution control, stream and river protection, artificial wetland construction, fish and wildlife enhancement.

Oceanography

It is the study of the ocean, its surroundings, and life within it.

Oceanography is not so much a science as a collection of scientists who find common cause in trying to understand the complex nature of the ocean. In the vast salty seas that encompass the earth, there is plenty of room for persons trained in physics, chemistry, biology, geology and engineering to practice their specialties. Thus, an oceanographer is any scientifically trained person who spends much of his (or her) career on ocean problems. Willard Bascom (1980).

Oceans cover 70.8% of earth's surface. Oceans are fringed by smaller, adjoining bodies of water such as seas, gulfs, bays, bights, and straits.

Oceans contain about 97% of all liquid water and moderate the earth's climates though they are too salty for human use. More than 90% of the earth's biomass is found in the oceans. In tropical seas, surface waters are warmed by the sun, diluted by rain water and run off from land and aerated by wave action. In higher latitudes, surface waters are cold and denser than waters

flowing from the equator. The dense water sinks to the bottom of deep ocean basins and flows towards the equator. The warm surface water of the tropics flows or stratifies on top of the cold dense water from higher latitudes. Sharp boundaries form between different densities, salinities and temperatures retarding mixing between layers.

Marine ecosystems cover approximately 71% of the Earth's surface. They generate 32% of the world's net primary production. They are distinguished from freshwater ecosystems by the presence of dissolved compounds especially salts in the water. Approximately 85% of the dissolved materials in seawater are sodium and chlorine. Seawater has an average salinity of 35 parts per thousand (ppt) of water but the actual salinity varies among different marine ecosystems.

The five principal oceans: Pacific, Atlantic, Indian, Arctic & Antarctic (Southern) Ocean

Study of oceanography can be sub-divided into the following category:

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1. Physical Oceanography: How and why ocean currents flow, air-sea interactions such as the generation of waves by wind.
2. Chemical Oceanography: Composition of sea water and the processing controlling and altering its composition, including marine pollution
3. Geological Oceanography: Study of rock structure in the ocean basin, properties of rocks such as magnetism, occurrence of earthquakes
4. Ocean Engineering: Design and installation of oceanographic instrumentation and vehicles
5. Biological Oceanography: Organisms that live in the ocean and their relationship to the environment

Characteristics of the World's Oceans

- Average depth of ocean- 4000m
- Average temperature- 39°F (3.9°C)
- Age of ocean- 4 billion years old
- Covers 71% of the Earth's surface
- Average salinity- 35 ‰
- Tallest Mountain: Hawaii's Mauna Kea rises 10,203 m (33,476 ft) from base to peak. Everest is only 8,848m, while the Deepest spot- Marianas Trench (10,912m)
- Age of the ocean basin: 200 my old
- The Earth has an area of approximately 197 million square miles of which 140 million square miles is water.
- The Earth is broken up into hemispheres...Northern and Southern.
- In the Northern Hemisphere, 61% is water and 39% is land, thus it is called the "Land Hemisphere".
- In the Southern Hemisphere, 81% is water and 19% is land, thus it is called the "Water Hemisphere".

Resources from Oceans and Seawater:

Some resources that humans currently use from seawater are:

1. sea salt (halite),
2. magnesium (a light metal used in the making of light bulbs),
3. bromine (gasoline additives/fireproofing materials).
4. Humans also desalinate seawater to get fresh water. This process is time-consuming, expensive and is not currently economically feasible. Many desalinization plants operate, but few are productive on a commercial scale.
5. Gold is present in very low concentrations, but not economic enough to actually "mine" for profit. Many tons of water and sediment would have to be processed in order to get a single ounce of gold.

Ocean Topography:

Oceanographers studying the oceans and ocean floor have delineated three (3) major units:

1. Continental margin (Shelf, Slope & Rise)
2. Ocean basin floor
3. Mid-oceanic ridges

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As technology develops, mapping improves. Currently, echo-sounding devices are used, which are slow and tedious.

1. Continental margin includes:

- > Continental shelf -- very gentle slope (submerged land)
- > Continental slope -- steep slope on edge of continental shelf.
- > Continental rise -- gentle slope where trenches do not exist

Deep sea fans exist where sediment is accumulated and falls off of the continental slope.

Mixture of sediment-laden heavy water forms submarine turbidity currents.

2. The Deep Sea includes:

- > Deep ocean trenches (deepest part of ocean) some as deep as 36,000 feet
- > Abyssal plains are the flat portions of the deep ocean, likely to be the flattest portions of the earth.
- > Isolated volcanic peaks (mantle hot spots) are referred to as "seamounts". Likewise volcanic cones near mid-oceanic ridges are called seamounts as well. Harry Hess called these "guyots".
- > Mid-oceanic ridges occur where divergent plate boundaries occur (Mid-Atlantic) and the narrow region at the ridge crest is called the rift zone.
- > Coral reefs and atolls develop usually in relatively shallow water where light is present and organisms can thrive on nutrients and food sources in a warm water setting. Atolls develop in response to a sinking of the oceanic crust.

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ORIGIN OF LAKES

A lake is a naturally occurring, relatively large and fixed body of water on the earth's surface.

It is localized in a basin or interconnected basins surrounded by dry land. Lakes lie completely

on land and are separate from the ocean, although, like the much larger oceans, they form part of the Earth's water cycle by serving as large standing pools of storage water. Most lakes are freshwater and account for almost all the world's surface freshwater, but some are salt lakes with salinities even higher than that of seawater. Lakes vary significantly in surface area and volume.

Lakes are typically much larger and deeper than ponds, which are also water-filled basins on land, although there are no official definitions or scientific criteria distinguishing the two. and lakes are also distinct from lagoons, which are shallow tidal pools dammed by sandbars at coastal regions of oceans or large lakes. Most lakes are fed by springs, and both fed and drained by creeks and rivers, but some lakes are endorheic (without any outflow), while volcanic lakes are filled directly by precipitation runoffs and do not have any inflow streams.

- ❖ Natural lakes are generally found in mountainous areas (i.e. alpine lakes),
- ❖ dormant volcanic craters,
- ❖ rift zones and
- ❖ areas with ongoing glaciation. Other lakes are found in
- ❖ depressed landforms or
- ❖ along the courses of mature rivers, where a river channel has widened over a basin formed by eroded floodplains and wetlands. Some parts of the world have many lakes formed by the chaotic drainage patterns left over from the last ice age.

All lakes are temporary over long periods of time, as they will slowly fill in with sediments or spill out of the basin containing them.

Artificially controlled lakes are known as reservoirs, and are usually constructed for the following purposes:

- ❖ Industrial uses
- ❖ Agricultural
- ❖ Hydro-electric power generation
- ❖ Supplying domestic drinking water
- ❖ Ecological uses
- ❖ Recreational purposes
- ❖ Other human activities.

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MORPHOLOGY AND ZONATION

In biology, morphology is the study of the shapes and arrangement of parts of organisms, in order to determine their function, development, and how they may have been shaped by evolution.

Lake Morphology refers to the shape and volume of a lake. It often relates to a lake's age, origins, chemical makeup, and the organisms that live within it. Lake shape is influenced by the surrounding landscape and controls much of what goes on physically and biologically underwater.

Zonation is the spatial arrangement of distinct ecological communities based on predominant flora and fauna across a gradient, influenced by both biotic and abiotic factors.

Zonation in Freshwater Lakes

Littoral, Limnetic, Profundal, and Benthic Zones

Freshwater lakes exhibit a distinct vertical zonation pattern influenced by light penetration, temperature, and depth.

- **Littoral Zone:** This is the shallow zone near the shore where light penetrates to the bottom, allowing for the growth of rooted plants. Aquatic plants, snails, and small fishes are common inhabitants, thriving in the warmer waters and benefiting from the abundance of nutrients.
- **Limnetic Zone:** Beyond the littoral is the limnetic zone, which encompasses the open water where light penetration supports phytoplankton but not rooted plants. Zooplankton and free-swimming fish species dominate this zone, relying on the phytoplankton for sustenance.
- **Profundal Zone:** Deeper still is the profundal zone, characterized by colder temperatures and minimal light penetration. Here, the absence of light means photosynthesis is non-existent, leading to lower oxygen levels. Organisms in this zone, such as certain fish and benthic invertebrates, are adapted to these low-oxygen conditions.
- **Benthic Zone:** This is the bottom-most layer, consisting of the lake bed. Decomposers, detritivores, and benthic invertebrates, which feed on detritus and other organic materials, inhabit this zone.

Importance of Zonation

Zonation, the division of ecosystems into distinct zones based on the distribution of species and environmental conditions, plays a pivotal role in understanding and maintaining ecological balance.

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1. **Biodiversity Conservation:** Zonation patterns highlight areas of rich biodiversity, enabling targeted conservation efforts. Different zones often harbor unique species adapted to specific conditions, contributing to the overall biodiversity of an area.
2. **Ecological Stability:** Each zone in an ecosystem plays a specific role in maintaining the ecological balance. For instance, the species in a particular zone might be crucial for nutrient cycling, soil stabilization, or providing habitat for other organisms.
3. **Resource Management:** Understanding zonation helps in the sustainable management of resources. For example, in coastal areas, different zones might have varying fish populations, guiding sustainable fishing practices.
4. **Predicting Ecological Responses:** Recognizing zonation patterns can help predict how ecosystems might respond to changes, such as climate change or human interventions. This is crucial for proactive environmental management.
5. **Scientific Research:** Zonation provides a structured framework for ecological research.

Scientists can study how species adapt to conditions in different zones, leading to insights into evolution, physiology, and behavior.

6. Environmental Education: Zonation serves as a tangible demonstration of ecological principles, making it an effective tool for environmental education. It offers a clear visual representation of how species distribution and abundance vary with changing conditions.

7. Cultural and Economic Significance: Many human communities rely on specific zones for their livelihood, be it through agriculture, fishing, or tourism. Understanding zonation can guide sustainable practices that benefit both the environment and local communities.

8. Habitat Restoration: In areas affected by human activities or natural disasters, knowledge of natural zonation patterns is crucial for effective habitat restoration. It guides the reintroduction of species and recovery efforts in a way that mimics natural systems.

9. Climate Change Mitigation: Zonation patterns can offer insights into areas more vulnerable to the impacts of climate change. This can guide mitigation efforts, such as the establishment of buffer zones or conservation areas.

10. Indicator of Environmental Health: Changes in zonation patterns can serve as early warning signs of environmental degradation or shifts in ecological balance. Monitoring these changes can prompt timely interventions.

In essence, zonation is not just a passive arrangement of species; it's a dynamic system that reflects the health, diversity, and adaptability of ecosystems. Recognizing its importance is crucial for informed ecological management and conservation efforts.

Why is zonation important in ecosystems?

Zonation is crucial as it reflects the adaptability and resilience of species to varying environmental conditions, ensuring biodiversity, ecological balance, and the proper functioning of ecosystems.

How does altitude affect zonation?

Altitude influences temperature, oxygen availability, and atmospheric pressure. As altitude increases, there's a shift in the species of flora and fauna due to these changing conditions, leading to distinct zones like subalpine, alpine, and nival.

What factors contribute to zonation in marine ecosystems?

Factors such as light penetration, salinity, temperature, wave action, and substrate type play significant roles in determining zonation in marine ecosystems.

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How is zonation observed in mangrove forests?

Mangrove forests display zonation with distinct zones like the seaward, middle, and landward zones. Each zone has specific mangrove species adapted to the tidal and salinity conditions of that zone.

Why do some species only exist in specific zones?

Some species are specially adapted to the environmental conditions of a particular zone. Their physiological, behavioral, and morphological adaptations restrict them to that zone for optimal survival.

Can human activities alter zonation patterns?

Yes, human activities like deforestation, pollution, and urbanization can disrupt natural zonation patterns, leading to habitat loss and reduced biodiversity.

What is the difference between the littoral and limnetic zones in freshwater lakes?

The littoral zone is the shallow area near the shore with rooted plants, while the limnetic zone is the open water area where light supports phytoplankton but not rooted plants.

How does competition among species influence zonation?

Competition for resources can lead to the dominance of certain species in specific zones. If one species outcompetes others, it can establish itself more prominently, influencing the zonation pattern.

Are zonation patterns permanent?

No, zonation patterns can change over time due to factors like climate change, natural disturbances, and human interventions. They reflect the dynamic nature of ecosystems and their adaptability to changing conditions.

The depth to which light can reach in lakes depends on turbidity, determined by the density and size of suspended particles. A particle is in suspension if its weight is less than the random turbidity forces acting upon it. These particles can be sedimentary or biological in origin and are responsible for the color of the water. Decaying plant matter, for instance, may be responsible for a yellow or brown color, while algae may cause greenish water. In very shallow water bodies, iron oxides make water reddish brown.

Biological particles include algae and detritus. Bottom-dwelling detritivorous fish can be responsible for turbid waters, because they stir the mud in search of food. Piscivorous fish contribute to turbidity by eating plant-eating (planktonivorous) fish, thus increasing the amount of algae (see aquatic trophic cascade). The light depth or transparency is measured by using a Secchi disk, a 20-centimeter (8 in) disk with alternating white and black quadrants. The depth at which the disk is no longer visible is the Secchi depth, a measure of transparency. The Secchi disk is commonly used to test for eutrophication.

A lake moderates the surrounding region's temperature and climate because water has a very high specific heat capacity ($4,186 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$). In the daytime, a lake can cool the land beside it with local winds, resulting in a sea breeze; in the night, it can warm it with a land breeze.

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HYDROLOGIC CYCLE

The water cycle (hydrologic or H₂O cycle): describes the continuous movement of water on, above and below the surface of the Earth.

Water can change states among liquid, vapor, and ice at various places in the water cycle. Although the balance of water on Earth remains fairly constant over time, individual water molecules can come and go.

The Hydrological Cycle

Description of the Hydrologic Cycle

The sun: drives the water cycle, heats water in oceans and seas.

Water evaporates as water vapor into the air. Ice and snow can sublime directly into water vapor.

Evapo-transpiration is water transpired from plants and evaporated from the soil. Rising air currents take the vapor up into the atmosphere where cooler temperatures cause it to condense into clouds. Air currents move water vapor around the globe; cloud particles collide, grow, and fall out of the sky as precipitation. Some precipitation falls as snow or hail, and can accumulate as ice caps and glaciers, which can store frozen water for thousands of years. Snowpacks can thaw and melt, and the melted water flows over land as snowmelt. Most water falls back into the oceans or onto land as rain, where the water flows over the ground as surface runoff. A portion of runoff enters rivers in valleys in the landscape, with streamflow moving water towards the oceans. Runoff and groundwater are stored as freshwater in lakes. Not all runoff flows into rivers, much of it soaks into the ground as infiltration. Some water infiltrates deep into the ground and replenishes aquifers, which store freshwater for long periods of time. Some infiltration stays close to the land surface and can seep back into surface-water bodies (and the ocean) as groundwater discharge. Some groundwater finds openings in the land surface and comes out as freshwater springs. Over time, the water returns to the ocean, where our water cycle started.

Different Processes

Precipitation: Condensed water vapor that falls to the Earth's surface. Most precipitation occurs as rain, but also includes snow, hail, fog drip, graupel, and sleet. Approximately 505,000 km³ (121,000 cu mi) of water fall as precipitation each year, 398,000 km³ (95,000 cu mi) of it over the oceans.

Canopy interception: The precipitation that is intercepted by plant foliage and eventually evaporates back to the atmosphere rather than falling to the ground.

Snowmelt : The runoff produced by melting snow.

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Runoff : The variety of ways by which water moves across the land. This includes both surface runoff and channel runoff. As it flows, the water may seep into the ground, evaporate into the air, become stored in lakes or reservoirs, or be extracted for agricultural or other human uses.

Infiltration: The flow of water from the ground surface into the ground. Once infiltrated, the water becomes soil moisture or groundwater.

Subsurface Flow : The flow of water underground, in the vadose zone and aquifers. Subsurface water may return to the surface (e.g. as a spring or by being pumped) or eventually seep into the oceans. Water returns to the land surface at lower elevation than where it infiltrated, under the force of gravity or gravity induced pressures. Groundwater tends to move slowly, and is replenished slowly, so it can remain in aquifers for thousands of years.

Evaporation: The transformation of water from liquid to gas phases as it moves from the ground or bodies of water into the overlying atmosphere. The source of energy for evaporation is primarily solar radiation. Evaporation often implicitly includes transpiration from plants, though together they are specifically referred to as evapotranspiration.

Sublimation: The state change directly from solid water (snow or ice) to water vapor.

Advection: The movement of water — in solid, liquid, or vapor states — through the atmosphere. Without advection, water that evaporated over the oceans could not precipitate over land.

Condensation: The transformation of water vapor to liquid water droplets in the air, creating clouds and fog.

Transpiration: The release of water vapor from plants and soil into the air. Water vapor is a gas that cannot be seen.

4.1.3 Residence times

The residence time of a reservoir within the hydrologic cycle is the average time a water molecule will spend in that reservoir (see table below). It is a measure of the average age of the water in that reservoir.

Average reservoir residence times

Reservoir Average residence times

Antarctica 20,000 years

Oceans 3,200 years

Glaciers 20-100 years

Seasonal; snow cover 2-6 months

Soil moisture 1-2 months

Groundwater (shallow) 100-200 years

Groundwater (deep) 10,000 years

Lakes 50-100 years

Rivers 2-6 months

Atmosphere 9days

Groundwater can spend over 10,000 years beneath Earth's surface before leaving. Particularly old groundwater is called fossil water. Water stored in the soil remains there very briefly, because it is spread thinly across the Earth, and is readily lost by evaporation, transpiration, stream flow, or groundwater recharge. After evaporating, the residence time in the atmosphere is about 9 days before condensing and falling to the Earth as precipitation.

The major ice sheets - Antarctica and Greenland - store ice for very long periods. Ice from Antarctica has been reliably dated to 650,000 years before present, though the average residence time is shorter.

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Human activities that alter the water cycle include:

- Agriculture
- Industry
- Alteration of the chemical composition of the atmosphere
- Construction of dams
- Deforestation and afforestation
- Removal of groundwater from wells

- Water abstraction from rivers
- Urbanization

Effects of Water Cycle

i. Effects on climate

The water cycle is powered from solar energy. 86% of the global evaporation occurs from the oceans, reducing their temperature by evaporative cooling. Without the cooling, the effect of evaporation on the greenhouse effect would lead to a much higher surface temperature of 67 °C (153 °F), and a warmer planet.

ii. Effects on biogeochemical cycling

While the water cycle is itself a biogeochemical cycle, flow of water over and beneath the Earth is a key component of the cycling of other biogeochemicals. Runoff is responsible for almost all of the transport of eroded sediment and phosphorus[14] from land to waterbodies. The salinity of the oceans is derived from erosion and transport of dissolved salts from the land. Cultural eutrophication of lakes is primarily due to phosphorus, applied in excess to agricultural fields in fertilizers, and then transported overland and down rivers. Both runoff and groundwater flow play significant roles in transporting nitrogen from the land to waterbodies.[15] The dead zone at the outlet of the Mississippi River is a consequence of nitrates from fertilizer being carried off agricultural fields and funnelled down the river system to the Gulf of Mexico. Runoff also plays a part in the carbon cycle, again through the transport of eroded rock and soil.

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THERMAL STRATIFICATION

Thermal stratification is a condition in which water of different temperatures or densities tend to exist in different layers.

For example, cold water usually floats below warm water and both layers do not mix easily. Stratification produces different layers called epilimnion, thermocline (or metalimnion) and hypolimnion. Wind is the major factor that affects the mixing, transport and exchange of materials or substances in water.

Definition

Temperature affects the rates of chemical and biological activities in water. This in turn affects the water quality. Warm temperatures increase the growth of algae, plants and animals; accelerate the decay of organic matter especially near the bottom of water. In the temperate region, there are four distinct seasons. Rising temperatures in spring warm the upper layers of the water. Warm water is lighter than cold water and the warm water floats over the cold and denser water. Temperature and density differences create distinct layers of water in the lake which do not mix easily. Warmer temperatures tend to increase the density differences and cause a separation between the water layers. This process is called thermal stratification.

Epilimnion: The upper layer of water receiving more sunlight is called epilimnion and most biological activities and growth occur here.

Hypolimnion: The colder, denser and darker bottom waters are called the hypolimnion where

dead plants and animals sink.

Thermocline: The metalimnion (thermocline) is the narrow band between the upper and lower waters where the temperature changes quickly with depth.

Fall turnover describes the mixing of the upper layers and lower layers of water in the cold season when the temperature and density differences between layers disappear. Shallow lakes mix readily and have greater potential to release nutrients from the lake bottoms which fuel algal blooms. The warm water fish ponds average about 2 meters in depth. Marked thermal stratification may develop in very shallow ponds because of turbid conditions resulting in rapid heating of surface waters on calm sunny days.

Stratification is less stable in ponds than large bodies of water. Ponds which stratify during daylight hours can de-stratify at night when the upper layers of water cool by conduction. Strong winds may supply enough energy to cause complete circulation of water. Disappearance of a heavy plankton bloom may allow heating to agreeable depth and also lead to complete mixture. Temperatures of the epilimnion in turbid waters are greater than those of clear waters because of greater absorption of heat by particulate matter.

The surface water of deep tropical lakes never reaches the temperature of maximum density; there is no process that makes the water mix. Bottom layers are depleted of oxygen but can be saturated with carbon dioxide, or other gases such as sulfur dioxide. Earthquakes or landslide
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can cause mixing which upturns the deep layers and may release a vast cloud of toxic gases. This is called a limnic eruption e.g. 1986 disaster at Lake Nyos in Cameroon.

Thermal stratification may cause fish kills due to thermal gradients, stagnation, and ice cover. Excess growth of plankton limits commercial and recreational activities in lakes and reduces drinking water quality. Thermal stratification can be reduced by aeration.

Destratification: Is a process in which the air or water is mixed in order to eliminate stratified layers of temperature, plant life or animal life.

Factors Affecting Lake Mixing, Transport and Exchange of Substances

i. Wind

Wind is the dominant external energy input responsible for mixing of water. Wind generates local currents that break lake boundaries and induce basin-scale motions. Wave breaking is an important mixing mechanism and enhances gas exchange with the atmosphere. Strong wind induces Langmuir circulations which are large-scale counter-rotating helical vortices. Langmuir circulation is responsible for mixing of the surface layer and deepening of the epilimnion. Processes that increase turbulence also decrease residence time and increase interaction with bottom water, reduce stratification and enhance mixing.

ii. Radiation

Two major types of radiation play a role in lake and reservoir stratification. These are short-wave ultraviolet and long-wave infrared radiations. The sun produces the short-wave radiation some of which is reflected at the water surface. The remaining radiation penetrates into the lake

and is absorbed in the water column then gets converted to heat. Different wavelengths of the radiation are absorbed at different depths of water. Blue light travels the farthest and heats the deepest layers of water. It can travel full circle and escape from the lake, giving the lake a blue colour.

Long-wave radiation originates from black-body radiation. The lake (water bodies) and the atmosphere emit black-body radiation. Long-wave atmospheric radiation is partially reflected at the lake surface, and the penetrating radiation is absorbed, causing heating. Black-body radiation results in a loss of thermal energy and cooling of the lake water.

iii. Evaporation and condensation

Evaporation is the conversion of liquid water to water vapor while condensation is the conversion of water vapor to liquid water. Both evaporation and condensation are accompanied by fluxes of heat. Evaporation from the lake surface extracts heat from the lake and results in cooling of the water surface. Condensation extracts heat from the atmosphere and adds it to the water surface, resulting in heating at the water surface. Evaporation and condensation are also accompanied by a flux of water and affect the total water budget of the lake.

iv. Direct Inflows and outflows

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Direct inflows to lakes and reservoirs can include surface inflow from rivers and streams, groundwater inflow, and precipitation. Outflows from the lake can be either surface or groundwater outflow. Each of these flows is accompanied by a flux of heat which may also add or remove kinetic energy. Among the inflows and outflows, surface flows have the greatest potential for kinetic energy input to a lake.

Below the mixing action of the wind and the penetration depth of the solar radiation, a strong temperature and accompanying density gradient develops. This region of strong gradients is called the thermocline, or sometimes pycnocline. Below the thermocline a weaker temperature gradient is observed and the water is cool and comparatively quiescent. The bottom region of the lake is called the hypolimnion. In very deep lakes, where solar radiation and thermal conduction cannot penetrate to the lake bottom, the bottom water will have a temperature near 4 °C, the temperature of maximum density of water. The strong density gradients in the thermocline inhibit exchange between the epilimnion and the hypolimnion. Bottom water in a stratified lake does not actively interact with the atmosphere and can easily become deprived of important dissolved gases.

As the air temperature gets cooler and the solar radiation input decreases in the fall, the surface water cools. The surface water and thermocline cool down to the temperature of the hypolimnion and the lake destratifies becomes mixed (turnover). This gives the bottom water an opportunity to aerate with the atmosphere.

If the air temperature goes below 4 °C so that the surface water can cool below this temperature, then the surface water becomes lighter than the bottom water and a so-called winter inverse stratification develops. The term inverse refers to the fact that the surface water is colder than the bottom water; however, the surface water remains less dense. When ice forms at the water

surface, wind mixing is not possible and the winter density profile may not exhibit a well-defined thermocline. The surface water heats up again in the spring, the lake will again reach a state of thermal homogeneity and in the presence of a light wind will turn over. As the surface waters become warmer, stratification sets in and we return to the summer stratification state. Lakes that experience this full cycle of stratification states are called dimictic because they stratify twice. Lakes that only stratify in the summer are called monomictic.

Temperatures of the epilimnion in turbid waters are greater than those of clear waters because of greater absorption of heat by particulate matter.

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PHYSICAL AND CHEMICAL PROPERTIES OF WATER

(O₂, N₂, NH₃), inorganic carbon, Alkalinity and pH, Bacteria and biogeochemistry, microbial loop, Iron and REDOX reactions, Phosphorus cycling, Nitrogen cycling;

Chemistry: Water chemistry between systems varies tremendously. The chemistry is foremost determined by inputs from the geology of its watershed, or catchment area, but can also be influenced by precipitation and the addition of pollutants from human sources. Large differences in chemistry do not usually exist within small lotic systems due to a high rate of mixing. In larger river systems, however, the concentrations of most nutrients, dissolved salts, and pH decrease as distance increases from the river's source .

Oxygen is likely the most important chemical constituent of lotic systems, as all aerobic organisms require it for survival. It enters the water mostly via diffusion at the water-air interface. Oxygen's solubility in water decreases as water temperature increases. Fast, turbulent streams expose more of the water's surface area to the air and tend to have low temperatures and thus more oxygen than slow, backwaters. Oxygen is a byproduct of photosynthesis, so systems with a high abundance of aquatic algae and plants may also have high concentrations of oxygen during the day. These levels can decrease significantly during the night when primary producers switch to respiration. Oxygen can be limiting if circulation between the surface and deeper layers is poor, if the activity of lotic animals is very high, or if there is a large amount of organic decay occurring.

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LAKE BIOTA

Bacteria: Bacteria are present in large numbers in lotic waters. Free-living forms are associated with decomposing organic material, biofilm on the surfaces of rocks and vegetation, in between particles that compose the substrate, and suspended in the water column. Other forms are also associated with the guts of lotic organisms as parasites or in commensal relationships. Bacteria play a large role in energy recycling, which will be discussed in the Trophic Relationships section.

Primary producers: Algae, consisting of phytoplankton and periphyton, are the most significant sources of primary production in most streams and rivers. Phytoplanktons float freely in the water column and thus are unable to maintain populations in fast flowing streams. They can, however, develop sizable populations in slow moving rivers and backwaters.

Periphyton are typically filamentous and tufted algae that can attach themselves to objects to avoid being washed away by fast current. In places where flow rates are negligible or absent, periphyton may form a gelatinous, unanchored floating mat.

Plants exhibit limited adaptations to fast flow and are most successful in reduced currents. More primitive plants, such as mosses and liverworts attach themselves to solid objects. This typically occurs in colder headwaters where the mostly rocky substrate offers attachment sites. Some plants are free floating at the water's surface in dense mats like duckweed or water hyacinth. Others are rooted and may be classified as submerged or emergent. Rooted plants usually occur in areas of slackened current where fine-grained soils are found. These rooted plants are flexible, with elongated leaves that offer minimal resistance to current.

Living in flowing water can be beneficial to plants and algae because the current is usually well aerated and it provides a continuous supply of nutrients. These organisms are limited by flow, light, water chemistry, substrate, and grazing pressure. Algae and plants are important to lotic systems as sources of energy, for forming microhabitats that shelter other fauna from predators and the current, and as a food resource.

Insects and other invertebrates: Up to 90% of invertebrates in some lotic systems are insects. These species exhibit tremendous diversity and can be found occupying almost every available habitat, including the surfaces of stones, deep below the substratum, adrift in the current, and in the surface film. Insects have developed several strategies for living in the diverse flows of lotic systems. Some avoid high current areas, inhabiting the substratum or the sheltered side of rocks. In stronger current, species have developed weighted cases, attachments to anchored pads of silk, recurved clinging claws, suction cup like devices, and flattened streamlined bodies. Additional invertebrate taxa common to flowing waters include mollusks such as snails, limpets, clams, mussels, as well as crustaceans like crayfish and crabs. Like most of the primary producers, lotic invertebrates often rely heavily on the current to bring them food and oxygen. Invertebrates, especially insects, are important as both consumers and prey items in lotic systems.

Fishes are probably the best-known inhabitants of lotic systems. The ability of a fish species to live in flowing waters depends upon the speed at which it can swim and the duration that its

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speed can be maintained. This ability can vary greatly between species and is tied to the habitat in which it can survive. Continuous swimming expends a tremendous amount of energy and, therefore, fishes spend only short periods in full current. Instead, individuals remain close to the bottom or the banks, behind obstacles, and sheltered from the current, swimming in the current only to feed or change locations. Some species have adapted to living only on the system bottom, never venturing into the open water flow. These fishes are dorso-ventrally flattened to reduce flow resistance and often have eyes on top of their heads to observe what is happening above them. Some also have sensory barrels positioned under the head to assist in the testing of substratum.

Lotic systems typically connect to each other, forming a path to the ocean (spring → stream →

river → ocean), and many fishes have life cycles that require stages in both fresh and salt water. Salmon, for example, are anadromous species that are born and develop in freshwater and then move to the ocean as adults. Eels are catadromous, and are born and develop in the ocean and then move into freshwater as adults.

Other vertebrate taxa that inhabit lotic systems include amphibians, such as salamanders, reptiles (e.g. snakes, turtles, crocodiles and alligators) various bird species, and mammals (e.g. otters, beavers, hippos, and river dolphins). With the exception of a few species, these vertebrates are not tied to water as fishes are, and spend part of their time in terrestrial habitat. Many fish species are important as consumers and as prey species to the larger vertebrates mentioned above.

Trophic relationships

Energy inputs

Energy sources can be autochthonous or allochthonous.

Autochthonous energy sources: are those derived from within the lotic system. During photosynthesis, for example, primary producers form organic carbon compounds out of carbon dioxide and inorganic matter. The energy they produce is important for the community because it may be transferred to higher trophic levels via consumption. Additionally, high rates of primary production can introduce dissolved organic matter (DOM) to the waters. Another form of autochthonous energy comes from the decomposition of dead organisms and feces that originate within the lotic system. In this case, bacteria decompose the detritus or coarse particulate organic material (CPOM; >1 mm pieces) into fine organic particulate matter (FPOM; <1 mm pieces) and then further into inorganic compounds that are required for photosynthesis. This process is discussed in more detail below.

Allochthonous energy sources: are those derived from outside the lotic system, that is, from the terrestrial environment. Leaves, twigs, fruits, etc. are typical forms of terrestrial CPOM that have entered the water by direct litterfall or lateral leaf blow. In addition, terrestrial animal-derived materials, such as feces or carcasses that have been added to the system are examples of allochthonous CPOM. The CPOM undergoes a specific process of degradation. An example of a leaf fallen into a stream. First, the soluble chemicals are dissolved and leached from the leaf upon its saturation with water. This adds to the DOM load in the system. Next, microbes such as bacteria and fungi colonize the leaf, softening it as the mycelium of the fungus grows into it. The composition of the microbial community is influenced by the species of tree from which the leaves are shed. This combination of bacteria, fungi, and leaf are a food source for shredding invertebrates, which leave only FPOM after consumption. These fine particles may be

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colonized by microbes again or serve as a food source for animals that consume FPOM. Organic matter can also enter the lotic system already in the FPOM stage by wind, surface runoff, bankerosion, or groundwater. Similarly, DOM can be introduced through canopy drip from rain or from surface flows.

Invertebrates

Invertebrates can be organized into many feeding guilds in lotic systems. Some species are shredders, which use large and powerful mouth parts to feed on non-woody CPOM and their associated microorganisms. Others are suspension feeders, which use their setae, filtering apparatus, nets, or even secretions to collect FPOM and microbes from the water. These species may be passive collectors, utilizing the natural flow of the system, or they may generate their own current to draw water, and also, FPOM. Members of the gatherer-collector guild actively search for FPOM under rocks and in other places where the stream flow has slackened enough to allow deposition. Grazing invertebrates utilize scraping, rasping, and browsing adaptations to feed on periphyton and detritus. Finally, several families are predatory, capturing and consuming animal prey. Both the number of species and the abundance of individuals within each guild is largely dependent upon food availability. Thus, these values may vary across both seasons and systems.

Fishes

Fishes can also be placed into feeding guilds. Planktivores pick plankton out of the water column. Herbivore-detritivores are bottom-feeding species that ingest both periphyton and detritus indiscriminately. Surface and water column feeders capture surface prey (mainly terrestrial and emerging insects) and drift (benthic invertebrates floating downstream). Benthic invertebrate feeders prey primarily on immature insects, but will also consume other benthic invertebrates. Top predators consume fishes and/or large invertebrates. Omnivores ingest a wide range of prey. These can be floral, faunal, and/or detrital in nature. Finally, parasites live off of host species, typically other fishes. Fishes are flexible in their feeding roles, capturing different prey with regard to seasonal availability and their own developmental stage. Thus, they may occupy multiple feeding guilds in their lifetime. The number of species in each guild can vary greatly between systems, with temperate warm water streams having the most benthic invertebrate feeders, and tropical systems having large numbers of detritus feeders due to high rates of allochthonous input.

Community patterns and diversity

Local species richness

Large rivers have comparatively more species than small streams. Many relate this pattern to the greater area and volume of larger systems, as well as an increase in habitat diversity. Some systems, however, show a poor fit between system size and species richness. In these cases, a combination of factors such as historical rates of speciation and extinction, type of substrate, microhabitat availability, water chemistry, temperature, and disturbance such as flooding seem to be important.

Human impacts

Pollution

Pollutant sources of lotic systems are hard to control because they derive, often in small amounts, over a very wide area and enter the system at many locations along its length. Agricultural fields often deliver large quantities of sediments, nutrients, and chemicals to nearby streams and rivers. Urban and residential areas can also add to this pollution when contaminants are accumulated on impervious surfaces such as roads and parking lots that then drain into the

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system. Elevated nutrient concentrations, especially nitrogen and phosphorus which are key components of fertilizers, can increase periphyton growth, which can be particularly dangerous in slow moving streams. Another pollutant, acid rain, forms from sulfur dioxide and nitrous oxide emitted from factories and power stations. These substances readily dissolve in atmospheric moisture and enter lotic systems through precipitation. This can lower the pH of these sites, affecting all trophic levels from algae to vertebrates. Mean species richness and total species numbers within a system decrease with decreasing pH. While direct pollution of lotic systems has been greatly reduced in the United States under the government's Clean Water Act, contaminants from diffuse, non-point sources, remain a large problem.

Dams alter the flow, temperature, and sediment regime of lotic systems. Additionally, many rivers are dammed at multiple locations, amplifying the impact. Dams can cause enhanced clarity and reduced variability in stream flow, which is due to an increase in periphyton abundance. Invertebrates immediately below a dam can show reductions in species richness due to an overall reduction in habitat heterogeneity. Also, thermal changes can affect insect development, with abnormally warm winter temperatures obscuring cues to break egg diapause and overly cool summer temperatures leaving too few acceptable days to complete growth. Finally, dams fragment river systems, isolating previously continuous populations, and preventing the migrations of anadromous and catadromous species.

Invasive species

Invasive species have been introduced to lotic systems through both purposeful events (e.g. stocking game and food species) as well as unintentional events (e.g. hitchhikers on boats or fishing waders). These organisms can affect natives via competition for prey or habitat, predation, habitat alteration, hybridization, or the introduction of harmful diseases and parasites. Once established, these species can be difficult to control or eradicate, particularly because of the connectivity of lotic systems. Invasive species can be especially harmful in areas that have endangered biota, such as mussels in the Southeast United States, or those that have localized endemic species, like lotic systems west of the Rocky Mountains, where many species evolved in isolation.

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LAKE FOOD CHAIN

Differences between aquatic and terrestrial organisms

Differences in chemical and physical characteristics of water and air expose aquatic and terrestrial organisms to different environmental conditions.

- i. Aquatic organisms do not need supporting body structures as strong as those for terrestrial organisms.
- ii. Water has a density close to that of many organisms living in it. The viscosity of water is about 50 times higher than air. This makes the moving energy cost for a comparable displacement higher for an aquatic than for terrestrial organisms.
- iii. The oxygen concentration is lower in water than in air and allows the aquatic organisms to

perform more work than the terrestrial ones to provide the same amount of oxygen for respiration.

iv. The concentration of free carbon dioxide is higher in water than in the atmosphere. Aquatic plants have more inorganic carbon available than the terrestrial ones in the surface layer.

v. Aquatic autotrophs access dissolved nutrients in water without requiring specific transporting structure.

In both aquatic and terrestrial ecosystems, matter and energy flow through a chain of organisms transforming it and constituting the food chain. The energy flowing through the food chain progressively degrades to heat and matter is recycled, passing through stages of variable molecular complexity.

Classification of Aquatic Organisms

A. Based on the Roles they Play in the Ecosystem

i. **Primary Producers:** These are autotrophic organisms which synthesise organic matter using energy from the sun and inorganic substances. The primary producers include unicellular algae, free floating phytoplankton or periphyton that adhere to substrates and aquatic plants (macrophytes) growing in the littoral zone. Autotrophic organisms also include photosynthetic and chemosynthetic bacteria.

ii. **Consumers:** depend on primary producers directly or indirectly for their food. These include primary (feeding on autotrophs), secondary (feeding on animals eating plants or autotrophs), tertiary consumers (feeding on secondary consumers), and so on. Carnivorous fish, birds or fish-eating mammals, occupies the last link in the aquatic consumer chain. The consumers are heterotrophs and use organic matter already produced. They cannot synthesise organic matter from inorganic compounds.

iii. **Detritivores:** These are organisms that feed on the remains of dead organisms and on the waste products of metabolism, degrading it to less complex chemical structures.

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iv. **Decomposers:** Decomposers are mainly the heterotrophic bacteria, oxidising (decomposing) the organic molecules till they are reduced to their inorganic constituents (mineralisation). These are made available for the autotrophs and are thus re-cycled.

B. Based on their Habitats

Plankton

Aquatic organisms consist of animals (zooplankton), plants (phytoplankton), bacteria and archaea (bacterioplankton) and virus (virioplancton). Plankton occupies the littoral and the pelagic zones of water bodies. Autotrophic phytoplankton and bacterioplankton, carry out photosynthesis, a light-dependent activity, and are limited to the euphotic zone (zone of light penetration). The freshwater zooplankton community consists of microscopic organisms belonging to different systematic groups, ranging from protozoa to crustaceans. Though some plankton classes have organs to move, the plankton cannot oppose the current movement and is transported by the water mass hosting it.

i. The Virioplankton Viruses: are the smallest known organisms. They have simple structures, made of nucleic acids as RNA or DNA, contained in a protein capsule called capsid. They range in size between 20 – 200nm and reproduce within living cells. They are obligate parasites. A virus develops through both extracellular and intracellular phases. The virus particle (virion) is stable, resistant to denaturation and floats freely in the water. When a virion meets a host, it adsorbs to it, protruding protein structures that are receptors of the host's cell surface. In the intracellular phase, the virus or viral DNA, enters the host cell where replicates. The newly formed viral particles lyse the host cell, not necessarily destroying it, and are released into the environment. This is called the lytic cycle. Viral replication can also occur through the lysogenic cycle. The viral genome becomes part of the host's genome, integrated in it or as an autonomous plasmid. The viral genome remains in a dormant stage (prophage) ending when the changed environmental conditions (UV radiation, stress, chemicals, etc.) induce the lytic phase. The abundance of the virioplanktonic community depends on the abundance of hosts.

ii. The Bacterioplankton: These are prokaryotes, between 0.2 and 2 µm in size. They can be autotrophs and heterotrophs. The autotrophs include cyanobacteria, photosynthetic and chemosynthetic bacteria. This group includes the bacteria that harvest light. Bacterial photosynthesis does not occur in the presence of oxygen but only in anoxic environments. Chemosynthetic bacteria meet their energy requirements through the oxidation of minerals substrates. Heterotrophic bacteria are more abundant than autotrophic bacteria and form the largest component of bacterioplankton. Archaea are genetically different from bacteria but are part of bacterioplankton. They are adapted to life in extreme environments such as hot springs and acidic water.

Roles of bacteria in the food chain

❖ Breakdown of the organic substances, first-hydrolysed by the exoenzymes produced by bacteria and then mineralized, thus making their mineral constituents available.

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❖ Recycling of dissolved organic matter, through the production of new bacterial cells usable by larger organisms, unable to use directly the dissolved organic matter.

iii. The phytoplankton The algal cells constituting the phytoplankton are at the beginning of the food chain. They synthesise the organic matter necessary to maintain the whole food chain working. The process ending with the production of organic compounds from inorganic substances and light is known as photosynthesis. The organic product of photosynthesis is glucose (C₆H₁₂O₆). The carbon and the oxygen necessary to build up the organic matter are provided by atmospheric CO₂. The equation summarising the process is: $6 \text{ CO}_2 + 6 \text{ H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ O}_2$

Photosynthesis occurs in two phases:

1. Light-dependent phase Light energy from the sun is harvested by special pigments such as chlorophyll (the most important). In the light-dependent reaction, an energy-rich electron of a chlorophyll molecule is passed along a chain of molecules known as the electron transfer

system. The energy lost by the electron as it moves along the chain is used to make NADP, (Nicotinamide or niacinamide Adenine Dinucleotide Phosphate) and ATP (Adenosine Triphosphate). These are very energy-rich but relatively unstable molecules.

2. The dark phase or Calvin cycle ATP and NADP are used to produce energy-rich carbohydrate molecules, which can be stored and used as a basis for the production of all other forms of organic molecules. The synthesis of new organic matter leads to an increase in cell size and a division in two daughter cells. These can be completely separated or they can remain connected though their cell walls, forming a colony.

Composition of Fresh Water Phytoplankton Communities

Freshwater phytoplankton communities consist of prokaryotes, Cyanobacteria, and eukaryotes from several taxonomic groups e.g. Diatoms, Chrysophytes (golden algae), Cryptophytes, Dynophytes, Euglenophytes and Chlorophytes (green algae).

- i. Prokaryotic cells: The chromosomes are dispersed in the protoplasm. Photosynthesis occurs in invaginations of the cell membrane.
- ii. Eukaryotic cells: The chromosomes are present in the nucleus and there are specific organelles, the chloroplasts, devoted to photosynthesis.
- iii. Cyanobacteria: Cyanophyta or blue - green algae are bacteria with the cell structure of prokaryotes. They differ from bacteria by having chlorophyll a, which is also found in eukaryotic algae. They contain phycocyanin and allophycocyanin (pigments) which make them blue-green. Examples of blue -green algae are genera *Anabaena*, *Microcystis* and *Planktothrix*. They form massive blooms of Cyanobacteria which produce toxins, reduce water quality and dissolved oxygen.
- iv. Diatoms are an important group of algae. They contain siliceous cell walls, called frustules. They contain chlorophyll (photosynthesis) and carotenoids (yellow-golden brown colour).

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- a. Chrysophytes (Chrysophyceae) are single-celled and typical of freshwater environments poor in calcium. They have two flagella. Examples include *Synura*, *Mallomonas*, *Ochromonas*, *Dinobryon* and *Uroglena*. Many species lack a true cell wall but may have a layer of scales, the lorica, made of cellulose and can be accompanied by limestone, silica or iron oxides. They contain photosynthetic pigment, chlorophyll.
- b. Cryptophytes (Cryptophyceae) have two flagella almost as long as the cell. The outer portion of the cell or periplast is composed of a plasmatic membrane and a series of plates placed beneath. The Cryptophytes contain chlorophyll, phycobilins phycocyanin and phycoerythrin (pigments)
- c. Dynophytes (division Dynophyta) are made by an upper part (the epicone) and a lower part (the hypocone) separated by a belt. They contain chlorophyll. The Dynophytes have an intermediate chromosome falling between that of prokaryotes and eukaryotes and are thus defined mesokaryotes.
- d. Euglenophytes (division: Euglenophyta) are flagellate algae present mostly in eutrophic

freshwater environments. The Euglenophytes have two basal bodies with one or two emerging flagella. Their chloroplast contains mainly chlorophyll a and b.

e. Chlorophytes, or green algae, (division Chlorophyta) are like higher plants. Their chloroplasts contain Chlorophylls a and b and they are green and contain no accessory pigments. They have two or more flagella similar in structure but different in length.

The Zooplankton

The zooplankton commonly found in freshwater are:

- i. Protozoa are unicellular organisms between tens and hundreds microns in size. They feed on bacteria or detritus particles which are incorporated in a vacuole where digestion occurs (phagocytosis). Planktonic protozoa move mostly by cilia. They transform dissolved organic matter into forms usable by higher organisms.
- ii. Cnidarians are diverse in form such as siphonophores, medusae, corals, feathery hydroids, and box jellies with complex eyes. They are all armed with stinging cells called nematocysts. Many cnidarians live in the ocean while a smaller number of species are found in fresh waters. There are four classes of Cnidarians namely Anthozoa, Cubozoa, Hydrozoa and Scyphozoa.
- iii. Rotifera are animals related to worms and are not segmented. They measure length 100 μm - 1 mm in length. They have a crown of cilia moving around their mouth to suck inside the food. They feed on algae, protozoa, bacteria, debris and some are carnivorous. Examples include the genera Asplanchna, Keratella, Kellicottia and Brachionus). They reproduce by parthenogenesis (without egg fertilization by males).
- iv. There are two groups of planktonic crustaceans namely Cladocerans e.g. Daphnia (water flea) and copepods. Daphnia and other cladocerans, are filter feeders because their limbs have filaments (setae) used to filter food particles from water. Cladocerans reproduce by

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parthenogenesis while sexual reproduction occurs in copepods and eggs are carried by females within ovigerous bags attached to the mother's body.

The transformation of organic matter by heterotrophs is called secondary production. The annual net secondary production of a zooplankton population is the sum of all biomass produced by the population during the year, taking into account the losses due to mortality, predation and emigration.

Benthos

This is the community living in the benthic zone, at the surface or inside the bottom sediment. It is a complex community consisting of both mobile and sessile organisms. The benthos includes fungi, bacteria, plant and animal forms which include algae, protozoa, rotifers, tardigrades, mollusks, worms, arthropods. Aquatic macroalgae and macrophytes grow in close relationship with the sediment as the benthos and account for a large portion of primary production in water.

- i. Phytobenthos or Periphyton (or "Aufwuchs") This is the community of microscopic autotrophs attached to submerged inorganic (epilithic algae) and organic surfaces, including living plants (epiphytic algae). They grow in areas of light penetration. Protozoa, sponges,

bryozoans and other animals are usually associated with the periphyton. Examples of periphyton are filamentous algae, unicellular or colonial algae belonging to the cyanobacteria, diatoms and chlorophytes.

ii. **Macrophytes or hydrophytes** These are plants growing in aquatic environments or on substrates that are fully or periodically submerged. Aquatic macrophytes may be emergent, floating or submerged. The emergent and submerged hydrophytes are always rooted in the substrate but the floating hydrophytes may be rooted in the substrate or freely floating. Examples of emergent macrophytes are *Phragmites* spp (reeds), *Scirpus*, *Typha*, *Sagittaria* sp. The floating macrophytes include *Nymphaea* sp., *Trapa natans* (water chestnut) and *Potamogeton* sp. The free floating macrophytes include *Lemna minor* and *Eicornia crassipes*. The shading effect of floating hydrophytes reduces the penetration of solar radiation, leading to a reduction in phytoplankton populations.

iii. **Nekton, Neuston and Psammon** The term nekton describes aquatic animals that can move actively in the water, independent of water movement. These include fish, mollusks, amphibians, reptiles and mammals. The fish community includes both littoral and pelagic species. The species composition of the fish population depends on aquatic morphology, which influences the thermal regime, on water chemistry (in particular on oxygen availability) and on the biological characteristics of the water, which determines the quality and quantity of available food (Matthews, 1998).

iv. **Neuston** consists of microscopic organisms, protozoa and bacteria, living in the surface film at the water-air interface. This group also includes larger organisms such as insects which can move on the water surface. **Psammon** refers to the animal community, including protozoa,

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rotifers, tardigrades, crustaceans, worms, larvae of some insects, living in interstitial water of sandy beaches.

WETLANDS

A wetland is a land area that is saturated with water, either permanently or seasonally, such that it takes on the characteristics of a distinct ecosystem.

It is a collective name for marshes, swamps, bogs, etc. It represents an ecosystem which is intermediate between an aquatic and terrestrial ecosystems.

Wetlands are areas where water covers the soil, or is present either at or near the surface of the soil all year or for varying periods of time during the year, including during the growing season. Water saturation (hydrology) largely determines how the soil develops and the types of plant and animal communities living in and on the soil. Wetlands may support both aquatic and terrestrial species. The prolonged presence of water creates conditions that favor the growth of specially adapted plants (hydrophytes) and promote the development of characteristic wetland (hydric) soils.

A patch of land that develops pools of water after a rain storm would not be considered a "wetland," even though the land is wet. A more concise definition is a community composed of

hydric and hydrophytes. Wetlands have also been described as ecotones, providing a transition between dry land and water bodies. Wetlands exist "...at the interface between truly terrestrial ecosystems and aquatic systems, making them inherently different from each other, yet highly dependent on both (Mitch et al. 2007).

A wetland is an ecosystem that arises when inundation by water produces soils dominated by anaerobic processes, which, in turn, forces the biota, particularly rooted plants, to adapt to flooding. Wetlands are places where land and water meets. The RAMSAR convention, an international convention concerning wetland conservation, defines wetlands as:

Ramsar Convention definition

Under the Ramsar International Wetland Conservation Treaty, wetlands are defined as follows:

- Wetlands are areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six metres (Ramsar Article 1.1).

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- Wetlands may incorporate riparian and coastal zones adjacent to the wetlands, and islands or bodies of marine water deeper than six metres at low tide lying within the wetlands (Ramsar Article 2.1)

The RAMSAR Convention is the name of the International Convention on Protection of Wetlands, particularly as waterfowl habitat. It was formed in 1971 by delegates who met at Ramsar, a town in Iran on the coast of the Caspian Sea. It is the only international treaty that seeks to conserve habitat rather than species. Over 100 countries have signed the Convention.

Wetlands are, thus, transitional habitats between dry land and deep water. They include marshes, swamps, peatlands (including bogs and fens), flood meadows, lakes and ponds, rivers and streams, estuaries and other coastal waters (including salt marshes, mangroves and even coral reefs).

Wetland Water Sources

- Precipitation
- Surface Flow
- Groundwater Discharge
- Ocean
- River
- Lake

Wetland Water Losses

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- Evaporation/Transpiration
- Outflows
- Groundwater Recharge

Indicators of Wetland Hydrology

Primary

- Visible surface water
- Saturated Surface Soils
- Saturated within 18 in.

Secondary

- Oxidized rhizospheres
- Water stained leaves
- Surface scouring
- Plant adaptations
- Hydric Soils

Classification of Wetlands

Classifying wetlands is difficult, due to the highly dynamic nature of the places, changing with the seasons and over time. Their precise boundaries are hard to define. Wetlands occur naturally on every continent except Antarctica, the largest including the Amazon River basin, the West Siberian Plain, and the Pantanal in South America. The water found in wetlands can be freshwater, brackish, or saltwater. The main wetland types include swamps, marshes, bogs, and fens; and sub-types include mangrove, carr, pocosin, and varzea. Wetlands vary widely due to local and regional differences in topography, hydrology, vegetation, and other factors, including human involvement.

Wetlands vary in size from micro-wetlands, possibly housing rare endemic species, to the vast water system of Lake Victoria and the Nile which has an estimated catchment area of 2.9 million square km (one tenth of the area of Africa Continent). They are dynamic systems, continually undergoing natural change due to flooding, subsidence, drought, sea-level rise or infilling with sediment or organic material. Lake Chad in the Sahel, for example, is rarely more than 7m deep but its area varies from 25,000 km² to less than 10,000 km² in times of drought. Many wetlands are only temporary features of the landscape and will change and eventually disappear, whilst new wetlands are naturally created elsewhere.

Wetlands also vary widely in type, with significant differences from one region to another. The grassland floodplains in Africa and South America may be contrasted with Sudan's huge swamps, the enormous mangrove forests in parts of Asia, coral reefs in the tropics and the high concentration of peatlands around the margins of the South China Sea. Some categories of wetland are very well known whilst others, such as karstic and subterranean wetlands, may be near-invisible to the general public, legislators and land managers.

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There are two main types of world wetlands: those found inland with freshwater (inland or non-tidal wetlands), and those on the coast with salty (marine or brackish or coastal or tidal wetlands and) water. There are about 30 recognizes natural wetland types and a further nine man-made ones.

Coastal wetlands, as their name suggests, they are found along the coasts. They are closely linked to estuaries, where sea water mixes with fresh water to form an environment of varying

salinities. The salt water and the fluctuating water levels (due to tidal action) combine to create a rather difficult environment for most plants. Consequently, many shallow coastal areas are unvegetated mud flats or sand flats. Some plants, however, have successfully adapted to this environment. Certain grasses and grasslike plants that adapt to the saline conditions form the tidal salt marshes that are found along the coasts. Mangrove swamps, with salt-loving shrubs or trees, are common in tropical climates, such as in southern parts of Nigeria. Some tidal freshwater wetlands form beyond the upper edges of tidal salt marshes where the influence of salt water ends. Some examples are listed below:

- Marine waters—permanent shallow waters less than six metres deep at low tide; includes sea bays, straits
- Sub-tidal aquatic beds; includes kelp beds, seagrasses, tropical marine meadows
- Coral reefs
- Rocky marine shores; includes rocky offshore islands, sea cliffs
- Sand, shingle or pebble beaches; includes sand bars, spits, sandy islets
- Intertidal mud, sand or salt flats
- Intertidal marshes; includes saltmarshes, salt meadows, saltings, raised salt marshes, tidal brackish and freshwater marshes
- Intertidal forested wetlands; includes mangrove swamps, nipa swamps, tidal freshwater swamp forests
- Brackish to saline lagoons and marshes with one or more relatively narrow connections with the sea
- Freshwater lagoons and marshes in the coastal zone
- Non-tidal freshwater forested wetlands

Inland wetlands are most common on floodplains along rivers and streams (riparian wetlands), in isolated depressions surrounded by dry land, along the margins of lakes and ponds, and in other low-lying areas where the groundwater intercepts the soil surface or where precipitation sufficiently saturates the soil (vernal pools and bogs). Inland wetlands include marshes and wet meadows dominated by herbaceous plants, swamps dominated by shrubs, and wooded swamps dominated by trees. Examples of inland wetlands are listed below:

- Permanent rivers and streams; includes waterfalls
- Seasonal and irregular rivers and streams
- Inland deltas (permanent)
- Riverine floodplains; includes river flats, flooded river basins, seasonally flooded grassland, savanna and palm savanna

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- Permanent freshwater lakes (> 8 ha); includes large oxbow lakes
- Seasonal/intermittent freshwater lakes (> 8 ha), floodplain lakes
- Permanent saline/brackish lakes
- Seasonal/intermittent saline lakes
- Permanent freshwater ponds (< 8 ha), marshes and swamps on inorganic soils; with

emergent vegetation waterlogged for at least most of the growing season

- Seasonal/intermittent freshwater ponds and marshes on inorganic soils; includes sloughs, potholes; seasonally flooded meadows, sedge marshes
- Permanent saline/brackish marshes
- Seasonal saline marshes
- Shrub swamps; shrub-dominated freshwater marsh, shrub carr, alder thicket on inorganic soils
- Freshwater swamp forest; seasonally flooded forest, wooded swamps; on inorganic soils
- Peatlands; forest, shrub or open bogs
- Alpine and tundra wetlands; includes alpine meadows, tundra pools, temporary waters from snow melt
- Freshwater springs, oases and rock pools
- Geothermal wetlands
- Inland, subterranean karst wetlands

Artificial (Human-made) Wetlands

Human made wetlands include irrigated lands, ponds for agriculture and livestock, aquaculture installations, rivers and reservoirs which have been dammed for water supply, irrigation or hydroelectric power generation, flooded excavations, certain wastewater treatment facilities and salt pans (salines). In some cases, the construction of an artificial wetland directly involves the destruction or degradation of a natural wetland, for example where mangrove ecosystems are converted to aquaculture ponds or where floodplains are drained and subsequently irrigated for intensive arable cultivation. Under some laws, the grant of planning permission to develop natural wetlands may be conditional on the developer funding the creation of new wetlands.

The common characteristic of artificial wetlands is that they are usually designed for a specific purpose and seldom have the full range of wetland functions and values provided by a natural wetland, even where they are of similar size. Whilst there is broad consensus that newly created or restored and rehabilitated wetlands cannot replace natural ecosystems, wetland creation and restoration are becoming an increasing priority in industrialised countries where wetland integrity has been seriously degraded over decades. The creation and restoration of wetlands is becoming technically and ecologically feasible for a limited range of habitat types, including freshwater marshes and tidal marshes on low energy coasts.

- Water storage areas; reservoirs, barrages, hydro-electric dams, impoundments (generally > 8 ha)
- Ponds, including farm ponds, stock ponds, small tanks (generally < 8 ha)
- Aquaculture ponds; fish ponds, shrimp ponds
- Salt exploitation; salt pans, salines
- Excavations; gravel pits, borrow pits, mining pools
- Wastewater treatment; sewage farms, settling ponds, oxidation basins
- Irrigated land and irrigation channels; rice fields, canals, ditches

- Seasonally flooded arable land, farm land

General Characteristics Features and Indicators of Wetlands

Wetlands have unique characteristics. They are generally distinguished from other water bodies or landforms based on their water level and on the types of plants that live within them. Specifically, wetlands are characterized as having a water table that stands at or near the land surface for a long enough period each year to support aquatic plants. Wetlands vary widely due to local and regional differences in topography, hydrology, vegetation, and other factors, including human involvement.

Some wetlands, such as swamps and marshes, are obvious to most people. Others, like seasonal wetlands or bogs, are not as easily recognized because they may dry out during part of the year or simply do not look very wet.

i. Evidence of Hydrology:

- Standing or flowing water for seven or more consecutive days during the growing season.
- Waterlogged soil: Determined by digging a 12-inch-deep hole and then checking for water in the hole; or by looking for soil that glistens with water; or squeezing water from a handful of soil.
- Water marks on trees or small piles of debris lodged in trees or piled against other objects in the direction of water movement near river systems.

ii. Wetland Soils:

- Check for a blue or gray color about a foot below the surface. Your local CD office may describe other color characteristics to look for.
- Look for organic matter such as peat or muck.
- Smell the soil for an odor like rotten eggs.

iii. Wetland Vegetation:

More than 5,000 different plants grow in wetlands. Some common wetland plants to look for include:

- Water lilies, cattails, arrowhead, smartweed, pondweed and other plants in standing water.
- Grasses such as reed canary grass, barnyard grass, and prairie cordgrass, or rushes and sedges.
- Trees such as willow, white cedar, cottonwood, silver and red maple, green ash, tamarack, pin oak, and elm.
- Shrubs such as buttonbush, Michigan holly, and red-osier dogwood.

Importance of Wetlands

Wetlands play a number of roles in the environment:

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- Wetlands are considered the most biologically diverse of all ecosystems, serving as home to a wide range of plant and animal life. Rivaling the likes of tropical rainforests and coral reefs, wetlands are among the most fertile, productive ecosystems in the world. Two thirds of

all fish consumed worldwide are dependent on coastal wetlands at some stage in their life cycle. Annual fish and seafood production in swamps and marshes worldwide has been estimated at an average of nine tons per square kilometer, 259 hectares or 640 acres.

ii. Reservoirs of biodiversity: Wetland systems' rich biodiversity is becoming a focal point at International Treaty Conventions and within the World Wildlife Fund (WWF) organization due to the high number of species present in wetlands, the small global geographic area of wetlands, the number of species which are endemic to wetlands, and the high productivity of wetland systems. Hundreds of thousands of animal species, 20,000 of them vertebrates, are living in wetland systems. The discovery rate of fresh water fish is at 200 new species per year. The Amazon holds 3,000 species of fresh water fish species within the boundaries of its basin whose function it is to disperse the seeds of trees. One of its key species, the Piramutaba catfish, *Brachyplatystoma vaillantii*, migrates more than 3,300 km (2,051 miles) from its nursery grounds near the mouth of the Amazon River to its spawning grounds in Andean tributaries (400 m or 437 yards above sea level) distributing plants seed along the route.

Intertidal mudflats have a similar productivity even while possessing a low number of species. iii. High biological productivity. Many wetlands are highly productive ecosystems in large part because they are rich in organic matter and nutrients. These nutrients support organisms within the marsh, but in many instances the nutrients are also transferred to nearby aquatic systems (lakes, rivers, and estuaries), enhancing the productivity of these systems and supporting human uses such as offshore commercial fisheries.

iv. Genetic diversity: Many species in wetland systems are unique due to the long period of time that the ecosystem has been physically isolated from other aquatic sources. The number of endemic species in Lake Baikal (Russia) classifies it as a hotspot for biodiversity and one of the most biodiverse wetlands in the entire world. Evidence from a research study suggest that the number of crustacean species endemic to Baikal Lake (>690 species and subspecies) exceeds the number of the same groups of animals inhabiting all the fresh water bodies of Eurasia together.

v. Flood protection: The wetland system of floodplains is formed from major rivers downstream from their headwaters. Notable river systems that produce large spans of floodplain include the Nile River, the Niger river inland delta, the Zambezi River flood plain, the Okavango River inland delta, the Kafue River flood plain, the Lake Bangweulu flood plain, Mississippi River (USA), Amazon River (South America), Yangtze River (China), Danube River(Central Europe) and Murray-Darling River (Australia).

vi. Water storage and reservoirs: They act like giant sponges, absorbing water from all around. Many wetlands help recharge underground aquifers that store 97% of the world's unfrozen freshwater. Groundwater is critical to billions of people as their only source of drinking water (almost one third of the human population). It is also the only source of water for many irrigation programmes (17% of the world's cropland is irrigated). "The floodplains of major rivers act as natural storage reservoirs, enabling excess water to spread out over a wide area, which reduces its depth and speed. Wetlands close to the headwaters of streams and rivers can

slow down rainwater runoff and spring snowmelt so that it doesn't run straight off the land into water courses. This can help prevent sudden, damaging floods downstream.

vii. Groundwater replenishment. The surface water which is the water visibly seen in wetland systems only represents a portion of the overall water cycle which also includes atmospheric water and groundwater. Wetland systems are directly linked to groundwater and a crucial regulator of both the quantity and quality of water found below the ground. Wetland systems that are made of permeable sediments like limestone or occur in areas with highly variable and fluctuating water tables especially have a role in groundwater replenishment or water recharge. Sediments that are porous allow water to filter down through the soil and overlying rock into aquifers which are the source of 95% of the world's drinking water. Wetlands can also act as recharge areas when the surrounding water table is low and as a discharge zone when it is too high.

viii. Shoreline stabilisation and storm protection: Tidal and inter-tidal wetland systems protect and stabilize coastal zones. Coral reefs provide a protective barrier to coastal shoreline. Mangroves stabilize the coastal zone from the interior and will migrate with the shoreline to remain adjacent to the boundary of the water. The main conservation benefit these systems have against storms and tidal waves is the ability to reduce the speed and height of waves and floodwaters.

ix. Erosion control. By dissipating wave energy and stabilizing shorelines, wetland vegetation buffers the adjacent upland from wave action and intensive erosion. Flood damage reduction. Wetlands intercept runoff and store stormwater, thereby changing rapid and high peak flows to slower and smaller discharges over longer periods of time. Because it is usually the peak flows that cause flood damage, the effect of wetlands is to reduce the danger of flooding. A classic study by the Corps in the Charles River Basin in Massachusetts estimated that the loss of 3,400 hectares (approximately 8,100 acres) of forested wetlands would increase downstream flood damage, costing millions of dollars annually.

x. Nutrient retention: Wetlands cycle both sediments and nutrients balancing terrestrial and aquatic ecosystems. A natural function of wetland vegetation is the up-take and storage of nutrients found in the surrounding soil and water. These nutrients are retained in the system until the plant dies or is harvested by animals or humans. Wetland vegetation productivity is linked to the climate, wetland type, and nutrient availability. The grasses of fertile floodplains such as the Nile produce the highest yield including plants such as *Arundodonax* (giant reed), *Cyperus papyrus* (papyrus), *Phragmites* (reed) and *Typha* (cattail, bulrush).

xi. Sediment traps: Rainfall run-off is responsible for moving sediment through waterways. These sediments move towards larger and more sizable waterways through a natural process that moves water towards oceans. All types of sediments which may be composed of clay, sand, silt, and rock can be carried into wetland systems through this process. Reedbeds or forests located in wetlands act as physical barriers to slow waterflow and trap sediment.

xii. Water purification: Many wetland systems possess biofilters, hydrophytes, and organisms that in addition to nutrient up-take abilities have the capacity to remove toxic substances that

have come from pesticides, industrial discharges, and mining activities. They slow water movement, allowing silt and sediment to settle. Wetland plants filter pollutants and remove high levels of phosphorus and nitrogen. Reed beds, for example, make wonderful natural filter

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systems to remove pollutants and organic waste from water. Reed stems block the passage of larger particles, and microbes in the root systems literally 'eat' certain noxious pollutants, breaking them down into harmless chemicals in the process.

xiii. As buffer zone. By storing water and slowing water movement, wetlands buffer surrounding areas from the worst effects of storms and floods. Wetland plants stabilize shorelines and wetland edges. Mangroves, for example, control the effects of floods; protect shorelines from erosion; trap sediments, toxins and nutrients, and act as wind breaks to buffer against storms. The value of intact mangrove swamps in Malaysia for storm protection and flood control alone has been estimated as US\$ 300,000 per kilometer - the cost of replacing them with rock walls. The value of one kilometer of coral reef ranged from US\$ 137,000 to US\$ 1.2 million over a 25 year period, based on the economic value of storm protection, fishing and tourism.

xiv. As producers. They are nature's larder and provider. Wetlands have supplied people with products for thousands of years. Throughout the world, people have found food, medicines, building materials, industrial oils and dyes, textiles and paper products in wetlands. Over half the human population depends on a wetland product - rice - for its staple diet. Wetlands also support the world's great fisheries and shellfish nurseries. We all use wetland products every day. Here are some of them. Prawn curry - tropical prawns form the basis of major fisheries in Asia. They depend on mangroves as nurseries. Other economically important shellfish worldwide include shrimps, crabs, clams, mussels and oysters.

xv. As habitats and sanctuary. Often thought of as wastelands, wetlands are, in fact, among the world's most productive environments and support high concentrations of life (biodiversity). Freshwater wetlands hold more than 40% of the world's known species, including 12% of all animal species. From northern rivers where salmon spawn to crocodile-rich tropical swamps, wetlands provide food and shelter for species of most major animal groups from microbes, plants and invertebrates (animals without backbones) to fish, amphibians, reptiles, birds and mammals. They are, for example, great havens for birds from storks to swans and ducks to divers. A 1991 study by James R. Chambers determined that approximately 75 percent (by weight) of commercially harvested fish and shellfish are dependent on estuaries and their wetlands.

xvi. Support for birds and other wildlife. Wetlands are probably best known for their value to waterfowl. The freshwater wetlands in the prairie pothole region of North America support an estimated 50 to 80 percent of the continental waterfowl production each year. The loss of wetlands in this region, which is estimated to be more than 50 percent of the original wetland acreage occurring at the time of settlement, has been considered a major factor in the decline in nesting success of duck populations in North America. Wetlands also support a wide diversity of other birds. Eighty percent of America's breeding bird population and more than 50 percent of the 800 species of protected migratory birds rely on wetlands. In addition to

birds, other wildlife makes its home in wetlands. Reptiles and amphibians are common wetland residents. Nearly all of the approximately 190 species of amphibians in North America depend on wetlands for breeding. Other wildlife associated with wetlands includes muskrat, beaver, mink, raccoon, marsh and swamp rabbits, numerous mice, voles, shrews, lemmings, and other small mammals. Large mammals also rely on wetlands. For example, moose often depend on wetlands such as white cedar swamps and other forested wetlands for winter shelter and food.

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xvii. Aesthetics and recreation. Many recreational activities take place in and around wetlands. Hunting and fishing are popular activities associated with wetlands. Other recreational activities in wetlands include hiking, nature observation and photography, canoeing, and other boating. Many people simply enjoy the beauty and sounds of nature and spend their leisure time near wetlands observing plant and animal life. Wetlands are also important places for outdoor study and for gaining an appreciation of natural history and ecology. Properties bordering wetlands often have higher property values than those that do not. Urban wetlands are typically some of the last remaining pieces of “natural habitat” providing residents some sense of wildness and open space. In 1991 wetland-related ecotourism activities such as hunting, fishing, bird-watching, and photography added approximately \$59 billion to the national economy. According to the Pacific Coast Federation of Fishermen’s Associations, almost \$79 billion per year is generated from wetland-dependent species, or about 71 percent of the nation’s entire \$111 billion commercial and recreational fishing industry in 1997.

Wetlands in Nigeria

Nigeria is one of the fifty-three countries that make up the African continent. Out of these 53 countries, 23 are Contracting Parties to the Ramsar Convention. Africa's size and diversity of landscape are striking: bordered by the Mediterranean Sea to the north, the Indian and Atlantic oceans to the east and west respectively, and the Antarctic in the south, it covers 70° of latitude, several climatic zones, and a considerable altitudinal range.

Various wetland types characterize the diverse and panoramic Nigerian environment, from mountains reaching an altitude of 6,000m in the middle belt (Jos Plateau) and eastern parts (Cameroun Mountains) of the country through deserts in the far north to coastal zones at sea level in the south. Wetlands constitute about 2.6% of Nigeria's total surface area. Relatively little scientific investigation has been undertaken in them in comparison to other ecosystems such as forest and savannah or to wetlands in other parts of the world, their important role in support of the region's biodiversity and the livelihood of large human populations.

The country is richly endowed with abundant wetlands ecosystem, and it represent 2.6% of the country’s area of about 923,768km². The majority of it lies in Niger Delta and is mainly freshwater. Niger Delta accounts for 55% of freshwater wetland in Nigeria and the largest wetland in Africa. The Niger Delta is located along the Atlantic coast forms the southern boundary of Nigeria. It is the entrance of Rivers Niger and Benue into the ocean through a web of rivers, creeks, and estuaries. It is the largest wetland in Africa and the third largest in the

world, with about 2370 square kilometers of rivers, creeks and estuaries. Its vegetation is predominantly of the forest type with 8600 square kilometers of swamp forest and about 1900 square kilometers of mangrove forests.

General Distribution and Diversity of Wetland Types

Nigeria's wetlands fall into two major categories;

1. The Coastal Wetlands (Mangrove Swamps), and
2. The Freshwater Wetlands (Floodplains).

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The mangrove swamps covers an area of 9,000km² in the coastal States of Akwa Ibom, Cross River, Rivers, Bayelsa, Delta, Edo, Ondo, Ogun and Lagos while the floodplains covers an area of 2,585 km² mostly along Niger/Benue River system . Other pockets of wetland are scattered round the country.

The wetland displays a richness and uniqueness in biodiversity. From the mangrove swamps and freshwater forests in the south to the massive floodplains scattered round the country, Nigeria's many wetland types support a great diversity of plants and animals, and their productivity provides the natural resources essential to the survival of a significant part the population.

Designated Ramsar Sites and Proposed Sites

The RAMSAR Convention is the name of the International Convention on Protection of Wetlands, particularly as waterfowl habitat. It was formed in 1971 by delegates who met at Ramsar, a town in Iran on the coast of the Caspian Sea. It is the only international treaty that seeks to conserve habitat rather than species. Over 100 countries have signed the Convention.

Nigeria presently has 11 sites designated as Wetlands of International Importance, with a surface area of 1,076,728 hectares. WWF Global Freshwater Programme has supported the Nigerian Ministry of Environment in developing these designations. The sites present a fascinating array of wetland types, including swamp forests, river floodplains, mangroves, and lakes and all of them are extremely important for their support for flora and fauna and for the ecosystem services they provide for the local communities. Most or all of them are formally state-owned but in practical terms under the customary control of local families and communities.

S/N Site Date of State Area Coordinates

Designation (Ha)

- 1 Nguru Lake (and Marma 2/10/2000 Jigawa 58,100 10°22'N
Channel) complex & Yobe 012°46'E
- 2 Apoi Creek Forests 30/04/2008 Bayelsa 29,213 05°47'N
004°42'E
- 3 Baturiya Wetland 30/04/2008 Kano 101,095 12°31'N
010°29'E
- 4 Dagona Sanctuary Lake 30/04/2008 Yobe 344 12°48'N
010°44'E

- 5 Foge Islands 30/04/2008 Kebbi & 4,229 10°30'N
Niger 004°33'E
- 6 Lake Chad Wetlands in 30/04/2008 Borno 607,354 13°04'N
Nigeria 013°48'E
- 7 Lower Kaduna-Middle 30/04/2008 Kwara & 229,054 08°51'N
Niger Floodplain Niger 005°45'E
- 8 Maladumba Lake 30/04/2008 Bauchi 1,860 10°24'N
009°51'E
- 9 Oguta Lake 30/04/2008 Imo 572 05°42'N
006°47'E
- 10 Pandam and Wase Lakes 30/04/2008 Nasarawa 19,742 08°42'N
39
008°58'E
- 11 Upper Orashi Forests 30/04/2008 Rivers 25,165 04°53'N
006°30'E

There is a draft plan by the nation to designate four more sites: Chingurme, Ibom/Cross River estuary, Wawan Rafi Wetlands and Akassa coastal wetland as Ramsar wetlands designated sites in Nigeria.

Threats to Wetland Biodiversity and Future Prospects

Nigeria still has a significant number of pristine wetlands when compared to Europe and other parts of North America. Despite their importance, human activities and the changing climate are degrading wetlands faster than any other ecosystem in Nigeria. Most of the government's national policies are influenced by policies from different sectors, such as agriculture, national resources and energy.

Wetlands in Nigeria are experiencing immense pressure from human activities. Some of the threats to wetland are

- Drainage for agriculture and settlement
- Excessive exploitation by local communities
- Improperly planned development activities.
- Increasing population in conjunction with efforts to increase food security is escalating pressure to expand agriculture within wetlands.
- Wetlands are actually considered by some as the 'new frontier' in agriculture. Many floodplains and tidal wetlands have been reduced in size and their productivity declined. IWMI indicates that 66% of all listed African wetland are used for agriculture, and this wetland agriculture is valued at US\$1.1 billion.
- Extraction of natural resource from wetlands.
- Industrial and urban water demand use of water.
- Oil spillage, gas flaring and uncontrolled exploration and exploitation of crude oil.
- Poorly planned and managed (oil) infrastructure development are heavily impacting the Niger Delta in Niger.

- Exploitation of forest resources from the wetlands.
- Over-fishing
- Climate changed related events, such as unpredictable and decreasing rainfall, are expected to reduce the water stored in wetlands and exacerbate their overexploitation and fuel further conversion into agricultural lands.
- Other threats to Nigerian wetlands include changes in wetland water quality due to the effects of industrial effluent and agricultural pesticides, siltation from highland catchment areas, and introduction of alien species of flora and fauna leading to

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colonization by single species and loss of endemic species diversity. Perhaps one of the biggest single catastrophes has been the introduction to the coastal waters, estuarine, creeks and rivers of alien plant species, the Water Hyacinth, *Eichhornia crassipes* and Water Lettuce *Pistia stratiotes*, which threatens the existence of endemic flora.

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Paper 5 of 7 — EMT 227 Overview of Oceanography Lecture note

Level 200 • Second Semester • Handout

INTRODUCTION

Overview of Oceanography

Oceanography, the scientific study of the ocean, encompasses a wide range of disciplines including biology, chemistry, geology, and physics. This field is essential for understanding the complex and dynamic nature of the world's oceans. Oceanographers study the physical properties of sea water, such as temperature, salinity, and currents, as well as the chemical composition, including dissolved gases and nutrients. They also investigate the geological structures on the ocean floor and the biological diversity within marine ecosystems. The insights gained from oceanography are crucial for predicting climate patterns, understanding marine biodiversity, and managing marine resources.

Relevance to Environmental Science and Marine Biology

The study of sea water is of paramount importance to both environmental science and marine biology. From an environmental science perspective, the ocean plays a critical role in the Earth's climate system. It acts as a major heat reservoir, regulating global temperatures and influencing weather patterns. The ocean also absorbs a significant amount of carbon dioxide, helping to mitigate the effects of climate change. Understanding the physical and chemical properties of sea water enables scientists to predict changes in the climate and devise strategies to protect the environment.

In marine biology, the composition and behavior of sea water directly affect the distribution and health of marine organisms. The temperature, salinity, and nutrient levels in sea water influence the growth and reproduction of marine life. For

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instance, certain species of plankton thrive in nutrient-rich waters, forming the base of the marine food web. The study of sea water chemistry helps marine biologists understand the ecological relationships and adaptations of marine organisms. Additionally, it aids in the conservation of marine biodiversity by identifying the impact of pollutants and other human activities on marine ecosystems.

Overall, the importance of studying sea water cannot be overstated. It provides critical insights into the functioning of our planet, informs environmental policies, and enhances our understanding of the diverse life forms that inhabit the

ocean. Through the interdisciplinary approach of oceanography, scientists are better equipped to address the challenges facing our oceans and to safeguard their future.

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PHYSICAL PROPERTIES OF SEA WATER

A. Temperature

The temperature of sea water is influenced by a variety of factors, each playing a crucial role in the thermal dynamics of the ocean:

1. Solar Radiation: The primary source of heat for the ocean is solar radiation.

Sunlight heats the surface layers of the ocean, and the amount of heat received depends on the angle of the sun and the duration of daylight. The equator receives more direct sunlight throughout the year, leading to warmer sea surface temperatures, while the poles receive less direct sunlight, resulting in cooler temperatures.

2. Geographical Location: Latitude significantly impacts sea water temperature. Regions near the equator experience consistently high temperatures, whereas polar regions remain cold. Coastal areas can also be influenced by the temperature of the adjacent land masses. For example, coastal waters can be warmer or cooler than open ocean areas depending on the season and land-sea interactions.

3. Ocean Currents: Ocean currents play a vital role in the redistribution of heat within the ocean. Warm currents, such as the Gulf Stream, transport warm water from the equator towards the poles, while cold currents, like the California Current, bring cooler water from polar regions towards the equator. The mixing of warm and cold currents can create regions with unique temperature profiles, affecting local climates and marine ecosystems.

4. Wind and Atmospheric Conditions: Wind patterns influence sea water temperature by driving surface currents and promoting mixing of surface

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waters with deeper layers. Strong winds can cause upwelling, where colder, nutrient-rich waters from the deep ocean are brought to the surface.

Atmospheric conditions, such as cloud cover and precipitation, also affect sea surface temperatures. Clouds can block sunlight, leading to cooler surface waters, while rain can cool or warm the sea surface depending on its temperature relative to the sea.

5. Seasonal Variations: Seasonal changes lead to variations in sea water temperature. During summer, increased solar radiation results in warmer sea surface temperatures, while in winter, reduced solar radiation and increased cooling from the atmosphere lead to lower temperatures. Seasonal variations are more pronounced at higher latitudes, where there is a significant difference in the amount of sunlight received between summer and winter.

Temperature Variation with Depth and Latitude

1. Vertical Temperature Distribution:

□ Surface Layer (Mixed Layer): The uppermost layer of the ocean, known as the mixed layer, is directly influenced by solar heating and atmospheric interactions. It is typically well-mixed due to wind and wave action, resulting in relatively uniform temperatures. The depth of the mixed layer can vary from a few meters to several hundred meters, depending on the intensity of wind and waves, as well as the season.

□ Thermocline: Below the mixed layer lies the thermocline, a region where temperature decreases rapidly with depth. The thermocline acts as a barrier to vertical mixing, separating the warm surface waters from the colder deep

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waters. The depth and steepness of the thermocline can vary with latitude and season. In tropical regions, the thermocline is typically deeper and more pronounced, while in polar regions, it may be less distinct.

□ Deep Ocean: The deep ocean, below the thermocline, experiences very little temperature variation and remains cold, typically around 0-3°C (32-37°F).

The temperature in this layer is largely influenced by the cold, dense water masses formed at high latitudes that sink and spread throughout the deep ocean.

2. Latitudinal Temperature Variation:

□ Tropical Regions: Near the equator, the sea surface temperature is consistently high, averaging around 25-30°C (77-86°F). The mixed layer is relatively deep, and the thermocline is well-defined. The stable warm temperatures support high biological productivity and diverse marine ecosystems.

□ Temperate Regions: In temperate regions, sea surface temperatures vary seasonally, ranging from about 10-25°C (50-77°F) in summer to 0-10°C (32-50°F) in winter. The mixed layer depth and thermocline can change

significantly with the seasons, influencing marine life and oceanographic processes.

□ Polar Regions: Polar regions experience consistently low sea surface temperatures, typically below 5°C (41°F) and often close to freezing. The mixed layer is shallow, and the thermocline is less pronounced. Seasonal ice cover can also affect temperature distribution and marine life.

B. Salinity

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Salinity refers to the concentration of dissolved salts in sea water. It is a key factor in determining the physical and chemical properties of ocean water, affecting its density, conductivity, and biological processes. Salinity is usually expressed in parts per thousand (ppt or ‰), indicating the number of grams of dissolved salts in one kilogram of sea water. The average salinity of ocean water is approximately 35 ppt, meaning there are 35 grams of dissolved salts in every kilogram of sea water.

Salinity is measured using several methods:

1. Refractometry: This method involves using a refractometer to measure the degree to which light is bent, or refracted, as it passes through sea water.

The refractive index is directly related to salinity.

2. Conductivity: Conductivity meters measure the ability of sea water to conduct an electric current. Since dissolved salts increase water's conductivity, these measurements can be used to calculate salinity.

3. Chemical Analysis: This method involves titrating sea water samples with silver nitrate to determine the concentration of chloride ions, which are a major component of sea water salinity. This is known as the Mohr method.

Factors Affecting Salinity

Several factors influence the salinity of sea water, including:

1. Evaporation: In regions with high evaporation rates, such as tropical and subtropical areas, salinity increases as water evaporates and leaves the salts behind. This process is particularly pronounced in enclosed or semi-enclosed seas, like the Red Sea and the Mediterranean Sea.

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2. Precipitation: Rainfall dilutes sea water, reducing salinity. Therefore, regions with high precipitation, such as the equatorial belt, generally have lower salinity levels.

3. River Inflow: Freshwater from rivers entering the ocean decreases salinity

in coastal areas. The influx of river water, carrying dissolved minerals and nutrients, can significantly alter local salinity levels.

4. Ice Formation and Melting: In polar regions, the formation of sea ice excludes salts, increasing the salinity of the remaining water. Conversely, melting ice dilutes sea water, decreasing salinity.

5. Ocean Currents: Ocean currents transport water masses with different salinities, mixing them and affecting local salinity levels. For example, the Gulf Stream carries high-salinity water from the subtropics to the North Atlantic, influencing salinity patterns.

6. Temperature: Temperature affects the solubility of salts and the rate of evaporation. Warmer water can hold more dissolved salts, and higher temperatures increase evaporation, both of which can raise salinity.

Salinity Distribution in Different Ocean Regions

Salinity varies significantly across different regions of the world's oceans:

1. Equatorial Regions: The equatorial belt typically has lower salinity, around 34-35 ppt, due to high rainfall and runoff from major river systems. However, areas with high evaporation rates, such as the central and eastern Pacific, may exhibit slightly higher salinity.

2. Subtropical Regions: Subtropical gyres, found between 20° and 30° latitude in both hemispheres, are characterized by high evaporation rates
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and low precipitation, leading to higher salinity levels, often exceeding 37 ppt. The Mediterranean Sea, with restricted inflow and high evaporation, can have salinity levels above 38 ppt.

3. Polar Regions: Polar regions generally have lower salinity, around 30-34 ppt, due to the input of freshwater from melting ice and relatively low evaporation rates. Seasonal ice formation and melting create significant salinity fluctuations.

4. Coastal Areas: Coastal salinity varies widely depending on the proximity to river mouths, the degree of freshwater influx, and the mixing with oceanic waters. Estuaries, where rivers meet the sea, often exhibit a gradient of salinity from freshwater to marine conditions.

5. Open Ocean: The open ocean has a more stable and predictable salinity pattern, typically around the average of 35 ppt. However, variations occur due to ocean currents, upwelling, and regional climatic conditions.

C. Conductivity

Conductivity in sea water refers to its ability to conduct an electric current.

This property is closely related to the concentration of dissolved salts, or salinity, in the water. The more salts present, the higher the conductivity because salts dissolve into ions, which are charged particles that facilitate the flow of electricity.

□ Ionization of Salts: When salts dissolve in water, they dissociate into positive and negative ions. For example, sodium chloride (NaCl) dissociates into sodium (Na⁺) and chloride (Cl⁻) ions. These ions increase the water's ability to conduct electricity.

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□ Direct Relationship: There is a direct correlation between salinity and conductivity. As salinity increases, the concentration of ions increases, leading to higher conductivity. Conversely, lower salinity results in lower conductivity.

□ Temperature Influence: Conductivity is also influenced by temperature; warmer water tends to have higher conductivity because increased temperature enhances ion mobility. Therefore, accurate measurements of conductivity must account for temperature variations.

Measurement Techniques

Several techniques are employed to measure the conductivity of sea water, each utilizing the relationship between salinity and conductivity to provide accurate salinity readings.

1. Conductivity Meters:

□ Principle: Conductivity meters work by applying an electrical current between two electrodes submerged in the water sample and measuring the current flow, which is directly proportional to the water's conductivity.

□ Compensation: These meters often include temperature sensors to adjust the conductivity readings for temperature variations, ensuring more accurate salinity calculations.

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□ Usage: Portable handheld conductivity meters are commonly used for field measurements, while more sophisticated, stationary meters are used in laboratories for precise analysis.

2. CTD (Conductivity, Temperature, Depth) Sensors:

□ Components: CTD sensors are comprehensive instruments that

simultaneously measure conductivity, temperature, and depth, providing a detailed profile of sea water characteristics.

□ Deployment: These sensors are typically deployed on research vessels, either lowered into the water column or attached to autonomous underwater vehicles. They record continuous data as they move through the water, offering a vertical profile of salinity, temperature, and depth.

□ Applications: CTD sensors are essential for oceanographic research, allowing scientists to study the structure and dynamics of ocean layers and understand how physical properties vary with depth.

3. Argo Floats:

□ Design: Argo floats are autonomous, free-drifting devices equipped with CTD sensors. They periodically descend to various depths, collect data, and then ascend to the surface to transmit the data via satellite.

□ Global Network: The Argo program comprises a global network of floats, providing real-time, high-resolution data on ocean salinity, temperature, and currents. This data is invaluable for climate research and ocean monitoring.

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4. In-Situ Probes:

□ Functionality: In-situ probes are specialized instruments that measure conductivity directly in the marine environment. They are often deployed on buoys, underwater observatories, or attached to long-term monitoring stations.

□ Continuous Monitoring: These probes offer continuous, long-term measurements of conductivity, which are crucial for observing changes in oceanographic conditions over time.

5. Laboratory Analysis:

□ Sample Collection: In some cases, water samples are collected in the field and analyzed in a laboratory setting. Conductivity measurements in the lab can be conducted using precise analytical instruments, ensuring high accuracy.

□ Calibration and Standards: Laboratory instruments are calibrated using standard solutions of known conductivity, ensuring that the measurements are reliable and consistent.

D. Viscosity

Viscosity is a measure of a fluid's resistance to deformation or flow. In simpler

terms, it describes how "thick" or "sticky" a fluid is. In the context of sea water, viscosity is an important physical property that influences how water moves and mixes within the ocean.

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□ **Dynamic Viscosity:** This refers to the internal friction within the fluid, affecting its ability to flow. It is usually measured in units of pascal-seconds (Pa·s) or poise (P).

□ **Kinematic Viscosity:** This takes into account the fluid's density and is defined as the ratio of dynamic viscosity to density. It is measured in square meters per second (m²/s).

Relevance to Sea Water Dynamics:

□ **Fluid Flow:** Viscosity plays a crucial role in determining the flow characteristics of sea water, affecting both large-scale ocean currents and small-scale turbulence.

□ **Mixing Processes:** Viscosity influences the mixing of water masses with different properties (e.g., temperature and salinity), which is essential for nutrient distribution, heat transfer, and the overall thermohaline circulation in the ocean.

□ **Boundary Layers:** Viscosity affects the formation and behavior of boundary layers near solid surfaces, such as the ocean floor and the hulls of ships.

These boundary layers influence drag and flow separation.

Impact on Marine Organisms and Transport Processes

Viscosity has significant implications for marine organisms and various transport processes within the ocean.

1 Impact on Marine Organisms:

□ **Locomotion:** The viscosity of sea water affects how marine organisms move. For small organisms, such as plankton, the viscous forces dominate over

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inertial forces, making their movement heavily influenced by the viscosity of the water. Larger organisms, like fish and marine mammals, experience a different balance of forces but still must contend with viscous drag.

□ **Feeding and Predator-Prey Interactions:** Viscosity influences the feeding mechanisms of filter feeders and the efficiency of predator-prey interactions. For example, higher viscosity can slow down the sinking rate of planktonic prey, making it easier for predators to capture them.

□ **Reproduction:** Many marine organisms rely on the viscosity of sea water for the dispersal of their gametes (eggs and sperm). The transport and mixing of these reproductive cells are influenced by the viscous properties of the surrounding water.

2 Impact on Transport Processes:

□ **Sediment Transport:** Viscosity affects the transport and deposition of sediments in the ocean. Fine particles, like silt and clay, are suspended and transported over long distances due to the viscosity of the water, while larger particles settle more quickly.

□ **Nutrient and Gas Exchange:** The viscosity of sea water plays a role in the diffusion and transport of nutrients and gases (such as oxygen and carbon dioxide) within the ocean. Efficient mixing and transport are crucial for maintaining the health of marine ecosystems and supporting biological productivity.

3 **Pollutant Dispersion:** The dispersion and dilution of pollutants, such as oil spills or chemical discharges, are influenced by the viscosity of sea water. Understanding

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viscosity helps in modeling and predicting the spread of pollutants in the marine environment.

E. Diffusion

Diffusion is the process by which molecules move from areas of higher concentration to areas of lower concentration, driven by their kinetic energy. In the context of sea water, diffusion plays a crucial role in the exchange of nutrients and gases, supporting marine life and influencing oceanic processes.

1 Mechanisms of Diffusion:

□ **Molecular Diffusion:** This occurs at the molecular level and is driven by the random motion of molecules. It is a relatively slow process but is essential for the transfer of nutrients and gases at microscopic scales.

□ **Turbulent Diffusion:** In the ocean, turbulent mixing caused by currents, waves, and wind-driven movements enhances the diffusion process. This type of diffusion is much faster than molecular diffusion and plays a significant role in distributing nutrients and gases over larger spatial scales.

□ **Fick's Laws of Diffusion:** The rate of diffusion is described by Fick's first and second laws. Fick's first law states that the flux of a substance is proportional to the concentration gradient. Fick's second law describes how

the concentration of a substance changes over time as it diffuses.

2 Significance in Nutrient and Gas Exchange:

□ Nutrient Transport: Diffusion is critical for transporting essential nutrients, such as nitrate, phosphate, and silicate, from areas of high concentration

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(e.g., nutrient-rich deep waters) to areas of lower concentration (e.g., the photic zone where photosynthesis occurs). This process supports primary production and the growth of phytoplankton, forming the base of the marine food web.

□ Gas Exchange: Diffusion facilitates the exchange of gases between the ocean and the atmosphere. Oxygen diffuses from the atmosphere into the ocean, supporting aerobic respiration in marine organisms. Similarly, carbon dioxide diffuses from the atmosphere into the ocean, playing a key role in the global carbon cycle and affecting ocean acidification.

□ Biological Processes: Diffusion is essential for cellular processes in marine organisms, including respiration, nutrient uptake, and waste removal. It ensures that cells receive the necessary nutrients and oxygen while removing metabolic waste products.

Diffusion Rates in Varying Conditions

The rate of diffusion in sea water is influenced by several factors, including temperature, salinity, and turbulence.

1. Temperature:

□ Effect on Molecular Diffusion: Higher temperatures increase the kinetic energy of molecules, leading to faster diffusion rates. For example, in warmer waters, the diffusion of oxygen and nutrients occurs more rapidly compared to colder waters.

□ Seasonal Variations: Seasonal temperature changes can affect diffusion rates, with higher rates in summer due to increased temperatures and lower rates in winter when temperatures decrease.

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2. Salinity:

□ Impact on Diffusion: Higher salinity increases the density and viscosity of sea water, which can slow down the rate of molecular diffusion. However, the impact is generally less significant compared to temperature.

□ Stratification: Salinity gradients, often found in estuaries and polar regions,

can create stratified layers in the ocean. These layers can act as barriers to diffusion, affecting the vertical transport of nutrients and gases.

3. Turbulence and Mixing:

□ Enhancing Diffusion: Turbulent mixing, driven by wind, waves, and currents, greatly enhances diffusion rates by breaking down concentration gradients and promoting the transport of molecules across larger distances. This type of diffusion is critical in the upper ocean, where physical processes create turbulence.

□ Boundary Layers: In regions with strong currents or wave action, the boundary layers near the sea surface or ocean floor can experience enhanced diffusion rates due to increased turbulence.

4. Pressure:

Deep Ocean Conditions: In the deep ocean, high pressure can affect the diffusion rates of gases, such as oxygen and carbon dioxide. While the overall impact is less significant than temperature and turbulence, pressure can influence the solubility and diffusion of gases in deep-sea environments.

5. Biological Activity:

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□ Bioturbation: The movement and activity of marine organisms, such as burrowing and feeding, can enhance diffusion rates by disturbing sediments and promoting the exchange of nutrients and gases between the water column and the seabed.

□ Microbial Processes: Microbial activity, particularly in sediments, can create microenvironments with varying diffusion rates. For example, microbial mats can form diffusion barriers, affecting the exchange of nutrients and gases.

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CHEMICAL PROPERTIES OF SEA WATER

A. Major Constituents

Overview of Dissolved Salts and Minerals

Sea water is a complex solution of dissolved salts, minerals, and organic materials. The major constituents, which make up over 99% of the dissolved solids in sea water, include:

1. Sodium (Na^+) and Chloride (Cl^-): These ions are the most abundant, accounting for approximately 85% of the total dissolved salts. Sodium and

chloride ions combine to form sodium chloride (table salt), which is the primary salt in sea water.

2. Magnesium (Mg^{2+}): Magnesium is the third most abundant ion in sea water.

It plays a crucial role in marine biological processes, including the formation of chlorophyll in plants and algae.

3. Calcium (Ca^{2+}): Calcium ions are essential for the formation of shells and skeletons of many marine organisms, such as corals and mollusks.

4. Potassium (K^{+}): Potassium is vital for cellular functions in marine organisms, including nerve transmission and muscle contraction.

5. Sulfate (SO_4^{2-}): Sulfate ions are important for the synthesis of organic compounds and the energy metabolism of certain bacteria.

6. Bicarbonate (HCO_3^{-}): Bicarbonate plays a key role in maintaining the pH balance of sea water and is involved in the carbon cycle, buffering changes in acidity.

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The concentration of these constituents can vary slightly depending on location, depth, and environmental conditions, but their relative proportions remain remarkably constant, a principle known as the rule of constant proportions.

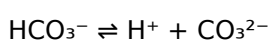
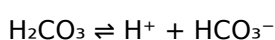
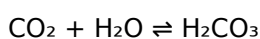
Chemical Equilibrium and Reactions

Chemical equilibrium in sea water involves a balance between the various chemical reactions that occur among dissolved salts, gases, and organic materials.

Key concepts include:

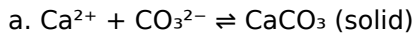
1. Carbonate System: The carbonate system is crucial for regulating the pH of sea water. It involves a series of equilibrium reactions among carbon dioxide (CO_2), carbonic acid (H_2CO_3), bicarbonate (HCO_3^{-}), and carbonate (CO_3^{2-}). This system acts as a buffer, helping to maintain stable pH levels despite variations in CO_2 concentrations.

Reactions:



2. Precipitation and Dissolution: The precipitation and dissolution of minerals, such as calcium carbonate (CaCO_3), influence the availability of ions in sea water. These processes are critical for the formation and maintenance of coral reefs and shellfish structures.

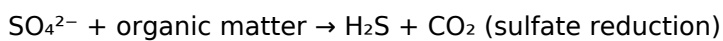
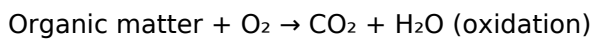
Reactions:



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3. Redox Reactions: Redox (reduction-oxidation) reactions involve the transfer of electrons between molecules and are essential for many biochemical processes in marine environments. For example, the oxidation of organic matter by bacteria affects the availability of nutrients and the cycling of elements like nitrogen and sulfur.

Reactions:



Understanding these chemical equilibria and reactions is crucial for predicting changes in sea water chemistry due to natural processes and human activities, such as pollution and climate change.

B. Trace Elements and Nutrients

Role in Marine Ecosystems

Trace elements and nutrients, though present in much smaller concentrations compared to the major constituents, are vital for the health and functioning of marine ecosystems.

1. Trace Elements: Trace elements, such as iron (Fe), manganese (Mn), zinc (Zn), and copper (Cu), are essential for various biochemical processes in marine organisms. They act as cofactors for enzymes, play roles in photosynthesis, respiration, and nitrogen fixation, and contribute to the structural integrity of proteins and other cellular components.

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2. Nutrients: Nutrients, including nitrogen (N), phosphorus (P), and silicon (Si), are key elements required for the growth and reproduction of marine plants and algae. They are fundamental to primary production, which forms the base of the marine food web.

□ Nitrogen: Present primarily as nitrate (NO_3^-), nitrite (NO_2^-), and ammonium (NH_4^+). It is essential for the synthesis of amino acids, proteins, and nucleic acids.

□ Phosphorus: Found mainly as phosphate (PO_4^{3-}). It is crucial for energy transfer (ATP) and the formation of DNA and RNA.

□ Silicon: Utilized by diatoms and some sponges for constructing their

silica-based skeletons.

Sources and Cycles of Trace Elements

Trace elements and nutrients enter the ocean through various pathways and undergo complex biogeochemical cycling:

1. Sources:

□ **Rivers and Runoff:** Rivers transport nutrients and trace elements from terrestrial environments to the ocean. Agricultural runoff, industrial discharges, and natural weathering of rocks contribute to the input of these elements.

□ **Atmospheric Deposition:** Wind-borne dust and aerosols can deposit trace elements and nutrients directly onto the ocean surface. For example, iron-21

rich dust from deserts can enhance primary productivity in iron-limited regions of the ocean.

□ **Hydrothermal Vents:** These undersea vents release trace elements, such as iron, manganese, and zinc, from the Earth's crust into the ocean. The unique chemistry of hydrothermal fluids supports specialized ecosystems.

2. Cycles:

□ **Biological Uptake and Recycling:** Marine organisms assimilate trace elements and nutrients from the water for growth and reproduction. When these organisms die, their decomposition releases the elements back into the water, making them available for other organisms.

□ **Sedimentation and Remineralization:** Some trace elements and nutrients settle to the ocean floor as part of particulate matter. Benthic organisms and microbial activity can recycle these elements from sediments back into the water column.

□ **Geochemical Cycling:** Trace elements undergo various geochemical processes, such as adsorption onto particles, precipitation, and complexation with organic and inorganic ligands. These processes affect their availability and distribution in the ocean.

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BIOLOGICAL ACTIVITIES AND DISTRIBUTION

A. Plankton

Classification: Phytoplankton and Zooplankton

Plankton are small organisms that drift with ocean currents and form the

base of the marine food web. They can be classified into two main groups: phytoplankton and zooplankton.

- Phytoplankton:

□ Characteristics: Phytoplankton are photosynthetic microorganisms, including diatoms, dinoflagellates, and cyanobacteria. They use sunlight to produce energy through photosynthesis, releasing oxygen and forming the primary source of organic matter in the ocean.

□ Role in Ecosystems: As primary producers, phytoplankton are crucial for the marine food web, providing energy and nutrients to higher trophic levels, including zooplankton, fish, and marine mammals.

- Zooplankton:

□ Characteristics: Zooplankton are heterotrophic organisms, meaning they consume other organisms for energy. This group includes a diverse range of organisms such as copepods, krill, jellyfish, and larval stages of larger animals.

□ Role in Ecosystems: Zooplankton feed on phytoplankton and other small particles, transferring energy up the food web. They are also prey for many larger marine species, playing a key role in nutrient cycling and energy transfer.

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Distribution Patterns and Ecological Roles

The distribution of plankton is influenced by various environmental factors, including light, temperature, nutrients, and water currents.

- Distribution Patterns:

□ Phytoplankton: Typically found in the euphotic zone, where sunlight penetrates the water and supports photosynthesis. Their distribution is highly variable, often forming blooms in nutrient-rich areas such as upwelling zones and coastal regions.

□ Zooplankton: Distributed throughout the water column, with some species undergoing diel vertical migration, moving to deeper waters during the day to avoid predators and returning to the surface at night to feed.

- Ecological Roles:

□ Phytoplankton: Besides being primary producers, phytoplankton influence global biogeochemical cycles, particularly the carbon cycle. They absorb carbon dioxide during photosynthesis, and their sinking particles contribute

to the ocean's biological pump, sequestering carbon in deep waters.

□ Zooplankton: Zooplankton help regulate phytoplankton populations through grazing. They also contribute to the vertical flux of organic matter by producing fecal pellets and other detritus that sink, supporting deep-sea ecosystems.

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B. Marine Fauna

Adaptations to Brackish Water Conditions

Brackish water environments, such as estuaries and coastal lagoons, exhibit varying salinity levels, requiring specialized adaptations from marine fauna.

- Physiological Adaptations:

□ Osmoregulation: Many brackish water species have developed mechanisms to regulate their internal salt concentration, such as efficient ion transport systems in their gills or specialized excretory organs.

□ Behavioural Adaptations: Some species may migrate between different salinity zones to optimize their physiological processes, such as spawning in lower salinity areas and feeding in higher salinity regions.

- Ecological Roles:

□ Nursery Habitats: Brackish waters often serve as crucial nursery grounds for juvenile stages of many marine species, providing a relatively safe and nutrient-rich environment for growth.

□ Biodiversity Hotspots: These areas support high biodiversity, including a mix of marine, freshwater, and estuarine species, contributing to the overall productivity and resilience of coastal ecosystems.

Diversity and Distribution in Various Marine Environments

Marine fauna exhibit remarkable diversity and are distributed across different marine habitats, from shallow coastal waters to the deep sea.

- Coastal Zones:

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□ Characteristics: Coastal zones are rich in nutrients due to runoff and upwelling, supporting diverse communities of fish, invertebrates, and marine mammals.

□ Habitats: Include coral reefs, mangroves, seagrass beds, and rocky shores, each hosting unique assemblages of species adapted to specific environmental conditions.

- Open Ocean:

□ Characteristics: The open ocean, or pelagic zone, is characterized by its vast expanse and varying depths. Nutrient availability is often lower compared to coastal areas.

□ Habitats: Includes the epipelagic zone (surface to 200 meters), where sunlight supports phytoplankton growth, and deeper zones like the mesopelagic and bathypelagic, inhabited by species adapted to low light and high pressure.

- Deep Sea:

□ Characteristics: The deep sea is one of the most extreme environments, with high pressure, low temperatures, and complete darkness. It includes habitats such as abyssal plains, hydrothermal vents, and deep-sea trenches.

□ Adaptations: Fauna in these areas have unique adaptations, such as bioluminescence, pressure-resistant body structures, and specialized feeding strategies to survive in the nutrient-sparse environment.

C. Physiological Adaptations

Mechanisms of Adaptation to Temperature, Salinity, and Pressure

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Marine organisms have evolved various physiological adaptations to cope with the diverse and often extreme conditions of the marine environment.

- Temperature Adaptations:

□ Ectothermy: Many marine organisms are ectothermic, meaning their body temperature fluctuates with the surrounding water. They may exhibit behavioral adaptations, such as seeking warmer or cooler waters, to regulate their body temperature.

□ Heat Shock Proteins: Some species produce heat shock proteins to protect their cells from temperature-induced damage during sudden changes in temperature.

- Salinity Adaptations:

□ Osmoregulation: As previously mentioned, osmoregulation is crucial for maintaining internal ionic balance. Marine fish, for example, drink seawater and excrete excess salts through specialized cells in their gills.

□ Euryhaline Species: These species can tolerate a wide range of salinities, allowing them to inhabit both marine and brackish waters.

- Pressure Adaptations:

□ Structural Reinforcements: Deep-sea organisms often have reinforced cell membranes and proteins that function optimally under high pressure.

□ Buoyancy Control: Many deep-sea species have adaptations to control buoyancy, such as oil-filled bladders or gas-filled sacs, enabling them to maintain their position in the water column.

Case Studies of Specific Marine Organisms

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Examining specific marine organisms provides insight into the diverse strategies employed to survive and thrive in various marine environments.

1. Emperor Penguin (*Aptenodytes forsteri*):

□ Adaptations: Emperor penguins have adaptations for extreme cold, including a thick layer of blubber, dense feathers for insulation, and the ability to slow their metabolism during long dives.

□ Habitat: They breed on the Antarctic ice and forage in the frigid Southern Ocean.

2. Deep-Sea Anglerfish (Family Lophiiformes):

□ Adaptations: These fish possess bioluminescent lures to attract prey in the dark depths of the ocean. They also have a specialized reproductive strategy where tiny males attach to the much larger females, providing a constant source of sperm.

□ Habitat: Found at depths of over 1,000 meters, in regions with high pressure and low light.

3. Mangrove Tree Crab (*Aratus pisonii*):

□ Adaptations: This crab has adapted to brackish water conditions in mangrove ecosystems. It can osmoregulate effectively and has behavioral adaptations to avoid predators and forage on both plant and animal matter.

□ Habitat: Inhabits mangrove forests along the Atlantic and Pacific coasts of the Americas.

4. Albatross (Family Diomedidae):

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□ Adaptations: Albatrosses have long wingspans that allow them to glide over vast ocean distances with minimal energy expenditure. They also have specialized salt glands to excrete excess salt from ingested seawater.

□ Habitat: These birds are found in open ocean environments, particularly in the Southern Hemisphere, often traveling thousands of kilometers for

foraging.

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PHYSICAL DYNAMICS OF THE SEA

A. Currents

Types and Causes of Ocean Currents

Ocean currents are continuous, directed movements of sea water generated by various forces acting upon the water, including wind, the Earth's rotation, and differences in water density.

- Surface Currents:

□ Wind-Driven Currents: These currents are primarily driven by wind patterns and are influenced by the Coriolis effect. Examples include the Gulf Stream, the Kuroshio Current, and the Antarctic Circumpolar Current.

□ Gyres: Large system of circular ocean currents formed by global wind patterns and forces created by the Earth's rotation. Major ocean gyres include the North Atlantic Gyre, the South Atlantic Gyre, the North Pacific Gyre, the South Pacific Gyre, and the Indian Ocean Gyre.

- Deep Ocean Currents:

□ Thermohaline Circulation: Also known as the "global conveyor belt," this system of deep ocean currents is driven by differences in water density, which is affected by temperature (thermo) and salinity (haline). Cold, salty water is denser and sinks, while warmer, less salty water rises, creating a global circulation pattern that helps regulate climate by transporting heat.

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Impact on Climate and Marine Navigation

Ocean currents play a crucial role in shaping global climate patterns and marine navigation.

- Climate Impact:

□ Heat Distribution: Ocean currents transport warm water from the equator towards the poles and cold water from the poles towards the equator, helping to regulate global climate and weather patterns. For instance, the Gulf Stream warms the climate of Western Europe.

□ Climate Events: Changes in ocean currents can lead to significant climate events. El Niño and La Niña are examples of such events, driven by changes in the Pacific Ocean currents, affecting global weather patterns.

- Marine Navigation:

□ **Navigation Routes:** Understanding ocean currents is essential for marine navigation. Ships can take advantage of favorable currents to reduce travel time and fuel consumption. Historical explorers, such as the Polynesians and European navigators, relied on knowledge of currents for their voyages.

□ **Safety and Efficiency:** Modern shipping routes are planned considering ocean currents to enhance safety and efficiency. Knowledge of currents also aids in search and rescue operations and the management of oil spills and marine debris.

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B. Tides

Causes and Effects of Tidal Movements

Tides are the periodic rise and fall of sea levels caused by the gravitational forces exerted by the Moon and the Sun, along with the rotation of the Earth.

- Causes:

□ **Gravitational Pull:** The gravitational pull of the Moon has the most significant effect on tides, creating two high tides and two low tides each day as the Earth rotates. The Sun also contributes to tides but to a lesser extent.

□ **Tidal Bulges:** The gravitational forces create bulges in the Earth's oceans—one facing the Moon and the other on the opposite side of the Earth. These bulges result in high tides, while areas between the bulges experience low tides.

- Effects:

□ **Coastal Flooding:** High tides can lead to coastal flooding, especially during storm surges or when combined with strong onshore winds.

□ **Intertidal Zones:** Tidal movements create intertidal zones, areas that are exposed at low tide and submerged at high tide. These zones are rich in biodiversity and serve as critical habitats for various marine species.

Tidal Patterns and Their Influence on Marine Life

Tidal patterns vary based on geographic location and local topography, influencing marine life and ecosystems.

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- Tidal Patterns:

□ **Diurnal Tides:** One high tide and one low tide each day.

□ **Semi-Diurnal Tides:** Two high tides and two low tides of approximately equal height each day.

☐ Mixed Tides: Two high tides and two low tides of different heights each day.

- Influence on Marine Life:

☐ Feeding and Reproduction: Many marine organisms, such as crabs, fish, and birds, time their feeding and reproductive activities with the tides. For example, grunion fish spawn on sandy beaches during high tides.

☐ Habitat Formation: Tidal movements shape coastal habitats, such as mudflats, salt marshes, and mangroves, providing essential breeding and feeding grounds for various species.

☐ Nutrient Transport: Tides help transport nutrients and oxygen into estuaries and coastal ecosystems, supporting high productivity and biodiversity.

C. Waves

Formation and Types of Waves

Waves are generated by the transfer of energy from wind to the water surface. The characteristics of waves depend on the wind speed, duration, and the distance over which the wind blows (fetch).

- Formation:

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☐ Wind-Generated Waves: Most common type, created by the friction between the wind and the water surface. As wind blows across the surface, it creates ripples that grow into larger waves.

☐ Seismic Waves (Tsunamis): Generated by underwater earthquakes, volcanic eruptions, or landslides. These waves can travel across entire ocean basins and cause significant destruction when they reach coastal areas.

- Types of Waves:

☐ Capillary Waves: Small, rippling waves created by light winds.

☐ Swell Waves: Long, rolling waves that have travelled out of their area of origin. They have longer wavelengths and can travel great distances.

☐ Breaking Waves: Occur when waves move into shallow water and become too steep to support themselves, causing them to collapse or break.

Effects on Coastal Areas and Marine Habitats

Waves have a profound impact on coastal areas and marine habitats, shaping the landscape and influencing ecological processes.

- Coastal Erosion and Deposition:

☐ Erosion:** Waves can erode coastlines, removing sediments and reshaping

shorelines. This process can lead to the loss of habitats and property.

□ Deposition:** Waves also transport and deposit sediments, forming features like beaches, sandbars, and dunes. These features provide habitats and protection against storm surges.

- Marine Habitats:

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□ Rocky Shores: Exposed to constant wave action, rocky shores host specialized organisms adapted to withstand strong forces and changing conditions.

□ Coral Reefs: Coral reefs benefit from wave action, which helps bring in nutrients and oxygen while removing waste products. However, excessive wave energy, particularly from storms, can damage reef structures.

□ Seagrass Beds and Kelp Forests: These habitats are found in areas with moderate wave action, which provides necessary water movement without causing physical damage.

D. Sound and Radiation

Underwater Acoustics and Its Applications

Sound travels much faster and farther in water than in air, making it a crucial tool for marine communication, navigation, and exploration.

- Underwater Acoustics:

□ Speed of Sound: In sea water, the speed of sound is approximately 1,500 meters per second, depending on temperature, salinity, and pressure.

□ Echo Sounding: Used for mapping the seafloor and detecting underwater objects. Ships send out sound pulses and measure the time it takes for the echoes to return, providing detailed bathymetric maps.

□ Sonar: Employed for navigation, communication, and detecting submarines and other underwater hazards. Both active sonar (sending out sound waves and listening for echoes) and passive sonar (listening for natural sounds) are used.

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- Applications:

□ Marine Research: Underwater acoustics are vital for studying marine mammals, fish populations, and underwater geological formations.

□ Submarine Communication: Sound waves are used for communication between submarines and other underwater devices.

□ Environmental Monitoring: Acoustic sensors monitor ocean conditions, including temperature and salinity profiles, which are crucial for understanding climate change impacts.

Light Penetration and Its Influence on Marine Ecosystems

Light penetration in the ocean is critical for photosynthesis and affects the distribution and behavior of marine organisms.

- Light Penetration:

□ Absorption and Scattering: Water absorbs and scatters light, with shorter wavelengths (blue and green) penetrating deeper than longer wavelengths (red and orange). This selective absorption creates the blue color of the deep ocean.

□ Euphotic Zone: The upper layer of the ocean where sunlight is sufficient for photosynthesis. Typically extends to a depth of about 100-200 meters, depending on water clarity.

- Influence on Marine Ecosystems:

□ Photosynthesis: Light penetration supports the growth of phytoplankton, which are the primary producers in marine ecosystems. The availability of light limits primary production to the euphotic zone.

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□ Behavioural Patterns: Many marine organisms exhibit vertical migration, moving up to the surface to feed at night and descending to deeper waters during the day to avoid predators. This behaviour is driven by changes in light levels.

□ Habitat Zonation: Light availability influences the vertical distribution of marine habitats. Coral reefs and seagrass beds thrive in shallow, well-lit waters, while deeper areas host organisms adapted to low-light conditions, such as bioluminescent species.

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SEA WATER BEHAVIOR AND DYNAMICS

A. Sea Water Circulation

Global Circulation Patterns

Global ocean circulation, also known as thermohaline circulation or the "global conveyor belt," is a complex system that drives the movement of sea water across the world's oceans. This circulation is crucial for distributing heat, nutrients, and gases around the globe.

- Thermohaline Circulation:

□ Definition: A large-scale circulation driven by differences in water density, which are controlled by temperature (thermo) and salinity (haline). Cold, salty water is denser and sinks, while warm, less salty water rises, creating a continuous global flow.

□ Pathway: The circulation begins in the North Atlantic, where cold, dense water sinks and flows southward, joining the Antarctic Circumpolar Current. It then moves into the Indian and Pacific Oceans, eventually rising to the surface and returning to the Atlantic via surface currents.

- Wind-Driven Circulation:

□ Surface Currents: Driven by wind patterns and the Coriolis effect, creating gyres in the major ocean basins. These currents include the Gulf Stream in the North Atlantic, the Kuroshio Current in the North Pacific, and the East Australian Current in the South Pacific.

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□ Equatorial Currents: Flow westward along the equator due to trade winds, with the Equatorial Counter current flowing eastward between the north and south equatorial currents.

Interaction Between Surface and Deep Water Currents

The interaction between surface and deep water currents is vital for maintaining the global thermohaline circulation and influencing regional climates.

- Upwelling and Downwelling:

□ Upwelling: Occurs when wind-driven surface water is replaced by deeper, nutrient-rich water. Common along western continental margins, upwelling zones are highly productive and support large fisheries.

□ Downwelling: Occurs when surface water converges and sinks, bringing oxygen and other surface properties to deeper layers. This process helps ventilate the deep ocean.

- Mixing Processes:

□ Vertical Mixing: Enhances the exchange of heat, nutrients, and gases between surface and deep waters. Turbulence, caused by wind, waves, and currents, plays a significant role in vertical mixing.

□ Horizontal Mixing: Driven by large-scale currents and eddies, it helps distribute substances laterally across ocean basins. Mesoscale eddies, which are circular currents spanning tens to hundreds of kilometers, contribute to

horizontal mixing.

B. Diffusion and Mixing

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Processes Influencing the Distribution of Substances

Diffusion and mixing are fundamental processes that determine the distribution of substances, such as nutrients, gases, and pollutants, in sea water.

□ **Molecular Diffusion:** The movement of molecules from areas of high concentration to areas of low concentration due to random molecular motion.

Although slow in sea water, molecular diffusion is crucial for the fine-scale distribution of substances. For Examples is Dissolved oxygen and carbon dioxide diffusing into and out of water masses, nutrient ions spreading from regions of high concentration (e.g., decaying organic matter) to surrounding waters.

□ **Turbulent Diffusion:** The mixing of water masses due to turbulence, which significantly enhances the rate of diffusion compared to molecular diffusion alone. Turbulent diffusion is caused by wind, waves, and currents. For examples is the mixing of fresh river water with saline sea water in estuaries, the dispersion of pollutants in coastal areas, and the spread of heat and nutrients in the open ocean.

Role in Nutrient Cycling and Marine Ecosystems

Diffusion and mixing are critical for nutrient cycling and supporting marine ecosystems by facilitating the distribution of essential elements.

- Nutrient Cycling:

□ **Upwelling:** Brings nutrient-rich deep water to the surface, fueling primary production by phytoplankton. This process supports high biological productivity and the growth of fisheries.

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□ **Nutrient Redistribution:** Diffusion and mixing spread nutrients horizontally and vertically, ensuring their availability throughout the water column. This redistribution supports diverse marine life at different depths and locations.

- Marine Ecosystems:

□ **Primary Production:** Phytoplankton, the primary producers in marine ecosystems, rely on the availability of nutrients, which are supplied through diffusion and mixing processes. Enhanced nutrient availability leads to phytoplankton blooms, which form the base of the marine food web.

□ **Habitat Connectivity:** Mixing processes create interconnected habitats by transporting larvae and spores, allowing species to colonize new areas and maintain genetic diversity. This connectivity is crucial for the resilience and stability of marine ecosystems.

□ **Oxygen Distribution:** Mixing ensures the distribution of dissolved oxygen throughout the water column, which is vital for the survival of aerobic organisms. Poor mixing can lead to hypoxic conditions, negatively impacting marine life.

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BRACKISH WATER CONDITIONS

A. Characteristics of Brackish Water

Definition and Comparison with Marine and Freshwater

Brackish water is a type of water with salinity levels between that of freshwater and seawater. It typically occurs in areas where freshwater from rivers or streams mixes with seawater from oceans or seas.

Brackish Water: Has a salinity range of approximately 0.5 to 30 parts per thousand (ppt), which is higher than freshwater but lower than seawater. The exact salinity can vary based on local conditions and the sources of freshwater and seawater.

Comparison with Marine and Freshwater:

□ **Marine Water:** Typically has a salinity of around 35 ppt, which is relatively constant and supports a wide range of marine organisms adapted to high salinity levels.

□ **Freshwater:** Has a salinity of less than 0.5 ppt. Freshwater environments include rivers, lakes, and ponds, and they support organisms adapted to very low salinity conditions.

Brackish Water Characteristics:

□ **Salinity Variation:** Brackish water environments often experience fluctuating salinity levels due to changes in freshwater inflow, tides, and evaporation rates. This variability can affect the composition and functioning of these ecosystems.

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□ **Water Chemistry:** In addition to salinity, brackish water may have unique chemical characteristics, such as varying levels of nutrients and organic matter, influenced by the mixing of freshwater and seawater.

Examples of Brackish Environments

Brackish water environments are commonly found in transitional zones between freshwater and marine systems. Examples include:

- **Estuaries:** Where rivers meet the sea, creating a mixing zone of freshwater and seawater. Estuaries are often highly productive and support diverse ecosystems. Example is the Chesapeake Bay in the United States and the Seine estuary in France.
- **Salt Marshes:** Coastal wetlands characterized by the presence of salt-tolerant plants and frequent tidal flooding, leading to brackish conditions. Example is the Everglades in Florida and the Wadden Sea in the Netherlands.
- **Mangrove Swamps:** Coastal areas with dense mangrove trees adapted to brackish water and tidal conditions. These areas provide critical habitats for various marine and terrestrial species. Example is the Sundarbans in India and Bangladesh and the Everglades in Florida.
- **Coastal Lagoons:** Shallow, brackish water bodies separated from the open sea by sandbars, barrier islands, or coral reefs. They often experience variable salinity due to tidal exchange and freshwater input. Example is the Laguna Madre in Texas and the Lagoons of New Caledonia.

B. Fauna of Brackish Water

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Adaptations of Organisms to Fluctuating Salinity

Organisms living in brackish water environments must be able to tolerate and adapt to fluctuating salinity levels. These adaptations are crucial for their survival and reproductive success.

- **Osmoregulation:** Brackish water fauna have developed specialized mechanisms to maintain internal osmotic balance despite changes in external salinity. This can include:
 - **Ion Transport:** Adjusting the uptake and excretion of ions through specialized cells or organs, such as gills or kidneys.
 - **Behavioral Adaptations:** Moving to areas with more stable salinity levels or altering activity patterns to avoid extreme conditions.
 - **Salinity Tolerance:** Some species are euryhaline, meaning they can tolerate a wide range of salinities. Others are stenohaline, adapted to a narrow range of salinities. Euryhaline species are often more common in brackish environments due to their adaptability.

□ Example: Euryhaline fish like the common mullet (*Mugil cephalus*) can thrive in both brackish and marine waters.

Case Studies of Brackish Water Species

1. American Eel (*Anguilla rostrata*):

□ Adaptations: The American eel is a catadromous fish, meaning it spawns in the ocean but spends most of its life in freshwater and brackish

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environments. It has specialized osmoregulatory mechanisms to cope with varying salinities.

□ Habitat: Found in estuaries and coastal rivers along the eastern coast of North America.

2. Common Snook (*Centropomus undecimalis*):

□ Adaptations: This fish can tolerate a wide range of salinities, from freshwater to full seawater. It has adapted to brackish environments by adjusting its osmoregulatory processes based on salinity.

□ Habitat: Commonly found in estuaries, mangroves, and coastal lagoons in the southeastern United States and the Caribbean.

3. Salt Marsh Cordgrass (*Spartina alterniflora*):

□ Adaptations: This plant is adapted to high salinity levels found in salt marshes. It has specialized structures, such as salt glands, to excrete excess salt and maintain water balance.

□ Habitat: Dominates salt marshes along the Atlantic and Gulf coasts of the United States.

4. Mangrove Tree Crab (*Aratus pisonii*):

□ Adaptations: This crab is well-adapted to the brackish and fluctuating salinity conditions of mangrove swamps. It can regulate its internal salt concentration and has behavioral adaptations to avoid extreme salinity.

□ Habitat: Found in mangrove forests along the Atlantic and Pacific coasts of the Americas.

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INTERRELATIONSHIPS IN MARINE ECOSYSTEMS

A. Trophic Interactions

Food Webs and Energy Flow

Marine ecosystems are characterized by complex trophic interactions that form food webs, illustrating how energy and nutrients flow through different levels

of the ecosystem.

- Food Webs:

□ Primary Producers: At the base of the food web, primary producers such as phytoplankton, seaweeds, and seagrasses convert solar energy into chemical energy through photosynthesis. These organisms form the foundation of marine food webs.

□ Primary Consumers: Herbivores, including zooplankton, small fish, and grazing invertebrates, feed on primary producers. They transfer energy up the food chain.

□ Secondary Consumers: Predators that consume herbivores, such as larger fish, squid, and some marine mammals, are secondary consumers. They are crucial for controlling prey populations and maintaining ecosystem balance.

□ Tertiary Consumers: Top predators, such as sharks, large marine mammals (e.g., whales), and apex predators, feed on secondary consumers. They play a key role in regulating the abundance and diversity of species at lower trophic levels.

- Energy Flow:

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□ Energy Transfer Efficiency: Energy transfer between trophic levels is relatively inefficient, with only about 10% of the energy passed from one level to the next. This inefficiency limits the number of trophic levels and the biomass of top predators.

□ Biomass Pyramids: Marine food webs can be represented as biomass pyramids, showing the decreasing biomass of organisms at each successive trophic level. This pyramid illustrates how energy decreases as it moves through the food web.

Impact of Environmental Changes on Marine Trophic Structures

Environmental changes can have profound effects on marine trophic structures, influencing the dynamics of food webs and the health of ecosystems.

- Climate Change:

□ Temperature Rise: Increasing ocean temperatures can affect the metabolism, distribution, and reproductive patterns of marine species.

Warmer temperatures may lead to shifts in species distributions, affecting predator-prey relationships.

□ Ocean Acidification: The absorption of excess CO₂ by the ocean leads to

acidification, which can impact calcifying organisms like corals, shellfish, and certain plankton species. This affects the entire food web, as these organisms are crucial for the structure and functioning of marine ecosystems.

- Pollution:

☐ Nutrient Enrichment: Excess nutrients from agricultural runoff can lead to eutrophication, causing algal blooms that deplete oxygen and create dead

47 zones. These blooms can disrupt food webs by reducing the availability of oxygen and altering the abundance of primary producers.

☐ Chemical Contaminants: Pollutants such as heavy metals, pesticides, and plastics can accumulate in marine organisms, leading to health issues and disruptions in food chains. For example, bio-magnification can cause toxic substances to concentrate in top predators.

- Overfishing:

☐ Species Depletion: Overfishing reduces the abundance of targeted species and can lead to the collapse of fisheries. This disruption affects the entire food web, altering predator-prey relationships and potentially causing cascading effects throughout the ecosystem.

☐ Bycatch: Non-target species caught unintentionally in fishing gear can lead to declines in populations of key marine species, further disrupting ecological balance.

B. Human Impact

Effects of Pollution, Climate Change, and Overfishing

Human activities have significant impacts on marine ecosystems, affecting their health, productivity, and resilience.

- Pollution:

☐ Marine Debris: Plastics and other debris in the ocean can entangle marine animals, cause ingestion issues, and damage habitats. For instance, sea turtles often mistake plastic bags for jellyfish, leading to internal injuries or death.

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☐ Chemical Pollution: Persistent organic pollutants (POPs) and heavy metals can accumulate in marine organisms, leading to health problems and impacting reproductive success. Examples include mercury contamination in fish and PCBs in marine mammals.

- Climate Change:

□ Sea Level Rise: Melting ice caps and thermal expansion contribute to rising sea levels, which can lead to the loss of coastal habitats such as mangroves, salt marshes, and coral reefs.

□ Coral Bleaching: Increased sea temperatures cause coral bleaching, where corals expel their symbiotic algae, leading to loss of colour and, if prolonged, coral death. Coral reefs provide essential habitats for many marine species, and their decline affects entire ecosystems.

- Overfishing:

□ Population Declines: Overfishing depletes fish stocks, reduces biodiversity, and disrupts food webs. This can lead to the collapse of important fisheries and loss of economic and nutritional resources for human communities.

□ Habitat Destruction: Destructive fishing practices, such as bottom trawling, damage seafloor habitats, including coral reefs and seagrass beds, which are critical for many marine species.

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Conservation Strategies and Management Practices

Effective conservation and management practices are essential for mitigating human impacts and ensuring the sustainability of marine ecosystems.

- Marine Protected Areas (MPAs):

□ Definition: Designated regions where human activities are regulated or restricted to protect marine habitats and species. MPAs can enhance biodiversity, protect critical habitats, and support fish stock recovery.

□ Examples: The Great Barrier Reef Marine Park in Australia and the Papahānaumokuākea Marine National Monument in the U.S. are examples of MPAs that contribute to marine conservation.

- Sustainable Fisheries Management:

□ Quotas and Regulations: Setting catch limits, implementing fishing quotas, and enforcing regulations to prevent overfishing and ensure the sustainability of fish stocks.

□ Bycatch Reduction: Developing and adopting fishing gear and practices that minimize bycatch and reduce the impact on non-target species.

- Pollution Control:

□ Legislation and Policies: Enforcing regulations to reduce pollutants and manage waste disposal, such as the Marine Pollution Convention (MARPOL)

and local policies on plastic use and disposal.

□ Cleanup Initiatives: Supporting efforts to clean up marine debris and restore polluted areas, including beach cleanups and the removal of derelict fishing gear.

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- Climate Change Mitigation:

□ Reduction of Greenhouse Gases: Implementing strategies to reduce greenhouse gas emissions, such as transitioning to renewable energy sources and improving energy efficiency.

□ Adaptation Strategies: Developing and implementing measures to help marine ecosystems adapt to changing conditions, including the restoration of habitats and the protection of climate-resilient species.

- Public Awareness and Education:

□ Engagement: Raising awareness about the importance of marine conservation and encouraging responsible behavior through education and outreach programs.

□ Community Involvement: Engaging local communities in conservation efforts, such as citizen science projects and coastal management initiatives.

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CONCLUSION

As we look towards the future, several key areas of research are likely to shape our understanding and management of marine ecosystems:

1. Climate Change and Ocean Acidification: Continued research into the effects of rising temperatures and ocean acidification on marine species and ecosystems is crucial. This includes understanding how these factors affect coral reefs, fisheries, and marine food webs.
2. Marine Conservation and Restoration: Advancements in conservation techniques, such as habitat restoration and the creation of marine protected areas, are needed to address biodiversity loss and ecosystem degradation.
3. Impact of Emerging Pollutants: Research into the effects of emerging pollutants, such as microplastics and pharmaceutical residues, will help us understand their impacts on marine organisms and ecosystems.
4. Sustainable Fisheries Management: Developing and implementing sustainable fishing practices and technologies will be vital for maintaining fish stocks and marine biodiversity.

5. Marine Spatial Planning: Enhanced marine spatial planning approaches are needed to balance human activities with conservation goals, ensuring the sustainable use of marine resources.

6. Technological Innovations: Advancements in marine technology, such as remote sensing, autonomous vehicles, and deep-sea exploration tools, will provide new insights into marine processes and habitats.

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Paper 6 of 7 — EMT 227 lecture Note for 2023.2024 session

Level 200 • Second Semester • Handout

INTRODUCTION

Overview of Oceanography

Oceanography, the scientific study of the ocean, encompasses a wide range of disciplines including biology, chemistry, geology, and physics. This field is essential for understanding the complex and dynamic nature of the world's oceans. Oceanographers study the physical properties of sea water, such as temperature, salinity, and currents, as well as the chemical composition, including dissolved gases and nutrients. They also investigate the geological structures on the ocean floor and the biological diversity within marine ecosystems. The insights gained from oceanography are crucial for predicting climate patterns, understanding marine biodiversity, and managing marine resources.

Relevance to Environmental Science and Marine Biology

The study of sea water is of paramount importance to both environmental science and marine biology. From an environmental science perspective, the ocean plays a critical role in the Earth's climate system. It acts as a major heat reservoir, regulating global temperatures and influencing weather patterns. The ocean also absorbs a significant amount of carbon dioxide, helping to mitigate the effects of climate change. Understanding the physical and chemical properties of sea water enables scientists to predict changes in the climate and devise strategies to protect the environment.

In marine biology, the composition and behavior of sea water directly affect the distribution and health of marine organisms. The temperature, salinity, and nutrient levels in sea water influence the growth and reproduction of marine life. For instance, certain species of plankton thrive in nutrient-rich waters, forming the base of the marine food web. The study of sea water chemistry helps marine biologists understand the ecological relationships and

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adaptations of marine organisms. Additionally, it aids in the conservation of marine biodiversity by identifying the impact of pollutants and other human activities on marine ecosystems.

Overall, the importance of studying sea water cannot be overstated. It provides critical insights into the functioning of our planet, informs environmental policies, and enhances our understanding of the diverse life forms that inhabit the ocean. Through the interdisciplinary approach of oceanography, scientists are better equipped to address the challenges facing our oceans and to safeguard their future.

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PHYSICAL PROPERTIES OF SEA WATER

A. Temperature

The temperature of sea water is influenced by a variety of factors, each playing a crucial role in the thermal dynamics of the ocean:

1. **Solar Radiation:** The primary source of heat for the ocean is solar radiation. Sunlight heats the surface layers of the ocean, and the amount of heat received depends on the angle of the sun and the duration of daylight. The equator receives more direct sunlight throughout the year, leading to warmer sea surface temperatures, while the poles receive less direct sunlight, resulting in cooler temperatures.
2. **Geographical Location:** Latitude significantly impacts sea water temperature. Regions near the equator experience consistently high temperatures, whereas polar regions remain cold. Coastal areas can also be influenced by the temperature of the adjacent land masses. For example, coastal waters can be warmer or cooler than open ocean areas depending on the season and land-sea interactions.
3. **Ocean Currents:** Ocean currents play a vital role in the redistribution of heat within the ocean. Warm currents, such as the Gulf Stream, transport warm water from the equator towards the poles, while cold currents, like the California Current, bring cooler water from polar regions towards the equator. The mixing of warm and cold currents can create regions with unique temperature profiles, affecting local climates and marine ecosystems.
4. **Wind and Atmospheric Conditions:** Wind patterns influence sea water temperature by driving surface currents and promoting mixing of surface waters with deeper layers. Strong winds can cause upwelling, where colder, nutrient-rich waters from the deep ocean are brought to the surface. Atmospheric conditions, such as cloud cover and precipitation, also affect sea surface temperatures. Clouds can block sunlight, leading to cooler surface waters, while rain can cool or warm the sea surface depending on its temperature relative to the sea.
5. **Seasonal Variations:** Seasonal changes lead to variations in sea water temperature. During summer, increased solar radiation results in warmer sea surface temperatures, while in winter, reduced solar radiation and increased cooling from the atmosphere lead to lower temperatures. Seasonal variations are more pronounced at higher latitudes, where there is a significant difference in the amount of sunlight received between summer and winter.

Temperature Variation with Depth and Latitude

1. Vertical Temperature Distribution:

- **Surface Layer (Mixed Layer):** The uppermost layer of the ocean, known as the mixed layer, is directly influenced by solar heating and atmospheric interactions. It is typically well-mixed due to wind and wave action, resulting in relatively uniform temperatures. The depth of the mixed layer can vary from a few meters to several hundred meters, depending on the intensity of wind and waves, as well as the season.
- **Thermocline:** Below the mixed layer lies the thermocline, a region where temperature decreases rapidly with depth. The thermocline acts as a barrier to vertical mixing, separating the warm surface waters from the colder deep waters. The depth and steepness of the thermocline can vary with latitude and season. In tropical regions, the thermocline is typically deeper and more pronounced, while in polar regions, it may be less distinct.

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- **Deep Ocean:** The deep ocean, below the thermocline, experiences very little temperature variation and remains cold, typically around 0-3°C (32-37°F). The temperature in this layer is largely influenced by the cold, dense water masses formed at high latitudes that sink and spread throughout the deep ocean.

2. Latitudinal Temperature Variation:

- **Tropical Regions:** Near the equator, the sea surface temperature is consistently high, averaging around 25-30°C (77-86°F). The mixed layer is relatively deep, and the thermocline is well-defined. The stable warm temperatures support high biological productivity and diverse marine ecosystems.
- **Temperate Regions:** In temperate regions, sea surface temperatures vary seasonally, ranging from about 10-25°C (50-77°F) in summer to 0-10°C (32-50°F) in winter. The mixed layer depth and thermocline can change significantly with the seasons, influencing marine life and oceanographic processes.
- **Polar Regions:** Polar regions experience consistently low sea surface temperatures, typically below 5°C (41°F) and often close to freezing. The mixed layer is shallow, and the thermocline is less pronounced. Seasonal ice cover can also affect temperature distribution and marine life.

B. Salinity

Salinity refers to the concentration of dissolved salts in sea water. It is a key factor in determining the physical and chemical properties of ocean water, affecting its density, conductivity, and biological processes. Salinity is usually expressed in parts per thousand (ppt or ‰), indicating the number of grams of dissolved salts in one kilogram of sea water. The

average salinity of ocean water is approximately 35 ppt, meaning there are 35 grams of dissolved salts in every kilogram of sea water.

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Salinity is measured using several methods:

1. **Refractometry:** This method involves using a refractometer to measure the degree to which light is bent, or refracted, as it passes through sea water. The refractive index is directly related to salinity.
2. **Conductivity:** Conductivity meters measure the ability of sea water to conduct an electric current. Since dissolved salts increase water's conductivity, these measurements can be used to calculate salinity.
3. **Chemical Analysis:** This method involves titrating sea water samples with silver nitrate to determine the concentration of chloride ions, which are a major component of sea water salinity. This is known as the Mohr method.

Factors Affecting Salinity

Several factors influence the salinity of sea water, including:

1. **Evaporation:** In regions with high evaporation rates, such as tropical and subtropical areas, salinity increases as water evaporates and leaves the salts behind. This process is particularly pronounced in enclosed or semi-enclosed seas, like the Red Sea and the Mediterranean Sea.
2. **Precipitation:** Rainfall dilutes sea water, reducing salinity. Therefore, regions with high precipitation, such as the equatorial belt, generally have lower salinity levels.
3. **River Inflow:** Freshwater from rivers entering the ocean decreases salinity in coastal areas. The influx of river water, carrying dissolved minerals and nutrients, can significantly alter local salinity levels.
4. **Ice Formation and Melting:** In polar regions, the formation of sea ice excludes salts, increasing the salinity of the remaining water. Conversely, melting ice dilutes sea water, decreasing salinity.

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5. **Ocean Currents:** Ocean currents transport water masses with different salinities, mixing them and affecting local salinity levels. For example, the Gulf Stream carries high-salinity water from the subtropics to the North Atlantic, influencing salinity patterns.
6. **Temperature:** Temperature affects the solubility of salts and the rate of evaporation. Warmer water can hold more dissolved salts, and higher temperatures increase evaporation, both of which can raise salinity.

Salinity Distribution in Different Ocean Regions

Salinity varies significantly across different regions of the world's oceans:

1. Equatorial Regions: The equatorial belt typically has lower salinity, around 34-35 ppt, due to high rainfall and runoff from major river systems. However, areas with high evaporation rates, such as the central and eastern Pacific, may exhibit slightly higher salinity.
2. Subtropical Regions: Subtropical gyres, found between 20° and 30° latitude in both hemispheres, are characterized by high evaporation rates and low precipitation, leading to higher salinity levels, often exceeding 37 ppt. The Mediterranean Sea, with restricted inflow and high evaporation, can have salinity levels above 38 ppt.
3. Polar Regions: Polar regions generally have lower salinity, around 30-34 ppt, due to the input of freshwater from melting ice and relatively low evaporation rates. Seasonal ice formation and melting create significant salinity fluctuations.
4. Coastal Areas: Coastal salinity varies widely depending on the proximity to river mouths, the degree of freshwater influx, and the mixing with oceanic waters. Estuaries, where rivers meet the sea, often exhibit a gradient of salinity from freshwater to marine conditions.

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5. Open Ocean: The open ocean has a more stable and predictable salinity pattern, typically around the average of 35 ppt. However, variations occur due to ocean currents, upwelling, and regional climatic conditions.

C. Conductivity

Conductivity in sea water refers to its ability to conduct an electric current. This property is closely related to the concentration of dissolved salts, or salinity, in the water. The more salts present, the higher the conductivity because salts dissolve into ions, which are charged particles that facilitate the flow of electricity.

- Ionization of Salts: When salts dissolve in water, they dissociate into positive and negative ions. For example, sodium chloride (NaCl) dissociates into sodium (Na⁺) and chloride (Cl⁻) ions. These ions increase the water's ability to conduct electricity.
- Direct Relationship: There is a direct correlation between salinity and conductivity. As salinity increases, the concentration of ions increases, leading to higher conductivity. Conversely, lower salinity results in lower conductivity.
- Temperature Influence: Conductivity is also influenced by temperature; warmer water tends to have higher conductivity because increased temperature enhances ion mobility. Therefore, accurate measurements of conductivity must account for temperature variations.

Measurement Techniques

Several techniques are employed to measure the conductivity of sea water, each utilizing the relationship between salinity and conductivity to provide accurate salinity readings.

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1. Conductivity Meters:

- **Principle:** Conductivity meters work by applying an electrical current between two electrodes submerged in the water sample and measuring the current flow, which is directly proportional to the water's conductivity.
- **Compensation:** These meters often include temperature sensors to adjust the conductivity readings for temperature variations, ensuring more accurate salinity calculations.
- **Usage:** Portable handheld conductivity meters are commonly used for field measurements, while more sophisticated, stationary meters are used in laboratories for precise analysis.

2. CTD (Conductivity, Temperature, Depth) Sensors:

- **Components:** CTD sensors are comprehensive instruments that simultaneously measure conductivity, temperature, and depth, providing a detailed profile of sea water characteristics.
- **Deployment:** These sensors are typically deployed on research vessels, either lowered into the water column or attached to autonomous underwater vehicles. They record continuous data as they move through the water, offering a vertical profile of salinity, temperature, and depth.
- **Applications:** CTD sensors are essential for oceanographic research, allowing scientists to study the structure and dynamics of ocean layers and understand how physical properties vary with depth.

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3. Argo Floats:

- **Design:** Argo floats are autonomous, free-drifting devices equipped with CTD sensors. They periodically descend to various depths, collect data, and then ascend to the surface to transmit the data via satellite.
- **Global Network:** The Argo program comprises a global network of floats, providing real-time, high-resolution data on ocean salinity, temperature, and currents. This data is invaluable for climate research and ocean monitoring.

4. In-Situ Probes:

- **Functionality:** In-situ probes are specialized instruments that measure conductivity

directly in the marine environment. They are often deployed on buoys, underwater observatories, or attached to long-term monitoring stations.

- **Continuous Monitoring:** These probes offer continuous, long-term measurements of conductivity, which are crucial for observing changes in oceanographic conditions over time.

5. Laboratory Analysis:

- **Sample Collection:** In some cases, water samples are collected in the field and analyzed in a laboratory setting. Conductivity measurements in the lab can be conducted using precise analytical instruments, ensuring high accuracy.

- **Calibration and Standards:** Laboratory instruments are calibrated using standard solutions of known conductivity, ensuring that the measurements are reliable and consistent.

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D. Viscosity

Viscosity is a measure of a fluid's resistance to deformation or flow. In simpler terms, it describes how "thick" or "sticky" a fluid is. In the context of sea water, viscosity is an important physical property that influences how water moves and mixes within the ocean.

- **Dynamic Viscosity:** This refers to the internal friction within the fluid, affecting its ability to flow. It is usually measured in units of pascal-seconds (Pa·s) or poise (P).
- **Kinematic Viscosity:** This takes into account the fluid's density and is defined as the ratio of dynamic viscosity to density. It is measured in square meters per second (m²/s).

Relevance to Sea Water Dynamics:

- **Fluid Flow:** Viscosity plays a crucial role in determining the flow characteristics of sea water, affecting both large-scale ocean currents and small-scale turbulence.

- **Mixing Processes:** Viscosity influences the mixing of water masses with different properties (e.g., temperature and salinity), which is essential for nutrient distribution, heat transfer, and the overall thermohaline circulation in the ocean.

- **Boundary Layers:** Viscosity affects the formation and behavior of boundary layers near solid surfaces, such as the ocean floor and the hulls of ships. These boundary layers influence drag and flow separation.

Impact on Marine Organisms and Transport Processes

Viscosity has significant implications for marine organisms and various transport processes within the ocean.

1 Impact on Marine Organisms:

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- **Locomotion:** The viscosity of sea water affects how marine organisms move. For small organisms, such as plankton, the viscous forces dominate over inertial forces, making their movement heavily influenced by the viscosity of the water. Larger organisms, like fish and marine mammals, experience a different balance of forces but still must contend with viscous drag.

- **Feeding and Predator-Prey Interactions:** Viscosity influences the feeding mechanisms of filter feeders and the efficiency of predator-prey interactions. For example, higher viscosity can slow down the sinking rate of planktonic prey, making it easier for predators to capture them.

- **Reproduction:** Many marine organisms rely on the viscosity of sea water for the dispersal of their gametes (eggs and sperm). The transport and mixing of these reproductive cells are influenced by the viscous properties of the surrounding water.

2 Impact on Transport Processes:

- **Sediment Transport:** Viscosity affects the transport and deposition of sediments in the ocean. Fine particles, like silt and clay, are suspended and transported over long distances due to the viscosity of the water, while larger particles settle more quickly.

- **Nutrient and Gas Exchange:** The viscosity of sea water plays a role in the diffusion and transport of nutrients and gases (such as oxygen and carbon dioxide) within the ocean. Efficient mixing and transport are crucial for maintaining the health of marine ecosystems and supporting biological productivity.

3 **Pollutant Dispersion:** The dispersion and dilution of pollutants, such as oil spills or chemical discharges, are influenced by the viscosity of sea water. Understanding viscosity helps in modeling and predicting the spread of pollutants in the marine environment.

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E. Diffusion

Diffusion is the process by which molecules move from areas of higher concentration to areas of lower concentration, driven by their kinetic energy. In the context of sea water, diffusion plays a crucial role in the exchange of nutrients and gases, supporting marine life and influencing oceanic processes.

1 Mechanisms of Diffusion:

- **Molecular Diffusion:** This occurs at the molecular level and is driven by the random motion of molecules. It is a relatively slow process but is essential for the transfer of nutrients and gases at microscopic scales.

- **Turbulent Diffusion:** In the ocean, turbulent mixing caused by currents, waves, and wind-driven movements enhances the diffusion process. This type of diffusion is much

faster than molecular diffusion and plays a significant role in distributing nutrients and gases over larger spatial scales.

■ **Fick's Laws of Diffusion:** The rate of diffusion is described by Fick's first and second laws. Fick's first law states that the flux of a substance is proportional to the concentration gradient. Fick's second law describes how the concentration of a substance changes over time as it diffuses.

2 Significance in Nutrient and Gas Exchange:

- **Nutrient Transport:** Diffusion is critical for transporting essential nutrients, such as nitrate, phosphate, and silicate, from areas of high concentration (e.g., nutrient-rich deep waters) to areas of lower concentration (e.g., the photic zone where photosynthesis occurs). This process supports primary production and the growth of phytoplankton, forming the base of the marine food web.

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- **Gas Exchange:** Diffusion facilitates the exchange of gases between the ocean and the atmosphere. Oxygen diffuses from the atmosphere into the ocean, supporting aerobic respiration in marine organisms. Similarly, carbon dioxide diffuses from the atmosphere into the ocean, playing a key role in the global carbon cycle and affecting ocean acidification.

- **Biological Processes:** Diffusion is essential for cellular processes in marine organisms, including respiration, nutrient uptake, and waste removal. It ensures that cells receive the necessary nutrients and oxygen while removing metabolic waste products.

Diffusion Rates in Varying Conditions

The rate of diffusion in sea water is influenced by several factors, including temperature, salinity, and turbulence.

1. Temperature:

- **Effect on Molecular Diffusion:** Higher temperatures increase the kinetic energy of molecules, leading to faster diffusion rates. For example, in warmer waters, the diffusion of oxygen and nutrients occurs more rapidly compared to colder waters.
- **Seasonal Variations:** Seasonal temperature changes can affect diffusion rates, with higher rates in summer due to increased temperatures and lower rates in winter when temperatures decrease.

2. Salinity:

- **Impact on Diffusion:** Higher salinity increases the density and viscosity of sea water, which can slow down the rate of molecular diffusion. However, the impact is generally less significant compared to temperature.

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- **Stratification:** Salinity gradients, often found in estuaries and polar regions, can create stratified layers in the ocean. These layers can act as barriers to diffusion, affecting the vertical transport of nutrients and gases.

3. Turbulence and Mixing:

- **Enhancing Diffusion:** Turbulent mixing, driven by wind, waves, and currents, greatly enhances diffusion rates by breaking down concentration gradients and promoting the transport of molecules across larger distances. This type of diffusion is critical in the upper ocean, where physical processes create turbulence.
- **Boundary Layers:** In regions with strong currents or wave action, the boundary layers near the sea surface or ocean floor can experience enhanced diffusion rates due to increased turbulence.

4. Pressure:

Deep Ocean Conditions: In the deep ocean, high pressure can affect the diffusion rates of gases, such as oxygen and carbon dioxide. While the overall impact is less significant than temperature and turbulence, pressure can influence the solubility and diffusion of gases in deep-sea environments.

5. Biological Activity:

- **Bioturbation:** The movement and activity of marine organisms, such as burrowing and feeding, can enhance diffusion rates by disturbing sediments and promoting the exchange of nutrients and gases between the water column and the seabed.
- **Microbial Processes:** Microbial activity, particularly in sediments, can create microenvironments with varying diffusion rates. For example, microbial mats can form diffusion barriers, affecting the exchange of nutrients and gases.

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CHEMICAL PROPERTIES OF SEA WATER

A. Major Constituents

Overview of Dissolved Salts and Minerals

Sea water is a complex solution of dissolved salts, minerals, and organic materials. The major constituents, which make up over 99% of the dissolved solids in sea water, include:

1. **Sodium (Na^+) and Chloride (Cl^-):** These ions are the most abundant, accounting for approximately 85% of the total dissolved salts. Sodium and chloride ions combine to form sodium chloride (table salt), which is the primary salt in sea water.
2. **Magnesium (Mg^{2+}):** Magnesium is the third most abundant ion in sea water. It plays a crucial role in marine biological processes, including the formation of chlorophyll in

plants and algae.

3. Calcium (Ca^{2+}): Calcium ions are essential for the formation of shells and skeletons of many marine organisms, such as corals and mollusks.

4. Potassium (K^+): Potassium is vital for cellular functions in marine organisms, including nerve transmission and muscle contraction.

5. Sulfate (SO_4^{2-}): Sulfate ions are important for the synthesis of organic compounds and the energy metabolism of certain bacteria.

6. Bicarbonate (HCO_3^-): Bicarbonate plays a key role in maintaining the pH balance of sea water and is involved in the carbon cycle, buffering changes in acidity.

The concentration of these constituents can vary slightly depending on location, depth, and environmental conditions, but their relative proportions remain remarkably constant, a principle known as the rule of constant proportions.

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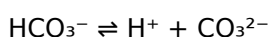
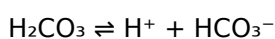
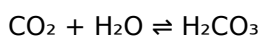
Chemical Equilibrium and Reactions

Chemical equilibrium in sea water involves a balance between the various chemical reactions that occur among dissolved salts, gases, and organic materials. Key concepts include:

1. Carbonate System: The carbonate system is crucial for regulating the pH of sea water.

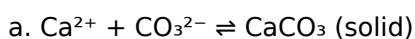
It involves a series of equilibrium reactions among carbon dioxide (CO_2), carbonic acid (H_2CO_3), bicarbonate (HCO_3^-), and carbonate (CO_3^{2-}). This system acts as a buffer, helping to maintain stable pH levels despite variations in CO_2 concentrations.

Reactions:



2. Precipitation and Dissolution: The precipitation and dissolution of minerals, such as calcium carbonate (CaCO_3), influence the availability of ions in sea water. These processes are critical for the formation and maintenance of coral reefs and shellfish structures.

Reactions:



3. Redox Reactions: Redox (reduction-oxidation) reactions involve the transfer of electrons between molecules and are essential for many biochemical processes in marine environments. For example, the oxidation of organic matter by bacteria affects the availability of nutrients and the cycling of elements like nitrogen and sulfur.

Reactions:

Organic matter + O₂ → CO₂ + H₂O (oxidation)

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SO₄²⁻ + organic matter → H₂S + CO₂ (sulfate reduction)

Understanding these chemical equilibria and reactions is crucial for predicting changes in sea water chemistry due to natural processes and human activities, such as pollution and climate change.

B. Trace Elements and Nutrients

Role in Marine Ecosystems

Trace elements and nutrients, though present in much smaller concentrations compared to the major constituents, are vital for the health and functioning of marine ecosystems.

1. Trace Elements: Trace elements, such as iron (Fe), manganese (Mn), zinc (Zn), and copper (Cu), are essential for various biochemical processes in marine organisms. They act as cofactors for enzymes, play roles in photosynthesis, respiration, and nitrogen fixation, and contribute to the structural integrity of proteins and other cellular components.

2. Nutrients: Nutrients, including nitrogen (N), phosphorus (P), and silicon (Si), are key elements required for the growth and reproduction of marine plants and algae. They are fundamental to primary production, which forms the base of the marine food web.

- Nitrogen: Present primarily as nitrate (NO₃⁻), nitrite (NO₂⁻), and ammonium (NH₄⁺).

It is essential for the synthesis of amino acids, proteins, and nucleic acids.

- Phosphorus: Found mainly as phosphate (PO₄³⁻). It is crucial for energy transfer (ATP) and the formation of DNA and RNA.

- Silicon: Utilized by diatoms and some sponges for constructing their silica-based skeletons.

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Sources and Cycles of Trace Elements

Trace elements and nutrients enter the ocean through various pathways and undergo complex biogeochemical cycling:

1. Sources:

- Rivers and Runoff: Rivers transport nutrients and trace elements from terrestrial environments to the ocean. Agricultural runoff, industrial discharges, and natural weathering of rocks contribute to the input of these elements.
- Atmospheric Deposition: Wind-borne dust and aerosols can deposit trace elements and nutrients directly onto the ocean surface. For example, iron-rich dust from deserts can enhance primary productivity in iron-limited regions of the ocean.

- **Hydrothermal Vents:** These undersea vents release trace elements, such as iron, manganese, and zinc, from the Earth's crust into the ocean. The unique chemistry of hydrothermal fluids supports specialized ecosystems.

2. Cycles:

- **Biological Uptake and Recycling:** Marine organisms assimilate trace elements and nutrients from the water for growth and reproduction. When these organisms die, their decomposition releases the elements back into the water, making them available for other organisms.
- **Sedimentation and Remineralization:** Some trace elements and nutrients settle to the ocean floor as part of particulate matter. Benthic organisms and microbial activity can recycle these elements from sediments back into the water column.

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- **Geochemical Cycling:** Trace elements undergo various geochemical processes, such as adsorption onto particles, precipitation, and complexation with organic and inorganic ligands. These processes affect their availability and distribution in the ocean.

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BIOLOGICAL ACTIVITIES AND DISTRIBUTION

A. Plankton

Classification: Phytoplankton and Zooplankton

Plankton are small organisms that drift with ocean currents and form the base of the marine food web. They can be classified into two main groups: phytoplankton and zooplankton.

- Phytoplankton:

- **Characteristics:** Phytoplankton are photosynthetic microorganisms, including diatoms, dinoflagellates, and cyanobacteria. They use sunlight to produce energy through photosynthesis, releasing oxygen and forming the primary source of organic matter in the ocean.
- **Role in Ecosystems:** As primary producers, phytoplankton are crucial for the marine food web, providing energy and nutrients to higher trophic levels, including zooplankton, fish, and marine mammals.

- Zooplankton:

- **Characteristics:** Zooplankton are heterotrophic organisms, meaning they consume other organisms for energy. This group includes a diverse range of organisms such as copepods, krill, jellyfish, and larval stages of larger animals.
- **Role in Ecosystems:** Zooplankton feed on phytoplankton and other small particles, transferring energy up the food web. They are also prey for many larger marine species,

playing a key role in nutrient cycling and energy transfer.

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Distribution Patterns and Ecological Roles

The distribution of plankton is influenced by various environmental factors, including light, temperature, nutrients, and water currents.

- Distribution Patterns:

- **Phytoplankton:** Typically found in the euphotic zone, where sunlight penetrates the water and supports photosynthesis. Their distribution is highly variable, often forming blooms in nutrient-rich areas such as upwelling zones and coastal regions.
- **Zooplankton:** Distributed throughout the water column, with some species undergoing diel vertical migration, moving to deeper waters during the day to avoid predators and returning to the surface at night to feed.

- Ecological Roles:

- **Phytoplankton:** Besides being primary producers, phytoplankton influence global biogeochemical cycles, particularly the carbon cycle. They absorb carbon dioxide during photosynthesis, and their sinking particles contribute to the ocean's biological pump, sequestering carbon in deep waters.
- **Zooplankton:** Zooplankton help regulate phytoplankton populations through grazing. They also contribute to the vertical flux of organic matter by producing fecal pellets and other detritus that sink, supporting deep-sea ecosystems.

B. Marine Fauna

Adaptations to Brackish Water Conditions

Brackish water environments, such as estuaries and coastal lagoons, exhibit varying salinity levels, requiring specialized adaptations from marine fauna.

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- Physiological Adaptations:

- **Osmoregulation:** Many brackish water species have developed mechanisms to regulate their internal salt concentration, such as efficient ion transport systems in their gills or specialized excretory organs.
- **Behavioural Adaptations:** Some species may migrate between different salinity zones to optimize their physiological processes, such as spawning in lower salinity areas and feeding in higher salinity regions.

- Ecological Roles:

- **Nursery Habitats:** Brackish waters often serve as crucial nursery grounds for juvenile stages of many marine species, providing a relatively safe and nutrient-rich environment

for growth.

- **Biodiversity Hotspots:** These areas support high biodiversity, including a mix of marine, freshwater, and estuarine species, contributing to the overall productivity and resilience of coastal ecosystems.

Diversity and Distribution in Various Marine Environments

Marine fauna exhibit remarkable diversity and are distributed across different marine habitats, from shallow coastal waters to the deep sea.

- Coastal Zones:

- **Characteristics:** Coastal zones are rich in nutrients due to runoff and upwelling, supporting diverse communities of fish, invertebrates, and marine mammals.
- **Habitats:** Include coral reefs, mangroves, seagrass beds, and rocky shores, each hosting unique assemblages of species adapted to specific environmental conditions.

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- Open Ocean:

- **Characteristics:** The open ocean, or pelagic zone, is characterized by its vast expanse and varying depths. Nutrient availability is often lower compared to coastal areas.
- **Habitats:** Includes the epipelagic zone (surface to 200 meters), where sunlight supports phytoplankton growth, and deeper zones like the mesopelagic and bathypelagic, inhabited by species adapted to low light and high pressure.

- Deep Sea:

- **Characteristics:** The deep sea is one of the most extreme environments, with high pressure, low temperatures, and complete darkness. It includes habitats such as abyssal plains, hydrothermal vents, and deep-sea trenches.
- **Adaptations:** Fauna in these areas have unique adaptations, such as bioluminescence, pressure-resistant body structures, and specialized feeding strategies to survive in the nutrient-sparse environment.

C. Physiological Adaptations

Mechanisms of Adaptation to Temperature, Salinity, and Pressure

Marine organisms have evolved various physiological adaptations to cope with the diverse and often extreme conditions of the marine environment.

- Temperature Adaptations:

- **Ectothermy:** Many marine organisms are ectothermic, meaning their body temperature fluctuates with the surrounding water. They may exhibit behavioral adaptations, such as seeking warmer or cooler waters, to regulate their body temperature.

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- **Heat Shock Proteins:** Some species produce heat shock proteins to protect their cells from temperature-induced damage during sudden changes in temperature.
- **Salinity Adaptations:**
 - **Osmoregulation:** As previously mentioned, osmoregulation is crucial for maintaining internal ionic balance. Marine fish, for example, drink seawater and excrete excess salts through specialized cells in their gills.
 - **Euryhaline Species:** These species can tolerate a wide range of salinities, allowing them to inhabit both marine and brackish waters.
- **Pressure Adaptations:**
 - **Structural Reinforcements:** Deep-sea organisms often have reinforced cell membranes and proteins that function optimally under high pressure.
 - **Buoyancy Control:** Many deep-sea species have adaptations to control buoyancy, such as oil-filled bladders or gas-filled sacs, enabling them to maintain their position in the water column.

Case Studies of Specific Marine Organisms

Examining specific marine organisms provides insight into the diverse strategies employed to survive and thrive in various marine environments.

1. Emperor Penguin (*Aptenodytes forsteri*):

- **Adaptations:** Emperor penguins have adaptations for extreme cold, including a thick layer of blubber, dense feathers for insulation, and the ability to slow their metabolism during long dives.
- **Habitat:** They breed on the Antarctic ice and forage in the frigid Southern Ocean.

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2. Deep-Sea Anglerfish (Family Lophiiformes):

- **Adaptations:** These fish possess bioluminescent lures to attract prey in the dark depths of the ocean. They also have a specialized reproductive strategy where tiny males attach to the much larger females, providing a constant source of sperm.
- **Habitat:** Found at depths of over 1,000 meters, in regions with high pressure and low light.

3. Mangrove Tree Crab (*Aratus pisonii*):

- **Adaptations:** This crab has adapted to brackish water conditions in mangrove ecosystems. It can osmoregulate effectively and has behavioral adaptations to avoid predators and forage on both plant and animal matter.
- **Habitat:** Inhabits mangrove forests along the Atlantic and Pacific coasts of the Americas.

4. Albatross (Family Diomedidae):

- Adaptations: Albatrosses have long wingspans that allow them to glide over vast ocean distances with minimal energy expenditure. They also have specialized salt glands to excrete excess salt from ingested seawater.
- Habitat: These birds are found in open ocean environments, particularly in the Southern Hemisphere, often traveling thousands of kilometers for foraging.

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PHYSICAL DYNAMICS OF THE SEA

A. Currents

Types and Causes of Ocean Currents

Ocean currents are continuous, directed movements of sea water generated by various forces acting upon the water, including wind, the Earth's rotation, and differences in water density.

- Surface Currents:

- Wind-Driven Currents: These currents are primarily driven by wind patterns and are influenced by the Coriolis effect. Examples include the Gulf Stream, the Kuroshio Current, and the Antarctic Circumpolar Current.
- Gyres: Large system of circular ocean currents formed by global wind patterns and forces created by the Earth's rotation. Major ocean gyres include the North Atlantic Gyre, the South Atlantic Gyre, the North Pacific Gyre, the South Pacific Gyre, and the Indian Ocean Gyre.

- Deep Ocean Currents:

- Thermohaline Circulation: Also known as the "global conveyor belt," this system of deep ocean currents is driven by differences in water density, which is affected by temperature (thermo) and salinity (haline). Cold, salty water is denser and sinks, while warmer, less salty water rises, creating a global circulation pattern that helps regulate climate by transporting heat.

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Impact on Climate and Marine Navigation

Ocean currents play a crucial role in shaping global climate patterns and marine navigation.

- Climate Impact:

- Heat Distribution: Ocean currents transport warm water from the equator towards the poles and cold water from the poles towards the equator, helping to regulate global climate and weather patterns. For instance, the Gulf Stream warms the climate of Western Europe.

- **Climate Events:** Changes in ocean currents can lead to significant climate events. El Niño and La Niña are examples of such events, driven by changes in the Pacific Ocean currents, affecting global weather patterns.
- **Marine Navigation:**
 - **Navigation Routes:** Understanding ocean currents is essential for marine navigation. Ships can take advantage of favorable currents to reduce travel time and fuel consumption. Historical explorers, such as the Polynesians and European navigators, relied on knowledge of currents for their voyages.
 - **Safety and Efficiency:** Modern shipping routes are planned considering ocean currents to enhance safety and efficiency. Knowledge of currents also aids in search and rescue operations and the management of oil spills and marine debris.

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B. Tides

Causes and Effects of Tidal Movements

Tides are the periodic rise and fall of sea levels caused by the gravitational forces exerted by the Moon and the Sun, along with the rotation of the Earth.

- Causes:

- **Gravitational Pull:** The gravitational pull of the Moon has the most significant effect on tides, creating two high tides and two low tides each day as the Earth rotates. The Sun also contributes to tides but to a lesser extent.
- **Tidal Bulges:** The gravitational forces create bulges in the Earth's oceans—one facing the Moon and the other on the opposite side of the Earth. These bulges result in high tides, while areas between the bulges experience low tides.

- Effects:

- **Coastal Flooding:** High tides can lead to coastal flooding, especially during storm surges or when combined with strong onshore winds.
- **Intertidal Zones:** Tidal movements create intertidal zones, areas that are exposed at low tide and submerged at high tide. These zones are rich in biodiversity and serve as critical habitats for various marine species.

Tidal Patterns and Their Influence on Marine Life

Tidal patterns vary based on geographic location and local topography, influencing marine life and ecosystems.

- Tidal Patterns:

- **Diurnal Tides:** One high tide and one low tide each day.

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- **Semi-Diurnal Tides:** Two high tides and two low tides of approximately equal height each day.
- **Mixed Tides:** Two high tides and two low tides of different heights each day.
- **Influence on Marine Life:**
 - **Feeding and Reproduction:** Many marine organisms, such as crabs, fish, and birds, time their feeding and reproductive activities with the tides. For example, grunion fish spawn on sandy beaches during high tides.
 - **Habitat Formation:** Tidal movements shape coastal habitats, such as mudflats, salt marshes, and mangroves, providing essential breeding and feeding grounds for various species.
 - **Nutrient Transport:** Tides help transport nutrients and oxygen into estuaries and coastal ecosystems, supporting high productivity and biodiversity.

C. Waves

Formation and Types of Waves

Waves are generated by the transfer of energy from wind to the water surface. The characteristics of waves depend on the wind speed, duration, and the distance over which the wind blows (fetch).

- Formation:

- **Wind-Generated Waves:** Most common type, created by the friction between the wind and the water surface. As wind blows across the surface, it creates ripples that grow into larger waves.

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- **Seismic Waves (Tsunamis):** Generated by underwater earthquakes, volcanic eruptions, or landslides. These waves can travel across entire ocean basins and cause significant destruction when they reach coastal areas.

- Types of Waves:

- **Capillary Waves:** Small, rippling waves created by light winds.
- **Swell Waves:** Long, rolling waves that have travelled out of their area of origin. They have longer wavelengths and can travel great distances.
- **Breaking Waves:** Occur when waves move into shallow water and become too steep to support themselves, causing them to collapse or break.

Effects on Coastal Areas and Marine Habitats

Waves have a profound impact on coastal areas and marine habitats, shaping the landscape and influencing ecological processes.

- Coastal Erosion and Deposition:

- **Erosion:** Waves can erode coastlines, removing sediments and reshaping shorelines.

This process can lead to the loss of habitats and property.

- **Deposition:** Waves also transport and deposit sediments, forming features like beaches, sandbars, and dunes. These features provide habitats and protection against storm surges.

- Marine Habitats:

- **Rocky Shores:** Exposed to constant wave action, rocky shores host specialized organisms adapted to withstand strong forces and changing conditions.

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- **Coral Reefs:** Coral reefs benefit from wave action, which helps bring in nutrients and oxygen while removing waste products. However, excessive wave energy, particularly from storms, can damage reef structures.

- **Seagrass Beds and Kelp Forests:** These habitats are found in areas with moderate wave action, which provides necessary water movement without causing physical damage.

D. Sound and Radiation

Underwater Acoustics and Its Applications

Sound travels much faster and farther in water than in air, making it a crucial tool for marine communication, navigation, and exploration.

- Underwater Acoustics:

- **Speed of Sound:** In sea water, the speed of sound is approximately 1,500 meters per second, depending on temperature, salinity, and pressure.

- **Echo Sounding:** Used for mapping the seafloor and detecting underwater objects.

Ships send out sound pulses and measure the time it takes for the echoes to return, providing detailed bathymetric maps.

- **Sonar:** Employed for navigation, communication, and detecting submarines and other underwater hazards. Both active sonar (sending out sound waves and listening for echoes) and passive sonar (listening for natural sounds) are used.

- Applications:

- **Marine Research:** Underwater acoustics are vital for studying marine mammals, fish populations, and underwater geological formations.

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- **Submarine Communication:** Sound waves are used for communication between submarines and other underwater devices.

- **Environmental Monitoring:** Acoustic sensors monitor ocean conditions, including

temperature and salinity profiles, which are crucial for understanding climate change impacts.

Light Penetration and Its Influence on Marine Ecosystems

Light penetration in the ocean is critical for photosynthesis and affects the distribution and behavior of marine organisms.

- Light Penetration:

- **Absorption and Scattering:** Water absorbs and scatters light, with shorter wavelengths (blue and green) penetrating deeper than longer wavelengths (red and orange). This selective absorption creates the blue color of the deep ocean.
- **Euphotic Zone:** The upper layer of the ocean where sunlight is sufficient for photosynthesis. Typically extends to a depth of about 100-200 meters, depending on water clarity.

- Influence on Marine Ecosystems:

- **Photosynthesis:** Light penetration supports the growth of phytoplankton, which are the primary producers in marine ecosystems. The availability of light limits primary production to the euphotic zone.
- **Behavioural Patterns:** Many marine organisms exhibit vertical migration, moving up to the surface to feed at night and descending to deeper waters during the day to avoid predators. This behaviour is driven by changes in light levels.

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- **Habitat Zonation:** Light availability influences the vertical distribution of marine habitats. Coral reefs and seagrass beds thrive in shallow, well-lit waters, while deeper areas host organisms adapted to low-light conditions, such as bioluminescent species.

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SEA WATER BEHAVIOR AND DYNAMICS

A. Sea Water Circulation

Global Circulation Patterns

Global ocean circulation, also known as thermohaline circulation or the "global conveyor belt," is a complex system that drives the movement of sea water across the world's oceans. This circulation is crucial for distributing heat, nutrients, and gases around the globe.

- Thermohaline Circulation:

- **Definition:** A large-scale circulation driven by differences in water density, which are controlled by temperature (thermo) and salinity (haline). Cold, salty water is denser and sinks, while warm, less salty water rises, creating a continuous global flow.
- **Pathway:** The circulation begins in the North Atlantic, where cold, dense water sinks and

flows southward, joining the Antarctic Circumpolar Current. It then moves into the Indian and Pacific Oceans, eventually rising to the surface and returning to the Atlantic via surface currents.

- Wind-Driven Circulation:

- Surface Currents: Driven by wind patterns and the Coriolis effect, creating gyres in the major ocean basins. These currents include the Gulf Stream in the North Atlantic, the Kuroshio Current in the North Pacific, and the East Australian Current in the South Pacific.
- Equatorial Currents: Flow westward along the equator due to trade winds, with the Equatorial Counter current flowing eastward between the north and south equatorial currents.

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Interaction Between Surface and Deep Water Currents

The interaction between surface and deep water currents is vital for maintaining the global thermohaline circulation and influencing regional climates.

- Upwelling and Downwelling:

- Upwelling: Occurs when wind-driven surface water is replaced by deeper, nutrient-rich water. Common along western continental margins, upwelling zones are highly productive and support large fisheries.
- Downwelling: Occurs when surface water converges and sinks, bringing oxygen and other surface properties to deeper layers. This process helps ventilate the deep ocean.

- Mixing Processes:

- Vertical Mixing: Enhances the exchange of heat, nutrients, and gases between surface and deep waters. Turbulence, caused by wind, waves, and currents, plays a significant role in vertical mixing.
- Horizontal Mixing: Driven by large-scale currents and eddies, it helps distribute substances laterally across ocean basins. Mesoscale eddies, which are circular currents spanning tens to hundreds of kilometers, contribute to horizontal mixing.

B. Diffusion and Mixing

Processes Influencing the Distribution of Substances

Diffusion and mixing are fundamental processes that determine the distribution of substances, such as nutrients, gases, and pollutants, in sea water.

- Molecular Diffusion: The movement of molecules from areas of high concentration to areas of low concentration due to random molecular motion. Although slow in sea water,

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molecular diffusion is crucial for the fine-scale distribution of substances. For Examples is Dissolved oxygen and carbon dioxide diffusing into and out of water masses, nutrient ions spreading from regions of high concentration (e.g., decaying organic matter) to surrounding waters.

- **Turbulent Diffusion:** The mixing of water masses due to turbulence, which significantly enhances the rate of diffusion compared to molecular diffusion alone. Turbulent diffusion is caused by wind, waves, and currents. For examples is the mixing of fresh river water with saline sea water in estuaries, the dispersion of pollutants in coastal areas, and the spread of heat and nutrients in the open ocean.

Role in Nutrient Cycling and Marine Ecosystems

Diffusion and mixing are critical for nutrient cycling and supporting marine ecosystems by facilitating the distribution of essential elements.

- Nutrient Cycling:

- **Upwelling:** Brings nutrient-rich deep water to the surface, fueling primary production by phytoplankton. This process supports high biological productivity and the growth of fisheries.

- **Nutrient Redistribution:** Diffusion and mixing spread nutrients horizontally and vertically, ensuring their availability throughout the water column. This redistribution supports diverse marine life at different depths and locations.

- Marine Ecosystems:

- **Primary Production:** Phytoplankton, the primary producers in marine ecosystems, rely on the availability of nutrients, which are supplied through diffusion and mixing

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processes. Enhanced nutrient availability leads to phytoplankton blooms, which form the base of the marine food web.

- **Habitat Connectivity:** Mixing processes create interconnected habitats by transporting larvae and spores, allowing species to colonize new areas and maintain genetic diversity. This connectivity is crucial for the resilience and stability of marine ecosystems.

- **Oxygen Distribution:** Mixing ensures the distribution of dissolved oxygen throughout the water column, which is vital for the survival of aerobic organisms. Poor mixing can lead to hypoxic conditions, negatively impacting marine life.

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BRACKISH WATER CONDITIONS

A. Characteristics of Brackish Water

Definition and Comparison with Marine and Freshwater

Brackish water is a type of water with salinity levels between that of freshwater and seawater. It typically occurs in areas where freshwater from rivers or streams mixes with seawater from oceans or seas.

Brackish Water: Has a salinity range of approximately 0.5 to 30 parts per thousand (ppt), which is higher than freshwater but lower than seawater. The exact salinity can vary based on local conditions and the sources of freshwater and seawater.

Comparison with Marine and Freshwater:

- Marine Water: Typically has a salinity of around 35 ppt, which is relatively constant and supports a wide range of marine organisms adapted to high salinity levels.
- Freshwater: Has a salinity of less than 0.5 ppt. Freshwater environments include rivers, lakes, and ponds, and they support organisms adapted to very low salinity conditions.

Brackish Water Characteristics:

- Salinity Variation: Brackish water environments often experience fluctuating salinity levels due to changes in freshwater inflow, tides, and evaporation rates. This variability can affect the composition and functioning of these ecosystems.
- Water Chemistry: In addition to salinity, brackish water may have unique chemical characteristics, such as varying levels of nutrients and organic matter, influenced by the mixing of freshwater and seawater.

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Examples of Brackish Environments

Brackish water environments are commonly found in transitional zones between freshwater and marine systems. Examples include:

- Estuaries: Where rivers meet the sea, creating a mixing zone of freshwater and seawater. Estuaries are often highly productive and support diverse ecosystems. Example is the Chesapeake Bay in the United States and the Seine estuary in France.
- Salt Marshes: Coastal wetlands characterized by the presence of salt-tolerant plants and frequent tidal flooding, leading to brackish conditions. Example is the Everglades in Florida and the Wadden Sea in the Netherlands.
- Mangrove Swamps: Coastal areas with dense mangrove trees adapted to brackish water and tidal conditions. These areas provide critical habitats for various marine and terrestrial species. Example is the Sundarbans in India and Bangladesh and the Everglades in Florida.
- Coastal Lagoons: Shallow, brackish water bodies separated from the open sea by sandbars, barrier islands, or coral reefs. They often experience variable salinity due to tidal exchange and freshwater input. Example is the Laguna Madre in Texas and the

Lagoons of New Caledonia.

B. Fauna of Brackish Water

Adaptations of Organisms to Fluctuating Salinity

Organisms living in brackish water environments must be able to tolerate and adapt to fluctuating salinity levels. These adaptations are crucial for their survival and reproductive success.

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- **Osmoregulation:** Brackish water fauna have developed specialized mechanisms to maintain internal osmotic balance despite changes in external salinity. This can include:

- **Ion Transport:** Adjusting the uptake and excretion of ions through specialized cells or organs, such as gills or kidneys.

- **Behavioral Adaptations:** Moving to areas with more stable salinity levels or altering activity patterns to avoid extreme conditions.

- **Salinity Tolerance:** Some species are euryhaline, meaning they can tolerate a wide range of salinities. Others are stenohaline, adapted to a narrow range of salinities.

Euryhaline species are often more common in brackish environments due to their adaptability.

- **Example:** Euryhaline fish like the common mullet (*Mugil cephalus*) can thrive in both brackish and marine waters.

Case Studies of Brackish Water Species

1. American Eel (*Anguilla rostrata*):

- **Adaptations:** The American eel is a catadromous fish, meaning it spawns in the ocean but spends most of its life in freshwater and brackish environments. It has specialized osmoregulatory mechanisms to cope with varying salinities.

- **Habitat:** Found in estuaries and coastal rivers along the eastern coast of North America.

2. Common Snook (*Centropomus undecimalis*):

- **Adaptations:** This fish can tolerate a wide range of salinities, from freshwater to full seawater. It has adapted to brackish environments by adjusting its osmoregulatory processes based on salinity.

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- **Habitat:** Commonly found in estuaries, mangroves, and coastal lagoons in the southeastern United States and the Caribbean.

3. Salt Marsh Cordgrass (*Spartina alterniflora*):

- **Adaptations:** This plant is adapted to high salinity levels found in salt marshes. It has specialized structures, such as salt glands, to excrete excess salt and maintain water

balance.

- Habitat: Dominates salt marshes along the Atlantic and Gulf coasts of the United States.

4. Mangrove Tree Crab (*Aratus pisonii*):

- Adaptations: This crab is well-adapted to the brackish and fluctuating salinity conditions of mangrove swamps. It can regulate its internal salt concentration and has behavioral adaptations to avoid extreme salinity.

- Habitat: Found in mangrove forests along the Atlantic and Pacific coasts of the Americas.

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INTERRELATIONSHIPS IN MARINE ECOSYSTEMS

A. Trophic Interactions

Food Webs and Energy Flow

Marine ecosystems are characterized by complex trophic interactions that form food webs, illustrating how energy and nutrients flow through different levels of the ecosystem.

- Food Webs:

- Primary Producers: At the base of the food web, primary producers such as phytoplankton, seaweeds, and seagrasses convert solar energy into chemical energy through photosynthesis. These organisms form the foundation of marine food webs.
- Primary Consumers: Herbivores, including zooplankton, small fish, and grazing invertebrates, feed on primary producers. They transfer energy up the food chain.
- Secondary Consumers: Predators that consume herbivores, such as larger fish, squid, and some marine mammals, are secondary consumers. They are crucial for controlling prey populations and maintaining ecosystem balance.
- Tertiary Consumers: Top predators, such as sharks, large marine mammals (e.g., whales), and apex predators, feed on secondary consumers. They play a key role in regulating the abundance and diversity of species at lower trophic levels.

- Energy Flow:

- Energy Transfer Efficiency: Energy transfer between trophic levels is relatively inefficient, with only about 10% of the energy passed from one level to the next. This inefficiency limits the number of trophic levels and the biomass of top predators.

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- Biomass Pyramids: Marine food webs can be represented as biomass pyramids, showing the decreasing biomass of organisms at each successive trophic level. This pyramid illustrates how energy decreases as it moves through the food web.

Impact of Environmental Changes on Marine Trophic Structures

Environmental changes can have profound effects on marine trophic structures, influencing the dynamics of food webs and the health of ecosystems.

- Climate Change:

- **Temperature Rise:** Increasing ocean temperatures can affect the metabolism, distribution, and reproductive patterns of marine species. Warmer temperatures may lead to shifts in species distributions, affecting predator-prey relationships.
- **Ocean Acidification:** The absorption of excess CO₂ by the ocean leads to acidification, which can impact calcifying organisms like corals, shellfish, and certain plankton species. This affects the entire food web, as these organisms are crucial for the structure and functioning of marine ecosystems.

- Pollution:

- **Nutrient Enrichment:** Excess nutrients from agricultural runoff can lead to eutrophication, causing algal blooms that deplete oxygen and create dead zones. These blooms can disrupt food webs by reducing the availability of oxygen and altering the abundance of primary producers.
- **Chemical Contaminants:** Pollutants such as heavy metals, pesticides, and plastics can accumulate in marine organisms, leading to health issues and disruptions in food chains. For example, bio-magnification can cause toxic substances to concentrate in top predators.

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- Overfishing:

- **Species Depletion:** Overfishing reduces the abundance of targeted species and can lead to the collapse of fisheries. This disruption affects the entire food web, altering predator-prey relationships and potentially causing cascading effects throughout the ecosystem.
- **Bycatch:** Non-target species caught unintentionally in fishing gear can lead to declines in populations of key marine species, further disrupting ecological balance.

B. Human Impact

Effects of Pollution, Climate Change, and Overfishing

Human activities have significant impacts on marine ecosystems, affecting their health, productivity, and resilience.

- Pollution:

- **Marine Debris:** Plastics and other debris in the ocean can entangle marine animals, cause ingestion issues, and damage habitats. For instance, sea turtles often mistake plastic bags for jellyfish, leading to internal injuries or death.

- **Chemical Pollution:** Persistent organic pollutants (POPs) and heavy metals can accumulate in marine organisms, leading to health problems and impacting reproductive success. Examples include mercury contamination in fish and PCBs in marine mammals.

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- **Climate Change:**

- **Sea Level Rise:** Melting ice caps and thermal expansion contribute to rising sea levels, which can lead to the loss of coastal habitats such as mangroves, salt marshes, and coral reefs.
- **Coral Bleaching:** Increased sea temperatures cause coral bleaching, where corals expel their symbiotic algae, leading to loss of colour and, if prolonged, coral death. Coral reefs provide essential habitats for many marine species, and their decline affects entire ecosystems.

- **Overfishing:**

- **Population Declines:** Overfishing depletes fish stocks, reduces biodiversity, and disrupts food webs. This can lead to the collapse of important fisheries and loss of economic and nutritional resources for human communities.
- **Habitat Destruction:** Destructive fishing practices, such as bottom trawling, damage seafloor habitats, including coral reefs and seagrass beds, which are critical for many marine species.

Conservation Strategies and Management Practices

Effective conservation and management practices are essential for mitigating human impacts and ensuring the sustainability of marine ecosystems.

- **Marine Protected Areas (MPAs):**

- **Definition:** Designated regions where human activities are regulated or restricted to protect marine habitats and species. MPAs can enhance biodiversity, protect critical habitats, and support fish stock recovery.

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- **Examples:** The Great Barrier Reef Marine Park in Australia and the Papahānaumokuākea Marine National Monument in the U.S. are examples of MPAs that contribute to marine conservation.

- **Sustainable Fisheries Management:**

- **Quotas and Regulations:** Setting catch limits, implementing fishing quotas, and enforcing regulations to prevent overfishing and ensure the sustainability of fish stocks.
- **Bycatch Reduction:** Developing and adopting fishing gear and practices that minimize

bycatch and reduce the impact on non-target species.

- Pollution Control:

- Legislation and Policies: Enforcing regulations to reduce pollutants and manage waste disposal, such as the Marine Pollution Convention (MARPOL) and local policies on plastic use and disposal.

- Cleanup Initiatives: Supporting efforts to clean up marine debris and restore polluted areas, including beach cleanups and the removal of derelict fishing gear.

- Climate Change Mitigation:

- Reduction of Greenhouse Gases: Implementing strategies to reduce greenhouse gas emissions, such as transitioning to renewable energy sources and improving energy efficiency.

- Adaptation Strategies: Developing and implementing measures to help marine ecosystems adapt to changing conditions, including the restoration of habitats and the protection of climate-resilient species.

- Public Awareness and Education:

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- Engagement: Raising awareness about the importance of marine conservation and encouraging responsible behavior through education and outreach programs.

- Community Involvement: Engaging local communities in conservation efforts, such as citizen science projects and coastal management initiatives.

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CONCLUSION

As we look towards the future, several key areas of research are likely to shape our understanding and management of marine ecosystems:

1. Climate Change and Ocean Acidification: Continued research into the effects of rising temperatures and ocean acidification on marine species and ecosystems is crucial. This includes understanding how these factors affect coral reefs, fisheries, and marine food webs.
2. Marine Conservation and Restoration: Advancements in conservation techniques, such as habitat restoration and the creation of marine protected areas, are needed to address biodiversity loss and ecosystem degradation.
3. Impact of Emerging Pollutants: Research into the effects of emerging pollutants, such as microplastics and pharmaceutical residues, will help us understand their impacts on marine organisms and ecosystems.
4. Sustainable Fisheries Management: Developing and implementing sustainable fishing

practices and technologies will be vital for maintaining fish stocks and marine biodiversity.

5. Marine Spatial Planning: Enhanced marine spatial planning approaches are needed to balance human activities with conservation goals, ensuring the sustainable use of marine resources.

6. Technological Innovations: Advancements in marine technology, such as remote sensing, autonomous vehicles, and deep-sea exploration tools, will provide new insights into marine processes and habitats.

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Paper 7 of 7 — EMT 227

Level 200 • Second Semester • Handout

Limnology & Oceanography

INTRODUCTION

Hydrobiology is the science of life and life processes in water.

Much of modern hydrobiology can be viewed as a sub-discipline of ecology but the sphere of hydrobiology includes taxonomy, economic biology, industrial biology, morphology, physiology etc. The one distinguishing aspect is that all relate to aquatic organisms. Much work is closely related to limnology and oceanography and can be divided into lotic system ecology (flowing waters) and lentic system ecology (still waters).

Hydrobiology can also be defined as the study of water populations, their interrelations with habitat and significance for the transformation of energy and matter, and the biological productivity of the ocean, seas, and inland waters.

Hydrobiology is primarily an ecological science. The living conditions in water are determined by the physico-geographic characteristics of the body of water. Many of these features—for example, the chemical composition of the water (in particular, the composition and amount of biogenic elements and dissolved gases, the nature of bottom sediments, and the transparency of the water)—are strongly influenced by aquatic organisms and are often determined by their life processes. Therefore, to the extent to which hydrobiology studies the role of living phenomena in the context of the aggregate of interdependent processes of the aquatic medium, it has problems in common with the combined disciplines of limnology and oceanography. Such matters as the biological structure of the ocean, the bio-limnological and bio-oceanological typology of bodies of water and water masses, and the regular patterns of the recycling of substances and the flow of energy are treated at this level of research.

Hydrobiology is greatly concerned with:

- ❖ Establishing a scientific basis for the rational exploitation of biological resources of the waters. This is bound up in many ways with the requirements of the marine and freshwater fishing industries, farm pond fishery, and the use of aquatic invertebrates and mammals (food-fish hydrobiology).
- ❖ The use of the continental surface fresh water for drinking purposes
- ❖ Industrial supply of water.
- ❖ The protection of natural water against pollution
- ❖ The self-purification of polluted water.
- ❖ The role of aquatic organisms as agents of self-purification
- ❖ The biological methods of treating waste waters.
- ❖ Presence of certain indicator organisms (biological analysis of water quality) through hydro-biological methods which is used to evaluate the extent of water pollution.
- ❖ Technical hydrobiology - concerns with the problems of biological interference with the water supply and operation of ships (e.g. the fouling by microorganisms and sessile animals)

of the hulls of ships, various apparatus and hydraulic structures, and the pipes and water supply lines of electro-thermal power plants; the overgrowth of aquatic plants in reservoirs, and the damage to ships and port installations by wood borers and rock borers).

- ❖ Navigational hydrobiology - the study of biological interference (for example, bioluminescence) with naval activities;
- ❖ Agricultural hydrobiology includes the study of the role of hydrobionts in the fertilization of rice paddies and the possibilities of fish breeding in these waters.

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- ❖ New problems arising from water supply, such as the need to determine the effect of plankton on the absorption and scattering of sound—information indispensable to specialists in underwater acoustics.

Hydrobiology used to be a branch of biology!

But the natural communities of aquatic organisms that constitute the water population did not receive systematic study until the latter part of the 19th century. This study led to the eventual separation of hydrobiology from botany and zoology, which had long been engaged in the study of both terrestrial and aquatic organisms.

Of great importance to the progress of hydrobiology as a science with its own objects of study, methods, and tasks were the first quantitative investigations of plankton (chiefly tiny water-dwelling organisms and a life-form specific to the aquatic environment), initiated in the 1880's by the German scientist W. Hansen. Using the example of Kiel Bay, he showed the need for quantitative data on plankton as a source of food for commercial fish and as the basis of the biological productivity of the sea. Quantitative studies were subsequently begun on benthos, organisms that live at the bottom of bodies of water (again primarily for the sake of promoting the fishing industry). The quantitative study of benthos became widespread after bottom grabs (sample-taking devices) came into use; these were first suggested for marine research in 1911 by the Danish scientist C. Petersen and for freshwater studies by the Swedish scientist S. Eckmann.

Quantitative methods for investigating natural communities of aquatic organisms in order to determine the number (density) of individuals of various species and their biomass have received very broad attention in hydrobiology. Many specialized hydrobiological devices, such as plankton nets, plankton catchers, plankton samplers, and bottom grabs of various designs, are used for this purpose.

Aquatic Organisms

1. Plankton: microscopic water-dwelling plants and animals
2. Benthos: organisms that live at the bottom of water bodies
3. Nekton: the larger free-swimming animals that are capable of resisting the current, such as fish, squid, aquatic mammals, etc.
4. Periphytons or epibionts: The communities of organisms living on the surface of submerged objects.
5. Neuston: communities of animals and plants characteristically of the surface of the water

6. Pleuston: semi-aquatic submerged organisms
7. Epineuston: organisms that skim over or lie upon the surface film
8. Hyponeuston: organisms that live under the surface film but remain in close contact with it.

The first, primarily floristic, faunistic, and biogeographical state in hydrobiological research arose from the need to study the composition and distribution of the species of organisms populating the seas and inland waters. This task, especially in regard to less known regions and taxonomic groups, is still important. The English marine expedition on the Challenger (December 1872 to May 1876) was of exceptional importance in laying the foundation for the study of life at great depths.

As information accumulated on the composition of aquatic populations, attention was focused on elucidating the ecological conditions under which particular biocenoses and habitats of individual species of aquatic organisms were formed.

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Much attention is devoted to furthering the understanding of the importance of biological phenomena in the classification of natural waters, as well as to the theory of biological productivity, the patterns of the biotic cycle, and the flow of energy in aquatic communities.

The next task of hydrobiological research is to clarify the function of aquatic organisms in the processes at work in their environment. This information is needed in order to control biological productivity and self-purification as well as to make rational use of biological resources. The functional characteristics of aquatic organisms can be determined only by experimental investigations of their metabolism, growth, nutrition, and chemical and biochemical composition.

The solution of a number of hydrobiological problems often requires research at the most varied levels—from the molecular, cellular, and organismic to the populational and biocenotic. For example, to determine the causes of excessive development of phytoplankton (so-called water-bloom), it is necessary to take into account both the interaction of different algal and microbial species through specific metabolites excreted into the water and the cycle of biogenic elements (for example, nitrogen and phosphorus). This cycle depends upon the properties of the body of water as a whole and upon the runoff from its drainage basin.

The regular pattern of the interdependence of all phenomena in a body of water, which is an integrated natural object, was emphasized at the turn of the century in the classic works of the Swiss limnologist F. Forel. In the 1920's, A. Thienemann of Germany and E. Naumann of Sweden demonstrated the possibility of subdividing lakes, as well as other bodies of water, into biolimnological types (for example, into oligotrophic and eutrophic). The problems of typology and the classification of natural bodies of water are still under study.

The great complexity and heterogeneity of the natural phenomena with which hydrobiology deals have led to the use of many research techniques. Examples of these techniques include measurement through radioactive carbon of the intensity of plankton photosynthesis (proposed by the Danish scientist E. Stiemann-Nielsen), a method which already has yielded data for gauging the primary production of the ocean and of the hydrosphere as a whole;

spectrophotometric methods of determining the chlorophyll content of plankton; and methods of studying the role of aquatic bacteria (principally by the Soviet scientists E. L. Isachenko, V. S. Butkevich, A. S. Razumov, S. I. Kuznetsov, Iu. and I. Sorokin). Sampling and observation in connection with marine and some freshwater hydrobiological research are aided by aqualungers; underwater television and photography are used at great depths, and the distribution of plankton and other aquatic organisms is traced with the help of echo-sounding devices. The latest physical methods are used in the study of deep-sea bioluminescence and to gain an understanding of the interconnections between the processes at work in aquatic ecosystems. Mathematical simulation and electronic computers are also employed.

The growing influence of theoretical studies upon the solution of practical problems is characteristic of hydrobiology. Hydrobiological knowledge and methods are widely used to evaluate food reserves of bodies of water as bases for their fish yield. These methods are also used in commercial prospecting and in fish breeding. The great progress of hydrobiology in the USSR has made it possible to begin to influence the biological productivity of bodies of water. Before the war, L. A. Zenkevich directed experimental work on enriching the bottom fauna of the Caspian Sea, to which the polychaeteworm *Nereis*, an important source of food for sturgeon, was introduced. Organisms used as foods (chiefly crustaceans, such as mysids) have been successfully adapted to many reservoirs and some lakes (for example, Lake Balkhash). As a result of hydrobiological research, new methods have been proposed for increasing the fish

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yield of ponds through the addition of inorganic fertilizers. These fertilizers are now regularly used in pond fish farming, the productivity of which has thereby been greatly increased.

The field of sanitary hydrobiology studies the effect upon aquatic organisms and their communities of the toxic substances in industrial effluent, the mechanism of the biological self-purification of water, and other matters relating to the pressing problem of supplying man's growing need for pure water.

Broad-scale research in marine hydrobiology is conducted by the Nigerian Institute of Oceanography Marine Research (NIOMR), Lagos.

The most important international organizations concerned with hydrobiology are the permanent International Council for the Exploration of the Sea (Copenhagen), which publishes the *Journal de Conseil* (since 1926), and the International Association of Theoretical and Applied Limnology, which was founded in 1922 and regularly sponsors congresses of limnologists (the 18th congress was held in 1971). The oldest international journal of hydrobiology is the *Archiv für Hydrobiologie* (Stuttgart, since 1906). Other publications include the *Internationale Revue der gesamten Hydrobiologie und Hydrographie* (Leipzig, since 1908) and *Limnology and Oceanography* (USA, since 1956).

Classification Of Waterbodies

Aquatic ecosystems can be classified into 2 based on the presence or absence of flow in its water. These are lentic and lotic aquatic ecosystems.

a. Based on presence or absence of flow in its water

- i. Lentic aquatic ecosystems are those which contain stagnant water e.g. ponds, lakes, etc.
- ii. Lotic aquatic ecosystems are those which contain flowing water e.g. streams, rivers, creeks, etc.

b. Based on salt content

- i. Freshwater e.g. rivers, stream, lakes, bogs, etc.
- ii. Marine water: Oligohaline, mesohaline & polyhaline

Water salinity

Fresh water Brackish water Saline water Brine

< 0.05% 0.05 – 3% 3 – 5% > 5%

< 0.5 ‰ 0.5 – 30 ‰ 30 – 50 ‰ > 50 ‰

Lentic Ecosystem

Lentic refers to standing or still water. It is derived from the Latin lentus, which means sluggish. Lentic systems are diverse, ranging from a small, temporary rainwater pool a few inches deep to Lake Baikal, which has a maximum depth of 1740 m. The general distinction between pools/ponds and lakes is vague, but Brown states that ponds and pools have their entire bottom surfaces exposed to light, while lakes do not. In addition, some lakes become seasonally stratified. Ponds and pools have two regions: the pelagic open water zone, and the benthic zone, which comprises the bottom and shore regions. Since lakes have deep bottom regions not exposed to light, these systems have an additional zone, the profundal. These three areas can

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have very different abiotic conditions and, hence, host species that are specifically adapted to live there.

Lotic Ecosystem

Lotic refers to flowing water, from the Latin lotus, past participle of lavare, to wash. Lotic ecosystems can be contrasted with lentic ecosystems, which involve relatively still terrestrial waters such as lakes and ponds. Together, these two fields form the more general study area of freshwater or aquatic ecology.

Lotic waters can be diverse in their form, ranging from a spring that is only a few centimeters wide to a major river that is kilometers in width. Despite these differences, the following unifying characteristics make the ecology of running waters unique from that of other aquatic habitats.

- Continuous unidirectional
- Presence of water current, with the velocity of water changing with changes in the water volume.
- Thorough mixing of contents, therefore stratification and stagnation are altogether absent.
- The physic-chemical parameters of water quality are in a state of perpetual change
- The amount of DO is usually less than that of ponds or lakes because they receive additional organic matter that adds to their BOD
- Turbidity usually limits light penetration to deeper zone of such systems
- Productivity is low

- Most algae grow attached to rocks and other organisms on the bottom

Importance of Water

1. Drinking
2. Domestic use (cooking, washing, etc)
3. Industrial use (cooling, washing, etc)
4. Transportation
5. Irrigation
6. Source of food (periwinkle, fishes, mollusks, etc)
7. Research
8. Relaxation (skiing, boating, yachting, water polo, swimming, etc)
9. Power generation (Hydro-electricity)
10. Major component of protoplasm
11. Lakes and sea also moderates the surrounding region's temperature and climate because water has a very high specific heat capacity ($4,186 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$). In the daytime, a lake can cool the land beside it with local winds, resulting in a sea breeze; in the night, it can warm it with a land breeze.

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FACETS OF HYDROBIOLOGY

Hydrobiology can be subdivided into two, Oceanography and Limnology. Oceanography is the study of the oceans, limnology deals with inland waters.

Limnology

Limnology is the study of inland bodies of water (lakes, ponds, reservoirs, streams, rivers, wetlands, estuaries) and their related ecosystems.

Limnology deals with freshwaters within continental boundaries. Limnology plays a major role in water use and distribution as well as in wildlife habitat protection. Limnologists work on lake and reservoir management, water pollution control, stream and river protection, artificial wetland construction, fish and wildlife enhancement.

Oceanography

It is the study of the ocean, its surroundings, and life within it.

Oceanography is not so much a science as a collection of scientists who find common cause in trying to understand the complex nature of the ocean. In the vast salty seas that encompass the earth, there is plenty of room for persons trained in physics, chemistry, biology, geology and engineering to practice their specialties. Thus, an oceanographer is any scientifically trained person who spends much of his (or her) career on ocean problems. Willard Bascom (1980).

Oceans cover 70.8% of earth's surface. Oceans are fringed by smaller, adjoining bodies of water such as seas, gulfs, bays, bights, and straits.

Oceans contain about 97% of all liquid water and moderate the earth's climates though they are too salty for human use. More than 90% of the earth's biomass is found in the oceans. In tropical seas, surface waters are warmed by the sun, diluted by rain water and run off from land and aerated by wave action. In higher latitudes, surface waters are cold and denser than waters

flowing from the equator. The dense water sinks to the bottom of deep ocean basins and flows towards the equator. The warm surface water of the tropics flows or stratifies on top of the cold dense water from higher latitudes. Sharp boundaries form between different densities, salinities and temperatures retarding mixing between layers.

Marine ecosystems cover approximately 71% of the Earth's surface. They generate 32% of the world's net primary production. They are distinguished from freshwater ecosystems by the presence of dissolved compounds especially salts in the water. Approximately 85% of the dissolved materials in seawater are sodium and chlorine. Seawater has an average salinity of 35 parts per thousand (ppt) of water but the actual salinity varies among different marine ecosystems.

The five principal oceans: Pacific, Atlantic, Indian, Arctic & Antarctic (Southern) Ocean

Study of oceanography can be sub-divided into the following category:

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1. Physical Oceanography: How and why ocean currents flow, air-sea interactions such as the generation of waves by wind.
2. Chemical Oceanography: Composition of sea water and the processing controlling and altering its composition, including marine pollution
3. Geological Oceanography: Study of rock structure in the ocean basin, properties of rocks such as magnetism, occurrence of earthquakes
4. Ocean Engineering: Design and installation of oceanographic instrumentation and vehicles
5. Biological Oceanography: Organisms that live in the ocean and their relationship to the environment

Characteristics of the World's Oceans

- Average depth of ocean- 4000m
- Average temperature- 39°F (3.9°C)
- Age of ocean- 4 billion years old
- Covers 71% of the Earth's surface
- Average salinity- 35 ‰
- Tallest Mountain: Hawaii's Mauna Kea rises 10,203 m (33,476 ft) from base to peak. Everest is only 8,848m, while the Deepest spot- Marianas Trench (10,912m)
- Age of the ocean basin: 200 my old
- The Earth has an area of approximately 197 million square miles of which 140 million square miles is water.
- The Earth is broken up into hemispheres...Northern and Southern.
- In the Northern Hemisphere, 61% is water and 39% is land, thus it is called the "Land Hemisphere".
- In the Southern Hemisphere, 81% is water and 19% is land, thus it is called the "Water Hemisphere".

Resources from Oceans and Seawater:

Some resources that humans currently use from seawater are:

1. sea salt (halite),
2. magnesium (a light metal used in the making of light bulbs),
3. bromine (gasoline additives/fireproofing materials).
4. Humans also desalinate seawater to get fresh water. This process is time-consuming, expensive and is not currently economically feasible. Many desalinization plants operate, but few are productive on a commercial scale.
5. Gold is present in very low concentrations, but not economic enough to actually "mine" for profit. Many tons of water and sediment would have to be processed in order to get a single ounce of gold.

Ocean Topography:

Oceanographers studying the oceans and ocean floor have delineated three (3) major units:

1. Continental margin (Shelf, Slope & Rise)
2. Ocean basin floor
3. Mid-oceanic ridges

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As technology develops, mapping improves. Currently, echo-sounding devices are used, which are slow and tedious.

1. Continental margin includes:

- > Continental shelf -- very gentle slope (submerged land)
- > Continental slope -- steep slope on edge of continental shelf.
- > Continental rise -- gentle slope where trenches do not exist

Deep sea fans exist where sediment is accumulated and falls off of the continental slope.

Mixture of sediment-laden heavy water forms submarine turbidity currents.

2. The Deep Sea includes:

- > Deep ocean trenches (deepest part of ocean) some as deep as 36,000 feet
- > Abyssal plains are the flat portions of the deep ocean, likely to be the flattest portions of the earth.
- > Isolated volcanic peaks (mantle hot spots) are referred to as "seamounts". Likewise volcanic cones near mid-oceanic ridges are called seamounts as well. Harry Hess called these "guyots".
- > Mid-oceanic ridges occur where divergent plate boundaries occur (Mid-Atlantic) and the narrow region at the ridge crest is called the rift zone.
- > Coral reefs and atolls develop usually in relatively shallow water where light is present and organisms can thrive on nutrients and food sources in a warm water setting. Atolls develop in response to a sinking of the oceanic crust.

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ORIGIN OF LAKES

A lake is a naturally occurring, relatively large and fixed body of water on the earth's surface.

It is localized in a basin or interconnected basins surrounded by dry land. Lakes lie completely

on land and are separate from the ocean, although, like the much larger oceans, they form part of the Earth's water cycle by serving as large standing pools of storage water. Most lakes are freshwater and account for almost all the world's surface freshwater, but some are salt lakes with salinities even higher than that of seawater. Lakes vary significantly in surface area and volume.

Lakes are typically much larger and deeper than ponds, which are also water-filled basins on land, although there are no official definitions or scientific criteria distinguishing the two. and lakes are also distinct from lagoons, which are shallow tidal pools dammed by sandbars at coastal regions of oceans or large lakes. Most lakes are fed by springs, and both fed and drained by creeks and rivers, but some lakes are endorheic (without any outflow), while volcanic lakes are filled directly by precipitation runoffs and do not have any inflow streams.

- ❖ Natural lakes are generally found in mountainous areas (i.e. alpine lakes),
- ❖ dormant volcanic craters,
- ❖ rift zones and
- ❖ areas with ongoing glaciation. Other lakes are found in
- ❖ depressed landforms or
- ❖ along the courses of mature rivers, where a river channel has widened over a basin formed by eroded floodplains and wetlands. Some parts of the world have many lakes formed by the chaotic drainage patterns left over from the last ice age.

All lakes are temporary over long periods of time, as they will slowly fill in with sediments or spill out of the basin containing them.

Artificially controlled lakes are known as reservoirs, and are usually constructed for the following purposes:

- ❖ Industrial uses
- ❖ Agricultural
- ❖ Hydro-electric power generation
- ❖ Supplying domestic drinking water
- ❖ Ecological uses
- ❖ Recreational purposes
- ❖ Other human activities.

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MORPHOLOGY AND ZONATION

In biology, morphology is the study of the shapes and arrangement of parts of organisms, in order to determine their function, development, and how they may have been shaped by evolution.

Lake Morphology refers to the shape and volume of a lake. It often relates to a lake's age, origins, chemical makeup, and the organisms that live within it. Lake shape is influenced by the surrounding landscape and controls much of what goes on physically and biologically underwater.

Zonation is the spatial arrangement of distinct ecological communities based on predominant flora and fauna across a gradient, influenced by both biotic and abiotic factors.

Zonation in Freshwater Lakes

Littoral, Limnetic, Profundal, and Benthic Zones

Freshwater lakes exhibit a distinct vertical zonation pattern influenced by light penetration, temperature, and depth.

- **Littoral Zone:** This is the shallow zone near the shore where light penetrates to the bottom, allowing for the growth of rooted plants. Aquatic plants, snails, and small fishes are common inhabitants, thriving in the warmer waters and benefiting from the abundance of nutrients.
- **Limnetic Zone:** Beyond the littoral is the limnetic zone, which encompasses the open water where light penetration supports phytoplankton but not rooted plants. Zooplankton and free-swimming fish species dominate this zone, relying on the phytoplankton for sustenance.
- **Profundal Zone:** Deeper still is the profundal zone, characterized by colder temperatures and minimal light penetration. Here, the absence of light means photosynthesis is non-existent, leading to lower oxygen levels. Organisms in this zone, such as certain fish and benthic invertebrates, are adapted to these low-oxygen conditions.
- **Benthic Zone:** This is the bottom-most layer, consisting of the lake bed. Decomposers, detritivores, and benthic invertebrates, which feed on detritus and other organic materials, inhabit this zone.

Importance of Zonation

Zonation, the division of ecosystems into distinct zones based on the distribution of species and environmental conditions, plays a pivotal role in understanding and maintaining ecological balance.

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1. **Biodiversity Conservation:** Zonation patterns highlight areas of rich biodiversity, enabling targeted conservation efforts. Different zones often harbor unique species adapted to specific conditions, contributing to the overall biodiversity of an area.
2. **Ecological Stability:** Each zone in an ecosystem plays a specific role in maintaining the ecological balance. For instance, the species in a particular zone might be crucial for nutrient cycling, soil stabilization, or providing habitat for other organisms.
3. **Resource Management:** Understanding zonation helps in the sustainable management of resources. For example, in coastal areas, different zones might have varying fish populations, guiding sustainable fishing practices.
4. **Predicting Ecological Responses:** Recognizing zonation patterns can help predict how ecosystems might respond to changes, such as climate change or human interventions. This is crucial for proactive environmental management.
5. **Scientific Research:** Zonation provides a structured framework for ecological research.

Scientists can study how species adapt to conditions in different zones, leading to insights into evolution, physiology, and behavior.

6. Environmental Education: Zonation serves as a tangible demonstration of ecological principles, making it an effective tool for environmental education. It offers a clear visual representation of how species distribution and abundance vary with changing conditions.

7. Cultural and Economic Significance: Many human communities rely on specific zones for their livelihood, be it through agriculture, fishing, or tourism. Understanding zonation can guide sustainable practices that benefit both the environment and local communities.

8. Habitat Restoration: In areas affected by human activities or natural disasters, knowledge of natural zonation patterns is crucial for effective habitat restoration. It guides the reintroduction of species and recovery efforts in a way that mimics natural systems.

9. Climate Change Mitigation: Zonation patterns can offer insights into areas more vulnerable to the impacts of climate change. This can guide mitigation efforts, such as the establishment of buffer zones or conservation areas.

10. Indicator of Environmental Health: Changes in zonation patterns can serve as early warning signs of environmental degradation or shifts in ecological balance. Monitoring these changes can prompt timely interventions.

In essence, zonation is not just a passive arrangement of species; it's a dynamic system that reflects the health, diversity, and adaptability of ecosystems. Recognizing its importance is crucial for informed ecological management and conservation efforts.

Why is zonation important in ecosystems?

Zonation is crucial as it reflects the adaptability and resilience of species to varying environmental conditions, ensuring biodiversity, ecological balance, and the proper functioning of ecosystems.

How does altitude affect zonation?

Altitude influences temperature, oxygen availability, and atmospheric pressure. As altitude increases, there's a shift in the species of flora and fauna due to these changing conditions, leading to distinct zones like subalpine, alpine, and nival.

What factors contribute to zonation in marine ecosystems?

Factors such as light penetration, salinity, temperature, wave action, and substrate type play significant roles in determining zonation in marine ecosystems.

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How is zonation observed in mangrove forests?

Mangrove forests display zonation with distinct zones like the seaward, middle, and landward zones. Each zone has specific mangrove species adapted to the tidal and salinity conditions of that zone.

Why do some species only exist in specific zones?

Some species are specially adapted to the environmental conditions of a particular zone. Their physiological, behavioral, and morphological adaptations restrict them to that zone for optimal survival.

Can human activities alter zonation patterns?

Yes, human activities like deforestation, pollution, and urbanization can disrupt natural zonation patterns, leading to habitat loss and reduced biodiversity.

What is the difference between the littoral and limnetic zones in freshwater lakes?

The littoral zone is the shallow area near the shore with rooted plants, while the limnetic zone is the open water area where light supports phytoplankton but not rooted plants.

How does competition among species influence zonation?

Competition for resources can lead to the dominance of certain species in specific zones. If one species outcompetes others, it can establish itself more prominently, influencing the zonation pattern.

Are zonation patterns permanent?

No, zonation patterns can change over time due to factors like climate change, natural disturbances, and human interventions. They reflect the dynamic nature of ecosystems and their adaptability to changing conditions.

The depth to which light can reach in lakes depends on turbidity, determined by the density and size of suspended particles. A particle is in suspension if its weight is less than the random turbidity forces acting upon it. These particles can be sedimentary or biological in origin and are responsible for the color of the water. Decaying plant matter, for instance, may be responsible for a yellow or brown color, while algae may cause greenish water. In very shallow water bodies, iron oxides make water reddish brown.

Biological particles include algae and detritus. Bottom-dwelling detritivorous fish can be responsible for turbid waters, because they stir the mud in search of food. Piscivorous fish contribute to turbidity by eating plant-eating (planktonivorous) fish, thus increasing the amount of algae (see aquatic trophic cascade). The light depth or transparency is measured by using a Secchi disk, a 20-centimeter (8 in) disk with alternating white and black quadrants. The depth at which the disk is no longer visible is the Secchi depth, a measure of transparency. The Secchi disk is commonly used to test for eutrophication.

A lake moderates the surrounding region's temperature and climate because water has a very high specific heat capacity ($4,186 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$). In the daytime, a lake can cool the land beside it with local winds, resulting in a sea breeze; in the night, it can warm it with a land breeze.

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HYDROLOGIC CYCLE

The water cycle (hydrologic or H₂O cycle): describes the continuous movement of water on, above and below the surface of the Earth.

Water can change states among liquid, vapor, and ice at various places in the water cycle. Although the balance of water on Earth remains fairly constant over time, individual water molecules can come and go.

The Hydrological Cycle

Description of the Hydrologic Cycle

The sun: drives the water cycle, heats water in oceans and seas.

Water evaporates as water vapor into the air. Ice and snow can sublime directly into water vapor.

Evapo-transpiration is water transpired from plants and evaporated from the soil. Rising air currents take the vapor up into the atmosphere where cooler temperatures cause it to condense into clouds. Air currents move water vapor around the globe; cloud particles collide, grow, and fall out of the sky as precipitation. Some precipitation falls as snow or hail, and can accumulate as ice caps and glaciers, which can store frozen water for thousands of years. Snowpacks can thaw and melt, and the melted water flows over land as snowmelt. Most water falls back into the oceans or onto land as rain, where the water flows over the ground as surface runoff. A portion of runoff enters rivers in valleys in the landscape, with streamflow moving water towards the oceans. Runoff and groundwater are stored as freshwater in lakes. Not all runoff flows into rivers, much of it soaks into the ground as infiltration. Some water infiltrates deep into the ground and replenishes aquifers, which store freshwater for long periods of time. Some infiltration stays close to the land surface and can seep back into surface-water bodies (and the ocean) as groundwater discharge. Some groundwater finds openings in the land surface and comes out as freshwater springs. Over time, the water returns to the ocean, where our water cycle started.

Different Processes

Precipitation: Condensed water vapor that falls to the Earth's surface. Most precipitation occurs as rain, but also includes snow, hail, fog drip, graupel, and sleet. Approximately 505,000 km³ (121,000 cu mi) of water fall as precipitation each year, 398,000 km³ (95,000 cu mi) of it over the oceans.

Canopy interception: The precipitation that is intercepted by plant foliage and eventually evaporates back to the atmosphere rather than falling to the ground.

Snowmelt : The runoff produced by melting snow.

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Runoff : The variety of ways by which water moves across the land. This includes both surface runoff and channel runoff. As it flows, the water may seep into the ground, evaporate into the air, become stored in lakes or reservoirs, or be extracted for agricultural or other human uses.

Infiltration: The flow of water from the ground surface into the ground. Once infiltrated, the water becomes soil moisture or groundwater.

Subsurface Flow : The flow of water underground, in the vadose zone and aquifers. Subsurface water may return to the surface (e.g. as a spring or by being pumped) or eventually seep into the oceans. Water returns to the land surface at lower elevation than where it infiltrated, under the force of gravity or gravity induced pressures. Groundwater tends to move slowly, and is replenished slowly, so it can remain in aquifers for thousands of years.

Evaporation: The transformation of water from liquid to gas phases as it moves from the ground or bodies of water into the overlying atmosphere. The source of energy for evaporation is primarily solar radiation. Evaporation often implicitly includes transpiration from plants, though together they are specifically referred to as evapotranspiration.

Sublimation: The state change directly from solid water (snow or ice) to water vapor.

Advection: The movement of water — in solid, liquid, or vapor states — through the atmosphere. Without advection, water that evaporated over the oceans could not precipitate over land.

Condensation: The transformation of water vapor to liquid water droplets in the air, creating clouds and fog.

Transpiration: The release of water vapor from plants and soil into the air. Water vapor is a gas that cannot be seen.

4.1.3 Residence times

The residence time of a reservoir within the hydrologic cycle is the average time a water molecule will spend in that reservoir (see table below). It is a measure of the average age of the water in that reservoir.

Average reservoir residence times

Reservoir Average residence times

Antarctica 20,000 years

Oceans 3,200 years

Glaciers 20-100 years

Seasonal; snow cover 2-6 months

Soil moisture 1-2 months

Groundwater (shallow) 100-200 years

Groundwater (deep) 10,000 years

Lakes 50-100 years

Rivers 2-6 months

Atmosphere 9days

Groundwater can spend over 10,000 years beneath Earth's surface before leaving. Particularly old groundwater is called fossil water. Water stored in the soil remains there very briefly, because it is spread thinly across the Earth, and is readily lost by evaporation, transpiration, stream flow, or groundwater recharge. After evaporating, the residence time in the atmosphere is about 9 days before condensing and falling to the Earth as precipitation.

The major ice sheets - Antarctica and Greenland - store ice for very long periods. Ice from Antarctica has been reliably dated to 650,000 years before present, though the average residence time is shorter.

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Human activities that alter the water cycle include:

- Agriculture
- Industry
- Alteration of the chemical composition of the atmosphere
- Construction of dams
- Deforestation and afforestation
- Removal of groundwater from wells

- Water abstraction from rivers
- Urbanization

Effects of Water Cycle

i. Effects on climate

The water cycle is powered from solar energy. 86% of the global evaporation occurs from the oceans, reducing their temperature by evaporative cooling. Without the cooling, the effect of evaporation on the greenhouse effect would lead to a much higher surface temperature of 67 °C (153 °F), and a warmer planet.

ii. Effects on biogeochemical cycling

While the water cycle is itself a biogeochemical cycle, flow of water over and beneath the Earth is a key component of the cycling of other biogeochemicals. Runoff is responsible for almost all of the transport of eroded sediment and phosphorus[14] from land to waterbodies. The salinity of the oceans is derived from erosion and transport of dissolved salts from the land. Cultural eutrophication of lakes is primarily due to phosphorus, applied in excess to agricultural fields in fertilizers, and then transported overland and down rivers. Both runoff and groundwater flow play significant roles in transporting nitrogen from the land to waterbodies.[15] The dead zone at the outlet of the Mississippi River is a consequence of nitrates from fertilizer being carried off agricultural fields and funnelled down the river system to the Gulf of Mexico. Runoff also plays a part in the carbon cycle, again through the transport of eroded rock and soil.

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THERMAL STRATIFICATION

Thermal stratification is a condition in which water of different temperatures or densities tend to exist in different layers.

For example, cold water usually floats below warm water and both layers do not mix easily. Stratification produces different layers called epilimnion, thermocline (or metalimnion) and hypolimnion. Wind is the major factor that affects the mixing, transport and exchange of materials or substances in water.

Definition

Temperature affects the rates of chemical and biological activities in water. This in turn affects the water quality. Warm temperatures increase the growth of algae, plants and animals; accelerate the decay of organic matter especially near the bottom of water. In the temperate region, there are four distinct seasons. Rising temperatures in spring warm the upper layers of the water. Warm water is lighter than cold water and the warm water floats over the cold and denser water. Temperature and density differences create distinct layers of water in the lake which do not mix easily. Warmer temperatures tend to increase the density differences and cause a separation between the water layers. This process is called thermal stratification.

Epilimnion: The upper layer of water receiving more sunlight is called epilimnion and most biological activities and growth occur here.

Hypolimnion: The colder, denser and darker bottom waters are called the hypolimnion where

dead plants and animals sink.

Thermocline: The metalimnion (thermocline) is the narrow band between the upper and lower waters where the temperature changes quickly with depth.

Fall turnover describes the mixing of the upper layers and lower layers of water in the cold season when the temperature and density differences between layers disappear. Shallow lakes mix readily and have greater potential to release nutrients from the lake bottoms which fuel algal blooms. The warm water fish ponds average about 2 meters in depth. Marked thermal stratification may develop in very shallow ponds because of turbid conditions resulting in rapid heating of surface waters on calm sunny days.

Stratification is less stable in ponds than large bodies of water. Ponds which stratify during daylight hours can de-stratify at night when the upper layers of water cool by conduction. Strong winds may supply enough energy to cause complete circulation of water. Disappearance of a heavy plankton bloom may allow heating to agreeable depth and also lead to complete mixture. Temperatures of the epilimnion in turbid waters are greater than those of clear waters because of greater absorption of heat by particulate matter.

The surface water of deep tropical lakes never reaches the temperature of maximum density; there is no process that makes the water mix. Bottom layers are depleted of oxygen but can be saturated with carbon dioxide, or other gases such as sulfur dioxide. Earthquakes or landslide
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can cause mixing which upturns the deep layers and may release a vast cloud of toxic gases. This is called a limnic eruption e.g. 1986 disaster at Lake Nyos in Cameroon.

Thermal stratification may cause fish kills due to thermal gradients, stagnation, and ice cover. Excess growth of plankton limits commercial and recreational activities in lakes and reduces drinking water quality. Thermal stratification can be reduced by aeration.

Destratification: Is a process in which the air or water is mixed in order to eliminate stratified layers of temperature, plant life or animal life.

Factors Affecting Lake Mixing, Transport and Exchange of Substances

i. Wind

Wind is the dominant external energy input responsible for mixing of water. Wind generates local currents that break lake boundaries and induce basin-scale motions. Wave breaking is an important mixing mechanism and enhances gas exchange with the atmosphere. Strong wind induces Langmuir circulations which are large-scale counter-rotating helical vortices. Langmuir circulation is responsible for mixing of the surface layer and deepening of the epilimnion. Processes that increase turbulence also decrease residence time and increase interaction with bottom water, reduce stratification and enhance mixing.

ii. Radiation

Two major types of radiation play a role in lake and reservoir stratification. These are short-wave ultraviolet and long-wave infrared radiations. The sun produces the short-wave radiation some of which is reflected at the water surface. The remaining radiation penetrates into the lake

and is absorbed in the water column then gets converted to heat. Different wavelengths of the radiation are absorbed at different depths of water. Blue light travels the farthest and heats the deepest layers of water. It can travel full circle and escape from the lake, giving the lake a blue colour.

Long-wave radiation originates from black-body radiation. The lake (water bodies) and the atmosphere emit black-body radiation. Long-wave atmospheric radiation is partially reflected at the lake surface, and the penetrating radiation is absorbed, causing heating. Black-body radiation results in a loss of thermal energy and cooling of the lake water.

iii. Evaporation and condensation

Evaporation is the conversion of liquid water to water vapor while condensation is the conversion of water vapor to liquid water. Both evaporation and condensation are accompanied by fluxes of heat. Evaporation from the lake surface extracts heat from the lake and results in cooling of the water surface. Condensation extracts heat from the atmosphere and adds it to the water surface, resulting in heating at the water surface. Evaporation and condensation are also accompanied by a flux of water and affect the total water budget of the lake.

iv. Direct Inflows and outflows

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Direct inflows to lakes and reservoirs can include surface inflow from rivers and streams, groundwater inflow, and precipitation. Outflows from the lake can be either surface or groundwater outflow. Each of these flows is accompanied by a flux of heat which may also add or remove kinetic energy. Among the inflows and outflows, surface flows have the greatest potential for kinetic energy input to a lake.

Below the mixing action of the wind and the penetration depth of the solar radiation, a strong temperature and accompanying density gradient develops. This region of strong gradients is called the thermocline, or sometimes pycnocline. Below the thermocline a weaker temperature gradient is observed and the water is cool and comparatively quiescent. The bottom region of the lake is called the hypolimnion. In very deep lakes, where solar radiation and thermal conduction cannot penetrate to the lake bottom, the bottom water will have a temperature near 4 °C, the temperature of maximum density of water. The strong density gradients in the thermocline inhibit exchange between the epilimnion and the hypolimnion. Bottom water in a stratified lake does not actively interact with the atmosphere and can easily become deprived of important dissolved gases.

As the air temperature gets cooler and the solar radiation input decreases in the fall, the surface water cools. The surface water and thermocline cool down to the temperature of the hypolimnion and the lake destratifies becomes mixed (turnover). This gives the bottom water an opportunity to aerate with the atmosphere.

If the air temperature goes below 4 °C so that the surface water can cool below this temperature, then the surface water becomes lighter than the bottom water and a so-called winter inverse stratification develops. The term inverse refers to the fact that the surface water is colder than the bottom water; however, the surface water remains less dense. When ice forms at the water

surface, wind mixing is not possible and the winter density profile may not exhibit a well-defined thermocline. The surface water heats up again in the spring, the lake will again reach a state of thermal homogeneity and in the presence of a light wind will turn over. As the surface waters become warmer, stratification sets in and we return to the summer stratification state. Lakes that experience this full cycle of stratification states are called dimictic because they stratify twice. Lakes that only stratify in the summer are called monomictic.

Temperatures of the epilimnion in turbid waters are greater than those of clear waters because of greater absorption of heat by particulate matter.

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PHYSICAL AND CHEMICAL PROPERTIES OF WATER

(O₂, N₂, NH₃), inorganic carbon, Alkalinity and pH, Bacteria and biogeochemistry, microbial loop, Iron and REDOX reactions, Phosphorus cycling, Nitrogen cycling;

Chemistry: Water chemistry between systems varies tremendously. The chemistry is foremost determined by inputs from the geology of its watershed, or catchment area, but can also be influenced by precipitation and the addition of pollutants from human sources. Large differences in chemistry do not usually exist within small lotic systems due to a high rate of mixing. In larger river systems, however, the concentrations of most nutrients, dissolved salts, and pH decrease as distance increases from the river's source .

Oxygen is likely the most important chemical constituent of lotic systems, as all aerobic organisms require it for survival. It enters the water mostly via diffusion at the water-air interface. Oxygen's solubility in water decreases as water temperature increases. Fast, turbulent streams expose more of the water's surface area to the air and tend to have low temperatures and thus more oxygen than slow, backwaters. Oxygen is a byproduct of photosynthesis, so systems with a high abundance of aquatic algae and plants may also have high concentrations of oxygen during the day. These levels can decrease significantly during the night when primary producers switch to respiration. Oxygen can be limiting if circulation between the surface and deeper layers is poor, if the activity of lotic animals is very high, or if there is a large amount of organic decay occurring.

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LAKE BIOTA

Bacteria: Bacteria are present in large numbers in lotic waters. Free-living forms are associated with decomposing organic material, biofilm on the surfaces of rocks and vegetation, in between particles that compose the substrate, and suspended in the water column. Other forms are also associated with the guts of lotic organisms as parasites or in commensal relationships. Bacteria play a large role in energy recycling, which will be discussed in the Trophic Relationships section.

Primary producers: Algae, consisting of phytoplankton and periphyton, are the most significant sources of primary production in most streams and rivers. Phytoplanktons float freely in the water column and thus are unable to maintain populations in fast flowing streams. They can, however, develop sizable populations in slow moving rivers and backwaters.

Periphyton are typically filamentous and tufted algae that can attach themselves to objects to avoid being washed away by fast current. In places where flow rates are negligible or absent, periphyton may form a gelatinous, unanchored floating mat.

Plants exhibit limited adaptations to fast flow and are most successful in reduced currents. More primitive plants, such as mosses and liverworts attach themselves to solid objects. This typically occurs in colder headwaters where the mostly rocky substrate offers attachment sites. Some plants are free floating at the water's surface in dense mats like duckweed or water hyacinth. Others are rooted and may be classified as submerged or emergent. Rooted plants usually occur in areas of slackened current where fine-grained soils are found. These rooted plants are flexible, with elongated leaves that offer minimal resistance to current.

Living in flowing water can be beneficial to plants and algae because the current is usually well aerated and it provides a continuous supply of nutrients. These organisms are limited by flow, light, water chemistry, substrate, and grazing pressure. Algae and plants are important to lotic systems as sources of energy, for forming microhabitats that shelter other fauna from predators and the current, and as a food resource.

Insects and other invertebrates: Up to 90% of invertebrates in some lotic systems are insects. These species exhibit tremendous diversity and can be found occupying almost every available habitat, including the surfaces of stones, deep below the substratum, adrift in the current, and in the surface film. Insects have developed several strategies for living in the diverse flows of lotic systems. Some avoid high current areas, inhabiting the substratum or the sheltered side of rocks. In stronger current, species have developed weighted cases, attachments to anchored pads of silk, recurved clinging claws, suction cup like devices, and flattened streamlined bodies. Additional invertebrate taxa common to flowing waters include mollusks such as snails, limpets, clams, mussels, as well as crustaceans like crayfish and crabs. Like most of the primary producers, lotic invertebrates often rely heavily on the current to bring them food and oxygen. Invertebrates, especially insects, are important as both consumers and prey items in lotic systems.

Fishes are probably the best-known inhabitants of lotic systems. The ability of a fish species to live in flowing waters depends upon the speed at which it can swim and the duration that its

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speed can be maintained. This ability can vary greatly between species and is tied to the habitat in which it can survive. Continuous swimming expends a tremendous amount of energy and, therefore, fishes spend only short periods in full current. Instead, individuals remain close to the bottom or the banks, behind obstacles, and sheltered from the current, swimming in the current only to feed or change locations. Some species have adapted to living only on the system bottom, never venturing into the open water flow. These fishes are dorso-ventrally flattened to reduce flow resistance and often have eyes on top of their heads to observe what is happening above them. Some also have sensory barrels positioned under the head to assist in the testing of substratum.

Lotic systems typically connect to each other, forming a path to the ocean (spring → stream →

river → ocean), and many fishes have life cycles that require stages in both fresh and salt water. Salmon, for example, are anadromous species that are born and develop in freshwater and then move to the ocean as adults. Eels are catadromous, and are born and develop in the ocean and then move into freshwater as adults.

Other vertebrate taxa that inhabit lotic systems include amphibians, such as salamanders, reptiles (e.g. snakes, turtles, crocodiles and alligators) various bird species, and mammals (e.g. otters, beavers, hippos, and river dolphins). With the exception of a few species, these vertebrates are not tied to water as fishes are, and spend part of their time in terrestrial habitat. Many fish species are important as consumers and as prey species to the larger vertebrates mentioned above.

Trophic relationships

Energy inputs

Energy sources can be autochthonous or allochthonous.

Autochthonous energy sources: are those derived from within the lotic system. During photosynthesis, for example, primary producers form organic carbon compounds out of carbon dioxide and inorganic matter. The energy they produce is important for the community because it may be transferred to higher trophic levels via consumption. Additionally, high rates of primary production can introduce dissolved organic matter (DOM) to the waters. Another form of autochthonous energy comes from the decomposition of dead organisms and feces that originate within the lotic system. In this case, bacteria decompose the detritus or coarse particulate organic material (CPOM; >1 mm pieces) into fine organic particulate matter (FPOM; <1 mm pieces) and then further into inorganic compounds that are required for photosynthesis. This process is discussed in more detail below.

Allochthonous energy sources: are those derived from outside the lotic system, that is, from the terrestrial environment. Leaves, twigs, fruits, etc. are typical forms of terrestrial CPOM that have entered the water by direct litterfall or lateral leaf blow. In addition, terrestrial animal-derived materials, such as feces or carcasses that have been added to the system are examples of allochthonous CPOM. The CPOM undergoes a specific process of degradation. An example of a leaf fallen into a stream. First, the soluble chemicals are dissolved and leached from the leaf upon its saturation with water. This adds to the DOM load in the system. Next, microbes such as bacteria and fungi colonize the leaf, softening it as the mycelium of the fungus grows into it. The composition of the microbial community is influenced by the species of tree from which the leaves are shed. This combination of bacteria, fungi, and leaf are a food source for shredding invertebrates, which leave only FPOM after consumption. These fine particles may be

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colonized by microbes again or serve as a food source for animals that consume FPOM. Organic matter can also enter the lotic system already in the FPOM stage by wind, surface runoff, bankerosion, or groundwater. Similarly, DOM can be introduced through canopy drip from rain or from surface flows.

Invertebrates

Invertebrates can be organized into many feeding guilds in lotic systems. Some species are shredders, which use large and powerful mouth parts to feed on non-woody CPOM and their associated microorganisms. Others are suspension feeders, which use their setae, filtering apparatus, nets, or even secretions to collect FPOM and microbes from the water. These species may be passive collectors, utilizing the natural flow of the system, or they may generate their own current to draw water, and also, FPOM. Members of the gatherer-collector guild actively search for FPOM under rocks and in other places where the stream flow has slackened enough to allow deposition. Grazing invertebrates utilize scraping, rasping, and browsing adaptations to feed on periphyton and detritus. Finally, several families are predatory, capturing and consuming animal prey. Both the number of species and the abundance of individuals within each guild is largely dependent upon food availability. Thus, these values may vary across both seasons and systems.

Fishes

Fishes can also be placed into feeding guilds. Planktivores pick plankton out of the water column. Herbivore-detritivores are bottom-feeding species that ingest both periphyton and detritus indiscriminately. Surface and water column feeders capture surface prey (mainly terrestrial and emerging insects) and drift (benthic invertebrates floating downstream). Benthic invertebrate feeders prey primarily on immature insects, but will also consume other benthic invertebrates. Top predators consume fishes and/or large invertebrates. Omnivores ingest a wide range of prey. These can be floral, faunal, and/or detrital in nature. Finally, parasites live off of host species, typically other fishes. Fishes are flexible in their feeding roles, capturing different prey with regard to seasonal availability and their own developmental stage. Thus, they may occupy multiple feeding guilds in their lifetime. The number of species in each guild can vary greatly between systems, with temperate warm water streams having the most benthic invertebrate feeders, and tropical systems having large numbers of detritus feeders due to high rates of allochthonous input.

Community patterns and diversity

Local species richness

Large rivers have comparatively more species than small streams. Many relate this pattern to the greater area and volume of larger systems, as well as an increase in habitat diversity. Some systems, however, show a poor fit between system size and species richness. In these cases, a combination of factors such as historical rates of speciation and extinction, type of substrate, microhabitat availability, water chemistry, temperature, and disturbance such as flooding seem to be important.

Human impacts

Pollution

Pollutant sources of lotic systems are hard to control because they derive, often in small amounts, over a very wide area and enter the system at many locations along its length. Agricultural fields often deliver large quantities of sediments, nutrients, and chemicals to nearby streams and rivers. Urban and residential areas can also add to this pollution when contaminants are accumulated on impervious surfaces such as roads and parking lots that then drain into the

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system. Elevated nutrient concentrations, especially nitrogen and phosphorus which are key components of fertilizers, can increase periphyton growth, which can be particularly dangerous in slow moving streams. Another pollutant, acid rain, forms from sulfur dioxide and nitrous oxide emitted from factories and power stations. These substances readily dissolve in atmospheric moisture and enter lotic systems through precipitation. This can lower the pH of these sites, affecting all trophic levels from algae to vertebrates. Mean species richness and total species numbers within a system decrease with decreasing pH. While direct pollution of lotic systems has been greatly reduced in the United States under the government's Clean Water Act, contaminants from diffuse, non-point sources, remain a large problem.

Dams alter the flow, temperature, and sediment regime of lotic systems. Additionally, many rivers are dammed at multiple locations, amplifying the impact. Dams can cause enhanced clarity and reduced variability in stream flow, which is due to an increase in periphyton abundance. Invertebrates immediately below a dam can show reductions in species richness due to an overall reduction in habitat heterogeneity. Also, thermal changes can affect insect development, with abnormally warm winter temperatures obscuring cues to break egg diapause and overly cool summer temperatures leaving too few acceptable days to complete growth. Finally, dams fragment river systems, isolating previously continuous populations, and preventing the migrations of anadromous and catadromous species.

Invasive species

Invasive species have been introduced to lotic systems through both purposeful events (e.g. stocking game and food species) as well as unintentional events (e.g. hitchhikers on boats or fishing waders). These organisms can affect natives via competition for prey or habitat, predation, habitat alteration, hybridization, or the introduction of harmful diseases and parasites. Once established, these species can be difficult to control or eradicate, particularly because of the connectivity of lotic systems. Invasive species can be especially harmful in areas that have endangered biota, such as mussels in the Southeast United States, or those that have localized endemic species, like lotic systems west of the Rocky Mountains, where many species evolved in isolation.

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LAKE FOOD CHAIN

Differences between aquatic and terrestrial organisms

Differences in chemical and physical characteristics of water and air expose aquatic and terrestrial organisms to different environmental conditions.

- i. Aquatic organisms do not need supporting body structures as strong as those for terrestrial organisms.
- ii. Water has a density close to that of many organisms living in it. The viscosity of water is about 50 times higher than air. This makes the moving energy cost for a comparable displacement higher for an aquatic than for terrestrial organisms.
- iii. The oxygen concentration is lower in water than in air and allows the aquatic organisms to

perform more work than the terrestrial ones to provide the same amount of oxygen for respiration.

iv. The concentration of free carbon dioxide is higher in water than in the atmosphere. Aquatic plants have more inorganic carbon available than the terrestrial ones in the surface layer.

v. Aquatic autotrophs access dissolved nutrients in water without requiring specific transporting structure.

In both aquatic and terrestrial ecosystems, matter and energy flow through a chain of organisms transforming it and constituting the food chain. The energy flowing through the food chain progressively degrades to heat and matter is recycled, passing through stages of variable molecular complexity.

Classification of Aquatic Organisms

A. Based on the Roles they Play in the Ecosystem

i. **Primary Producers:** These are autotrophic organisms which synthesise organic matter using energy from the sun and inorganic substances. The primary producers include unicellular algae, free floating phytoplankton or periphyton that adhere to substrates and aquatic plants (macrophytes) growing in the littoral zone. Autotrophic organisms also include photosynthetic and chemosynthetic bacteria.

ii. **Consumers:** depend on primary producers directly or indirectly for their food. These include primary (feeding on autotrophs), secondary (feeding on animals eating plants or autotrophs), tertiary consumers (feeding on secondary consumers), and so on. Carnivorous fish, birds or fish-eating mammals, occupies the last link in the aquatic consumer chain. The consumers are heterotrophs and use organic matter already produced. They cannot synthesise organic matter from inorganic compounds.

iii. **Detritivores:** These are organisms that feed on the remains of dead organisms and on the waste products of metabolism, degrading it to less complex chemical structures.

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iv. **Decomposers:** Decomposers are mainly the heterotrophic bacteria, oxidising (decomposing) the organic molecules till they are reduced to their inorganic constituents (mineralisation). These are made available for the autotrophs and are thus re-cycled.

B. Based on their Habitats

Plankton

Aquatic organisms consist of animals (zooplankton), plants (phytoplankton), bacteria and archaea (bacterioplankton) and virus (virioplankton). Plankton occupies the littoral and the pelagic zones of water bodies. Autotrophic phytoplankton and bacterioplankton, carry out photosynthesis, a light-dependent activity, and are limited to the euphotic zone (zone of light penetration). The freshwater zooplankton community consists of microscopic organisms belonging to different systematic groups, ranging from protozoa to crustaceans. Though some plankton classes have organs to move, the plankton cannot oppose the current movement and is transported by the water mass hosting it.

i. The Virioplankton Viruses: are the smallest known organisms. They have simple structures, made of nucleic acids as RNA or DNA, contained in a protein capsule called capsid. They range in size between 20 – 200nm and reproduce within living cells. They are obligate parasites. A virus develops through both extracellular and intracellular phases. The virus particle (virion) is stable, resistant to denaturation and floats freely in the water. When a virion meets a host, it adsorbs to it, protruding protein structures that are receptors of the host's cell surface. In the intracellular phase, the virus or viral DNA, enters the host cell where replicates. The newly formed viral particles lyse the host cell, not necessarily destroying it, and are released into the environment. This is called the lytic cycle. Viral replication can also occur through the lysogenic cycle. The viral genome becomes part of the host's genome, integrated in it or as an autonomous plasmid. The viral genome remains in a dormant stage (prophage) ending when the changed environmental conditions (UV radiation, stress, chemicals, etc.) induce the lytic phase. The abundance of the virioplanktonic community depends on the abundance of hosts.

ii. The Bacterioplankton: These are prokaryotes, between 0.2 and 2 µm in size. They can be autotrophs and heterotrophs. The autotrophs include cyanobacteria, photosynthetic and chemosynthetic bacteria. This group includes the bacteria that harvest light. Bacterial photosynthesis does not occur in the presence of oxygen but only in anoxic environments. Chemosynthetic bacteria meet their energy requirements through the oxidation of minerals substrates. Heterotrophic bacteria are more abundant than autotrophic bacteria and form the largest component of bacterioplankton. Archaea are genetically different from bacteria but are part of bacterioplankton. They are adapted to life in extreme environments such as hot springs and acidic water.

Roles of bacteria in the food chain

❖ Breakdown of the organic substances, first-hydrolysed by the exoenzymes produced by bacteria and then mineralized, thus making their mineral constituents available.

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❖ Recycling of dissolved organic matter, through the production of new bacterial cells usable by larger organisms, unable to use directly the dissolved organic matter.

iii. The phytoplankton The algal cells constituting the phytoplankton are at the beginning of the food chain. They synthesise the organic matter necessary to maintain the whole food chain working. The process ending with the production of organic compounds from inorganic substances and light is known as photosynthesis. The organic product of photosynthesis is glucose (C₆H₁₂O₆). The carbon and the oxygen necessary to build up the organic matter are provided by atmospheric CO₂. The equation summarising the process is: $6 \text{ CO}_2 + 6 \text{ H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ O}_2$

Photosynthesis occurs in two phases:

1. Light-dependent phase Light energy from the sun is harvested by special pigments such as chlorophyll (the most important). In the light-dependent reaction, an energy-rich electron of a chlorophyll molecule is passed along a chain of molecules known as the electron transfer

system. The energy lost by the electron as it moves along the chain is used to make NADP, (Nicotinamide or niacinamide Adenine Dinucleotide Phosphate) and ATP (Adenosine Triphosphate). These are very energy-rich but relatively unstable molecules.

2. The dark phase or Calvin cycle ATP and NADP are used to produce energy-rich carbohydrate molecules, which can be stored and used as a basis for the production of all other forms of organic molecules. The synthesis of new organic matter leads to an increase in cell size and a division in two daughter cells. These can be completely separated or they can remain connected though their cell walls, forming a colony.

Composition of Fresh Water Phytoplankton Communities

Freshwater phytoplankton communities consist of prokaryotes, Cyanobacteria, and eukaryotes from several taxonomic groups e.g. Diatoms, Chrysophytes (golden algae), Cryptophytes, Dynophytes, Euglenophytes and Chlorophytes (green algae).

- i. Prokaryotic cells: The chromosomes are dispersed in the protoplasm. Photosynthesis occurs in invaginations of the cell membrane.
- ii. Eukaryotic cells: The chromosomes are present in the nucleus and there are specific organelles, the chloroplasts, devoted to photosynthesis.
- iii. Cyanobacteria: Cyanophyta or blue - green algae are bacteria with the cell structure of prokaryotes. They differ from bacteria by having chlorophyll a, which is also found in eukaryotic algae. They contain phycocyanin and allophycocyanin (pigments) which make them blue-green. Examples of blue -green algae are genera *Anabaena*, *Microcystis* and *Planktothrix*. They form massive blooms of Cyanobacteria which produce toxins, reduce water quality and dissolved oxygen.
- iv. Diatoms are an important group of algae. They contain siliceous cell walls, called frustules. They contain chlorophyll (photosynthesis) and carotenoids (yellow-golden brown colour).

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- a. Chrysophytes (Chrysophyceae) are single-celled and typical of freshwater environments poor in calcium. They have two flagella. Examples include *Synura*, *Mallomonas*, *Ochromonas*, *Dinobryon* and *Uroglena*. Many species lack a true cell wall but may have a layer of scales, the lorica, made of cellulose and can be accompanied by limestone, silica or iron oxides. They contain photosynthetic pigment, chlorophyll.
- b. Cryptophytes (Cryptophyceae) have two flagella almost as long as the cell. The outer portion of the cell or periplast is composed of a plasmatic membrane and a series of plates placed beneath. The Cryptophytes contain chlorophyll, phycobilins phycocyanin and phycoerythrin (pigments)
- c. Dynophytes (division Dynophyta) are made by an upper part (the epicone) and a lower part (the hypocone) separated by a belt. They contain chlorophyll. The Dynophytes have an intermediate chromosome falling between that of prokaryotes and eukaryotes and are thus defined mesokaryotes.
- d. Euglenophytes (division: Euglenophyta) are flagellate algae present mostly in eutrophic

freshwater environments. The Euglenophytes have two basal bodies with one or two emerging flagella. Their chloroplast contains mainly chlorophyll a and b.

e. Chlorophytes, or green algae, (division Chlorophyta) are like higher plants. Their chloroplasts contain Chlorophylls a and b and they are green and contain no accessory pigments. They have two or more flagella similar in structure but different in length.

The Zooplankton

The zooplankton commonly found in freshwater are:

- i. Protozoa are unicellular organisms between tens and hundreds microns in size. They feed on bacteria or detritus particles which are incorporated in a vacuole where digestion occurs (phagocytosis). Planktonic protozoa move mostly by cilia. They transform dissolved organic matter into forms usable by higher organisms.
- ii. Cnidarians are diverse in form such as siphonophores, medusae, corals, feathery hydroids, and box jellies with complex eyes. They are all armed with stinging cells called nematocysts. Many cnidarians live in the ocean while a smaller number of species are found in fresh waters. There are four classes of Cnidarians namely Anthozoa, Cubozoa, Hydrozoa and Scyphozoa.
- iii. Rotifera are animals related to worms and are not segmented. They measure length 100 μm - 1 mm in length. They have a crown of cilia moving around their mouth to suck inside the food. They feed on algae, protozoa, bacteria, debris and some are carnivorous. Examples include the genera Asplanchna, Keratella, Kellicottia and Brachionus). They reproduce by parthenogenesis (without egg fertilization by males).
- iv. There are two groups of planktonic crustaceans namely Cladocerans e.g. Daphnia (water flea) and copepods. Daphnia and other cladocerans, are filter feeders because their limbs have filaments (setae) used to filter food particles from water. Cladocerans reproduce by

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parthenogenesis while sexual reproduction occurs in copepods and eggs are carried by females within ovigerous bags attached to the mother's body.

The transformation of organic matter by heterotrophs is called secondary production. The annual net secondary production of a zooplankton population is the sum of all biomass produced by the population during the year, taking into account the losses due to mortality, predation and emigration.

Benthos

This is the community living in the benthic zone, at the surface or inside the bottom sediment. It is a complex community consisting of both mobile and sessile organisms. The benthos includes fungi, bacteria, plant and animal forms which include algae, protozoa, rotifers, tardigrades, mollusks, worms, arthropods. Aquatic macroalgae and macrophytes grow in close relationship with the sediment as the benthos and account for a large portion of primary production in water.

- i. Phytobenthos or Periphyton (or "Aufwuchs") This is the community of microscopic autotrophs attached to submerged inorganic (epilithic algae) and organic surfaces, including living plants (epiphytic algae). They grow in areas of light penetration. Protozoa, sponges,

bryozoans and other animals are usually associated with the periphyton. Examples of periphyton are filamentous algae, unicellular or colonial algae belonging to the cyanobacteria, diatoms and chlorophytes.

ii. **Macrophytes or hydrophytes** These are plants growing in aquatic environments or on substrates that are fully or periodically submerged. Aquatic macrophytes may be emergent, floating or submerged. The emergent and submerged hydrophytes are always rooted in the substrate but the floating hydrophytes may be rooted in the substrate or freely floating. Examples of emergent macrophytes are *Phragmites* spp (reeds), *Scirpus*, *Typha*, *Sagittaria* sp. The floating macrophytes include *Nymphaea* sp., *Trapa natans* (water chestnut) and *Potamogeton* sp. The free floating macrophytes include *Lemna minor* and *Eicornia crassipes*. The shading effect of floating hydrophytes reduces the penetration of solar radiation, leading to a reduction in phytoplankton populations.

iii. **Nekton, Neuston and Psammon** The term nekton describes aquatic animals that can move actively in the water, independent of water movement. These include fish, mollusks, amphibians, reptiles and mammals. The fish community includes both littoral and pelagic species. The species composition of the fish population depends on aquatic morphology, which influences the thermal regime, on water chemistry (in particular on oxygen availability) and on the biological characteristics of the water, which determines the quality and quantity of available food (Matthews, 1998).

iv. **Neuston** consists of microscopic organisms, protozoa and bacteria, living in the surface film at the water-air interface. This group also includes larger organisms such as insects which can move on the water surface. **Psammon** refers to the animal community, including protozoa,

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rotifers, tardigrades, crustaceans, worms, larvae of some insects, living in interstitial water of sandy beaches.

WETLANDS

A wetland is a land area that is saturated with water, either permanently or seasonally, such that it takes on the characteristics of a distinct ecosystem.

It is a collective name for marshes, swamps, bogs, etc. It represents an ecosystem which is intermediate between an aquatic and terrestrial ecosystems.

Wetlands are areas where water covers the soil, or is present either at or near the surface of the soil all year or for varying periods of time during the year, including during the growing season. Water saturation (hydrology) largely determines how the soil develops and the types of plant and animal communities living in and on the soil. Wetlands may support both aquatic and terrestrial species. The prolonged presence of water creates conditions that favor the growth of specially adapted plants (hydrophytes) and promote the development of characteristic wetland (hydric) soils.

A patch of land that develops pools of water after a rain storm would not be considered a "wetland," even though the land is wet. A more concise definition is a community composed of

hydric and hydrophytes. Wetlands have also been described as ecotones, providing a transition between dry land and water bodies. Wetlands exist "...at the interface between truly terrestrial ecosystems and aquatic systems, making them inherently different from each other, yet highly dependent on both (Mitch et al. 2007).

A wetland is an ecosystem that arises when inundation by water produces soils dominated by anaerobic processes, which, in turn, forces the biota, particularly rooted plants, to adapt to flooding. Wetlands are places where land and water meets. The RAMSAR convention, an international convention concerning wetland conservation, defines wetlands as:

Ramsar Convention definition

Under the Ramsar International Wetland Conservation Treaty, wetlands are defined as follows:

- Wetlands are areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six metres (Ramsar Article 1.1).

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- Wetlands may incorporate riparian and coastal zones adjacent to the wetlands, and islands or bodies of marine water deeper than six metres at low tide lying within the wetlands (Ramsar Article 2.1)

The RAMSAR Convention is the name of the International Convention on Protection of Wetlands, particularly as waterfowl habitat. It was formed in 1971 by delegates who met at Ramsar, a town in Iran on the coast of the Caspian Sea. It is the only international treaty that seeks to conserve habitat rather than species. Over 100 countries have signed the Convention.

Wetlands are, thus, transitional habitats between dry land and deep water. They include marshes, swamps, peatlands (including bogs and fens), flood meadows, lakes and ponds, rivers and streams, estuaries and other coastal waters (including salt marshes, mangroves and even coral reefs).

Wetland Water Sources

- Precipitation
- Surface Flow
- Groundwater Discharge
- Ocean
- River
- Lake

Wetland Water Losses

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- Evaporation/Transpiration
- Outflows
- Groundwater Recharge

Indicators of Wetland Hydrology

Primary

- Visible surface water
- Saturated Surface Soils
- Saturated within 18 in.

Secondary

- Oxidized rhizospheres
- Water stained leaves
- Surface scouring
- Plant adaptations
- Hydric Soils

Classification of Wetlands

Classifying wetlands is difficult, due to the highly dynamic nature of the places, changing with the seasons and over time. Their precise boundaries are hard to define. Wetlands occur naturally on every continent except Antarctica, the largest including the Amazon River basin, the West Siberian Plain, and the Pantanal in South America. The water found in wetlands can be freshwater, brackish, or saltwater. The main wetland types include swamps, marshes, bogs, and fens; and sub-types include mangrove, carr, pocosin, and varzea. Wetlands vary widely due to local and regional differences in topography, hydrology, vegetation, and other factors, including human involvement.

Wetlands vary in size from micro-wetlands, possibly housing rare endemic species, to the vast water system of Lake Victoria and the Nile which has an estimated catchment area of 2.9 million square km (one tenth of the area of Africa Continent). They are dynamic systems, continually undergoing natural change due to flooding, subsidence, drought, sea-level rise or infilling with sediment or organic material. Lake Chad in the Sahel, for example, is rarely more than 7m deep but its area varies from 25,000 km² to less than 10,000 km² in times of drought. Many wetlands are only temporary features of the landscape and will change and eventually disappear, whilst new wetlands are naturally created elsewhere.

Wetlands also vary widely in type, with significant differences from one region to another. The grassland floodplains in Africa and South America may be contrasted with Sudan's huge swamps, the enormous mangrove forests in parts of Asia, coral reefs in the tropics and the high concentration of peatlands around the margins of the South China Sea. Some categories of wetland are very well known whilst others, such as karstic and subterranean wetlands, may be near-invisible to the general public, legislators and land managers.

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There are two main types of world wetlands: those found inland with freshwater (inland or non-tidal wetlands), and those on the coast with salty (marine or brackish or coastal or tidal wetlands and) water. There are about 30 recognizes natural wetland types and a further nine man-made ones.

Coastal wetlands, as their name suggests, they are found along the coasts. They are closely linked to estuaries, where sea water mixes with fresh water to form an environment of varying

salinities. The salt water and the fluctuating water levels (due to tidal action) combine to create a rather difficult environment for most plants. Consequently, many shallow coastal areas are unvegetated mud flats or sand flats. Some plants, however, have successfully adapted to this environment. Certain grasses and grasslike plants that adapt to the saline conditions form the tidal salt marshes that are found along the coasts. Mangrove swamps, with salt-loving shrubs or trees, are common in tropical climates, such as in southern parts of Nigeria. Some tidal freshwater wetlands form beyond the upper edges of tidal salt marshes where the influence of salt water ends. Some examples are listed below:

- Marine waters—permanent shallow waters less than six metres deep at low tide; includes sea bays, straits
- Sub-tidal aquatic beds; includes kelp beds, seagrasses, tropical marine meadows
- Coral reefs
- Rocky marine shores; includes rocky offshore islands, sea cliffs
- Sand, shingle or pebble beaches; includes sand bars, spits, sandy islets
- Intertidal mud, sand or salt flats
- Intertidal marshes; includes saltmarshes, salt meadows, saltings, raised salt marshes, tidal brackish and freshwater marshes
- Intertidal forested wetlands; includes mangrove swamps, nipa swamps, tidal freshwater swamp forests
- Brackish to saline lagoons and marshes with one or more relatively narrow connections with the sea
- Freshwater lagoons and marshes in the coastal zone
- Non-tidal freshwater forested wetlands

Inland wetlands are most common on floodplains along rivers and streams (riparian wetlands), in isolated depressions surrounded by dry land, along the margins of lakes and ponds, and in other low-lying areas where the groundwater intercepts the soil surface or where precipitation sufficiently saturates the soil (vernal pools and bogs). Inland wetlands include marshes and wet meadows dominated by herbaceous plants, swamps dominated by shrubs, and wooded swamps dominated by trees. Examples of inland wetlands are listed below:

- Permanent rivers and streams; includes waterfalls
- Seasonal and irregular rivers and streams
- Inland deltas (permanent)
- Riverine floodplains; includes river flats, flooded river basins, seasonally flooded grassland, savanna and palm savanna

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- Permanent freshwater lakes (> 8 ha); includes large oxbow lakes
- Seasonal/intermittent freshwater lakes (> 8 ha), floodplain lakes
- Permanent saline/brackish lakes
- Seasonal/intermittent saline lakes
- Permanent freshwater ponds (< 8 ha), marshes and swamps on inorganic soils; with

emergent vegetation waterlogged for at least most of the growing season

- Seasonal/intermittent freshwater ponds and marshes on inorganic soils; includes sloughs, potholes; seasonally flooded meadows, sedge marshes
- Permanent saline/brackish marshes
- Seasonal saline marshes
- Shrub swamps; shrub-dominated freshwater marsh, shrub carr, alder thicket on inorganic soils
- Freshwater swamp forest; seasonally flooded forest, wooded swamps; on inorganic soils
- Peatlands; forest, shrub or open bogs
- Alpine and tundra wetlands; includes alpine meadows, tundra pools, temporary waters from snow melt
- Freshwater springs, oases and rock pools
- Geothermal wetlands
- Inland, subterranean karst wetlands

Artificial (Human-made) Wetlands

Human made wetlands include irrigated lands, ponds for agriculture and livestock, aquaculture installations, rivers and reservoirs which have been dammed for water supply, irrigation or hydroelectric power generation, flooded excavations, certain wastewater treatment facilities and salt pans (salines). In some cases, the construction of an artificial wetland directly involves the destruction or degradation of a natural wetland, for example where mangrove ecosystems are converted to aquaculture ponds or where floodplains are drained and subsequently irrigated for intensive arable cultivation. Under some laws, the grant of planning permission to develop natural wetlands may be conditional on the developer funding the creation of new wetlands.

The common characteristic of artificial wetlands is that they are usually designed for a specific purpose and seldom have the full range of wetland functions and values provided by a natural wetland, even where they are of similar size. Whilst there is broad consensus that newly created or restored and rehabilitated wetlands cannot replace natural ecosystems, wetland creation and restoration are becoming an increasing priority in industrialised countries where wetland integrity has been seriously degraded over decades. The creation and restoration of wetlands is becoming technically and ecologically feasible for a limited range of habitat types, including freshwater marshes and tidal marshes on low energy coasts.

- Water storage areas; reservoirs, barrages, hydro-electric dams, impoundments (generally > 8 ha)
- Ponds, including farm ponds, stock ponds, small tanks (generally < 8 ha)
- Aquaculture ponds; fish ponds, shrimp ponds
- Salt exploitation; salt pans, salines
- Excavations; gravel pits, borrow pits, mining pools
- Wastewater treatment; sewage farms, settling ponds, oxidation basins
- Irrigated land and irrigation channels; rice fields, canals, ditches

- Seasonally flooded arable land, farm land

General Characteristics Features and Indicators of Wetlands

Wetlands have unique characteristics. They are generally distinguished from other water bodies or landforms based on their water level and on the types of plants that live within them. Specifically, wetlands are characterized as having a water table that stands at or near the land surface for a long enough period each year to support aquatic plants. Wetlands vary widely due to local and regional differences in topography, hydrology, vegetation, and other factors, including human involvement.

Some wetlands, such as swamps and marshes, are obvious to most people. Others, like seasonal wetlands or bogs, are not as easily recognized because they may dry out during part of the year or simply do not look very wet.

i. Evidence of Hydrology:

- Standing or flowing water for seven or more consecutive days during the growing season.
- Waterlogged soil: Determined by digging a 12-inch-deep hole and then checking for water in the hole; or by looking for soil that glistens with water; or squeezing water from a handful of soil.
- Water marks on trees or small piles of debris lodged in trees or piled against other objects in the direction of water movement near river systems.

ii. Wetland Soils:

- Check for a blue or gray color about a foot below the surface. Your local CD office may describe other color characteristics to look for.
- Look for organic matter such as peat or muck.
- Smell the soil for an odor like rotten eggs.

iii. Wetland Vegetation:

More than 5,000 different plants grow in wetlands. Some common wetland plants to look for include:

- Water lilies, cattails, arrowhead, smartweed, pondweed and other plants in standing water.
- Grasses such as reed canary grass, barnyard grass, and prairie cordgrass, or rushes and sedges.
- Trees such as willow, white cedar, cottonwood, silver and red maple, green ash, tamarack, pin oak, and elm.
- Shrubs such as buttonbush, Michigan holly, and red-osier dogwood.

Importance of Wetlands

Wetlands play a number of roles in the environment:

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- Wetlands are considered the most biologically diverse of all ecosystems, serving as home to a wide range of plant and animal life. Rivaling the likes of tropical rainforests and coral reefs, wetlands are among the most fertile, productive ecosystems in the world. Two thirds of

all fish consumed worldwide are dependent on coastal wetlands at some stage in their life cycle. Annual fish and seafood production in swamps and marshes worldwide has been estimated at an average of nine tons per square kilometer, 259 hectares or 640 acres.

ii. Reservoirs of biodiversity: Wetland systems' rich biodiversity is becoming a focal point at International Treaty Conventions and within the World Wildlife Fund (WWF) organization due to the high number of species present in wetlands, the small global geographic area of wetlands, the number of species which are endemic to wetlands, and the high productivity of wetland systems. Hundreds of thousands of animal species, 20,000 of them vertebrates, are living in wetland systems. The discovery rate of fresh water fish is at 200 new species per year. The Amazon holds 3,000 species of fresh water fish species within the boundaries of its basin whose function it is to disperse the seeds of trees. One of its key species, the Piramutaba catfish, *Brachyplatystoma vaillantii*, migrates more than 3,300 km (2,051 miles) from its nursery grounds near the mouth of the Amazon River to its spawning grounds in Andean tributaries (400 m or 437 yards above sea level) distributing plants seed along the route.

Intertidal mudflats have a similar productivity even while possessing a low number of species. iii. High biological productivity. Many wetlands are highly productive ecosystems in large part because they are rich in organic matter and nutrients. These nutrients support organisms within the marsh, but in many instances the nutrients are also transferred to nearby aquatic systems (lakes, rivers, and estuaries), enhancing the productivity of these systems and supporting human uses such as offshore commercial fisheries.

iv. Genetic diversity: Many species in wetland systems are unique due to the long period of time that the ecosystem has been physically isolated from other aquatic sources. The number of endemic species in Lake Baikal (Russia) classifies it as a hotspot for biodiversity and one of the most biodiverse wetlands in the entire world. Evidence from a research study suggest that the number of crustacean species endemic to Baikal Lake (>690 species and subspecies) exceeds the number of the same groups of animals inhabiting all the fresh water bodies of Eurasia together.

v. Flood protection: The wetland system of floodplains is formed from major rivers downstream from their headwaters. Notable river systems that produce large spans of floodplain include the Nile River, the Niger river inland delta, the Zambezi River flood plain, the Okavango River inland delta, the Kafue River flood plain, the Lake Bangweulu flood plain, Mississippi River (USA), Amazon River (South America), Yangtze River (China), Danube River (Central Europe) and Murray-Darling River (Australia).

vi. Water storage and reservoirs: They act like giant sponges, absorbing water from all around. Many wetlands help recharge underground aquifers that store 97% of the world's unfrozen freshwater. Groundwater is critical to billions of people as their only source of drinking water (almost one third of the human population). It is also the only source of water for many irrigation programmes (17% of the world's cropland is irrigated). "The floodplains of major rivers act as natural storage reservoirs, enabling excess water to spread out over a wide area, which reduces its depth and speed. Wetlands close to the headwaters of streams and rivers can

slow down rainwater runoff and spring snowmelt so that it doesn't run straight off the land into water courses. This can help prevent sudden, damaging floods downstream.

vii. Groundwater replenishment. The surface water which is the water visibly seen in wetland systems only represents a portion of the overall water cycle which also includes atmospheric water and groundwater. Wetland systems are directly linked to groundwater and a crucial regulator of both the quantity and quality of water found below the ground. Wetland systems that are made of permeable sediments like limestone or occur in areas with highly variable and fluctuating water tables especially have a role in groundwater replenishment or water recharge. Sediments that are porous allow water to filter down through the soil and overlying rock into aquifers which are the source of 95% of the world's drinking water. Wetlands can also act as recharge areas when the surrounding water table is low and as a discharge zone when it is too high.

viii. Shoreline stabilisation and storm protection: Tidal and inter-tidal wetland systems protect and stabilize coastal zones. Coral reefs provide a protective barrier to coastal shoreline. Mangroves stabilize the coastal zone from the interior and will migrate with the shoreline to remain adjacent to the boundary of the water. The main conservation benefit these systems have against storms and tidal waves is the ability to reduce the speed and height of waves and floodwaters.

ix. Erosion control. By dissipating wave energy and stabilizing shorelines, wetland vegetation buffers the adjacent upland from wave action and intensive erosion. Flood damage reduction. Wetlands intercept runoff and store stormwater, thereby changing rapid and high peak flows to slower and smaller discharges over longer periods of time. Because it is usually the peak flows that cause flood damage, the effect of wetlands is to reduce the danger of flooding. A classic study by the Corps in the Charles River Basin in Massachusetts estimated that the loss of 3,400 hectares (approximately 8,100 acres) of forested wetlands would increase downstream flood damage, costing millions of dollars annually.

x. Nutrient retention: Wetlands cycle both sediments and nutrients balancing terrestrial and aquatic ecosystems. A natural function of wetland vegetation is the up-take and storage of nutrients found in the surrounding soil and water. These nutrients are retained in the system until the plant dies or is harvested by animals or humans. Wetland vegetation productivity is linked to the climate, wetland type, and nutrient availability. The grasses of fertile floodplains such as the Nile produce the highest yield including plants such as *Arundodonax* (giant reed), *Cyperus papyrus* (papyrus), *Phragmites* (reed) and *Typha* (cattail, bulrush).

xi. Sediment traps: Rainfall run-off is responsible for moving sediment through waterways. These sediments move towards larger and more sizable waterways through a natural process that moves water towards oceans. All types of sediments which may be composed of clay, sand, silt, and rock can be carried into wetland systems through this process. Reedbeds or forests located in wetlands act as physical barriers to slow waterflow and trap sediment.

xii. Water purification: Many wetland systems possess biofilters, hydrophytes, and organisms that in addition to nutrient up-take abilities have the capacity to remove toxic substances that

have come from pesticides, industrial discharges, and mining activities. They slow water movement, allowing silt and sediment to settle. Wetland plants filter pollutants and remove high levels of phosphorus and nitrogen. Reed beds, for example, make wonderful natural filter

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systems to remove pollutants and organic waste from water. Reed stems block the passage of larger particles, and microbes in the root systems literally 'eat' certain noxious pollutants, breaking them down into harmless chemicals in the process.

xiii. As buffer zone. By storing water and slowing water movement, wetlands buffer surrounding areas from the worst effects of storms and floods. Wetland plants stabilize shorelines and wetland edges. Mangroves, for example, control the effects of floods; protect shorelines from erosion; trap sediments, toxins and nutrients, and act as wind breaks to buffer against storms. The value of intact mangrove swamps in Malaysia for storm protection and flood control alone has been estimated as US\$ 300,000 per kilometer - the cost of replacing them with rock walls. The value of one kilometer of coral reef ranged from US\$ 137,000 to US\$ 1.2 million over a 25 year period, based on the economic value of storm protection, fishing and tourism.

xiv. As producers. They are nature's larder and provider. Wetlands have supplied people with products for thousands of years. Throughout the world, people have found food, medicines, building materials, industrial oils and dyes, textiles and paper products in wetlands. Over half the human population depends on a wetland product - rice - for its staple diet. Wetlands also support the world's great fisheries and shellfish nurseries. We all use wetland products every day. Here are some of them. Prawn curry - tropical prawns form the basis of major fisheries in Asia. They depend on mangroves as nurseries. Other economically important shellfish worldwide include shrimps, crabs, clams, mussels and oysters.

xv. As habitats and sanctuary. Often thought of as wastelands, wetlands are, in fact, among the world's most productive environments and support high concentrations of life (biodiversity). Freshwater wetlands hold more than 40% of the world's known species, including 12% of all animal species. From northern rivers where salmon spawn to crocodile-rich tropical swamps, wetlands provide food and shelter for species of most major animal groups from microbes, plants and invertebrates (animals without backbones) to fish, amphibians, reptiles, birds and mammals. They are, for example, great havens for birds from storks to swans and ducks to divers. A 1991 study by James R. Chambers determined that approximately 75 percent (by weight) of commercially harvested fish and shellfish are dependent on estuaries and their wetlands.

xvi. Support for birds and other wildlife. Wetlands are probably best known for their value to waterfowl. The freshwater wetlands in the prairie pothole region of North America support an estimated 50 to 80 percent of the continental waterfowl production each year. The loss of wetlands in this region, which is estimated to be more than 50 percent of the original wetland acreage occurring at the time of settlement, has been considered a major factor in the decline in nesting success of duck populations in North America. Wetlands also support a wide diversity of other birds. Eighty percent of America's breeding bird population and more than 50 percent of the 800 species of protected migratory birds rely on wetlands. In addition to

birds, other wildlife makes its home in wetlands. Reptiles and amphibians are common wetland residents. Nearly all of the approximately 190 species of amphibians in North America depend on wetlands for breeding. Other wildlife associated with wetlands includes muskrat, beaver, mink, raccoon, marsh and swamp rabbits, numerous mice, voles, shrews, lemmings, and other small mammals. Large mammals also rely on wetlands. For example, moose often depend on wetlands such as white cedar swamps and other forested wetlands for winter shelter and food.

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xvii. Aesthetics and recreation. Many recreational activities take place in and around wetlands. Hunting and fishing are popular activities associated with wetlands. Other recreational activities in wetlands include hiking, nature observation and photography, canoeing, and other boating. Many people simply enjoy the beauty and sounds of nature and spend their leisure time near wetlands observing plant and animal life. Wetlands are also important places for outdoor study and for gaining an appreciation of natural history and ecology. Properties bordering wetlands often have higher property values than those that do not. Urban wetlands are typically some of the last remaining pieces of “natural habitat” providing residents some sense of wildness and open space. In 1991 wetland-related ecotourism activities such as hunting, fishing, bird-watching, and photography added approximately \$59 billion to the national economy. According to the Pacific Coast Federation of Fishermen’s Associations, almost \$79 billion per year is generated from wetland-dependent species, or about 71 percent of the nation’s entire \$111 billion commercial and recreational fishing industry in 1997.

Wetlands in Nigeria

Nigeria is one of the fifty-three countries that make up the African continent. Out of these 53 countries, 23 are Contracting Parties to the Ramsar Convention. Africa's size and diversity of landscape are striking: bordered by the Mediterranean Sea to the north, the Indian and Atlantic oceans to the east and west respectively, and the Antarctic in the south, it covers 70° of latitude, several climatic zones, and a considerable altitudinal range.

Various wetland types characterize the diverse and panoramic Nigerian environment, from mountains reaching an altitude of 6,000m in the middle belt (Jos Plateau) and eastern parts (Cameroun Mountains) of the country through deserts in the far north to coastal zones at sea level in the south. Wetlands constitute about 2.6% of Nigeria's total surface area. Relatively little scientific investigation has been undertaken in them in comparison to other ecosystems such as forest and savannah or to wetlands in other parts of the world, their important role in support of the region's biodiversity and the livelihood of large human populations.

The country is richly endowed with abundant wetlands ecosystem, and it represent 2.6% of the country’s area of about 923,768km². The majority of it lies in Niger Delta and is mainly freshwater. Niger Delta accounts for 55% of freshwater wetland in Nigeria and the largest wetland in Africa. The Niger Delta is located along the Atlantic coast forms the southern boundary of Nigeria. It is the entrance of Rivers Niger and Benue into the ocean through a web of rivers, creeks, and estuaries. It is the largest wetland in Africa and the third largest in the

world, with about 2370 square kilometers of rivers, creeks and estuaries. Its vegetation is predominantly of the forest type with 8600 square kilometers of swamp forest and about 1900 square kilometers of mangrove forests.

General Distribution and Diversity of Wetland Types

Nigeria's wetlands fall into two major categories;

1. The Coastal Wetlands (Mangrove Swamps), and
2. The Freshwater Wetlands (Floodplains).

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The mangrove swamps covers an area of 9,000km² in the coastal States of Akwa Ibom, Cross River, Rivers, Bayelsa, Delta, Edo, Ondo, Ogun and Lagos while the floodplains covers an area of 2,585 km² mostly along Niger/Benue River system . Other pockets of wetland are scattered round the country.

The wetland displays a richness and uniqueness in biodiversity. From the mangrove swamps and freshwater forests in the south to the massive floodplains scattered round the country, Nigeria's many wetland types support a great diversity of plants and animals, and their productivity provides the natural resources essential to the survival of a significant part the population.

Designated Ramsar Sites and Proposed Sites

The RAMSAR Convention is the name of the International Convention on Protection of Wetlands, particularly as waterfowl habitat. It was formed in 1971 by delegates who met at Ramsar, a town in Iran on the coast of the Caspian Sea. It is the only international treaty that seeks to conserve habitat rather than species. Over 100 countries have signed the Convention.

Nigeria presently has 11 sites designated as Wetlands of International Importance, with a surface area of 1,076,728 hectares. WWF Global Freshwater Programme has supported the Nigerian Ministry of Environment in developing these designations. The sites present a fascinating array of wetland types, including swamp forests, river floodplains, mangroves, and lakes and all of them are extremely important for their support for flora and fauna and for the ecosystem services they provide for the local communities. Most or all of them are formally state-owned but in practical terms under the customary control of local families and communities.

S/N Site Date of State Area Coordinates

Designation (Ha)

- 1 Nguru Lake (and Marma 2/10/2000 Jigawa 58,100 10°22'N
Channel) complex & Yobe 012°46'E
- 2 Apoi Creek Forests 30/04/2008 Bayelsa 29,213 05°47'N
004°42'E
- 3 Baturiya Wetland 30/04/2008 Kano 101,095 12°31'N
010°29'E
- 4 Dagona Sanctuary Lake 30/04/2008 Yobe 344 12°48'N
010°44'E

- 5 Foge Islands 30/04/2008 Kebbi & 4,229 10°30'N
Niger 004°33'E
- 6 Lake Chad Wetlands in 30/04/2008 Borno 607,354 13°04'N
Nigeria 013°48'E
- 7 Lower Kaduna-Middle 30/04/2008 Kwara & 229,054 08°51'N
Niger Floodplain Niger 005°45'E
- 8 Maladumba Lake 30/04/2008 Bauchi 1,860 10°24'N
009°51'E
- 9 Oguta Lake 30/04/2008 Imo 572 05°42'N
006°47'E
- 10 Pandam and Wase Lakes 30/04/2008 Nasarawa 19,742 08°42'N
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008°58'E
- 11 Upper Orashi Forests 30/04/2008 Rivers 25,165 04°53'N
006°30'E

There is a draft plan by the nation to designate four more sites: Chingurme, Ibom/Cross River estuary, Wawan Rafi Wetlands and Akassa coastal wetland as Ramsar wetlands designated sites in Nigeria.

Threats to Wetland Biodiversity and Future Prospects

Nigeria still has a significant number of pristine wetlands when compared to Europe and other parts of North America. Despite their importance, human activities and the changing climate are degrading wetlands faster than any other ecosystem in Nigeria. Most of the government's national policies are influenced by policies from different sectors, such as agriculture, national resources and energy.

Wetlands in Nigeria are experiencing immense pressure from human activities. Some of the threats to wetland are

- Drainage for agriculture and settlement
- Excessive exploitation by local communities
- Improperly planned development activities.
- Increasing population in conjunction with efforts to increase food security is escalating pressure to expand agriculture within wetlands.
- Wetlands are actually considered by some as the 'new frontier' in agriculture. Many floodplains and tidal wetlands have been reduced in size and their productivity declined. IWMI indicates that 66% of all listed African wetland are used for agriculture, and this wetland agriculture is valued at US\$1.1 billion.
- Extraction of natural resource from wetlands.
- Industrial and urban water demand use of water.
- Oil spillage, gas flaring and uncontrolled exploration and exploitation of crude oil.
- Poorly planned and managed (oil) infrastructure development are heavily impacting the Niger Delta in Niger.

- Exploitation of forest resources from the wetlands.
- Over-fishing
- Climate changed related events, such as unpredictable and decreasing rainfall, are expected to reduce the water stored in wetlands and exacerbate their overexploitation and fuel further conversion into agricultural lands.
- Other threats to Nigerian wetlands include changes in wetland water quality due to the effects of industrial effluent and agricultural pesticides, siltation from highland catchment areas, and introduction of alien species of flora and fauna leading to

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colonization by single species and loss of endemic species diversity. Perhaps one of the biggest single catastrophes has been the introduction to the coastal waters, estuarine, creeks and rivers of alien plant species, the Water Hyacinth, *Eichhornia crassipes* and Water Lettuce *Pistia stratiotes*, which threatens the existence of endemic flora.

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